BALEXMETAL

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PIR SANDWICH PANELS

TECHNICAL CATALOG

BALEXMETAL

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PIR PLUS wall panel
PIR STANDARD wall panel
PIR ROOF roof panel

Sandwich panels with polyurethane core

EN-2026-08-24

The content of this folder does not constitute a commercial offer in the understanding of the Civil Code regulations. Information included in this paper demonstrates only sample solutions that require consultation and specification by the designer of a particular building according to individual clients' needs. Balex Metal does not bear any responsibility in case of any technical irregularities or errors resulting from inappropriate application of information included in this paper.

TABLE OF CONTENTS

I. TECHNICAL INFORMATION CONCERNING ENCLOSURE MADE OF BALEXMETAL SANDWICH PANELS WITH A POLYURETHANE CORE

1. General information – about the company 7
2. Balex Metal's sandwich panels 7
3. PIR sandwich panels construction 8
4. PIR sandwich panels production technology 9
5. PIR panels types 9
6. The scope of panel application 10
7. PIR panels contact types 10
8. PIR STANDARD wall panel 11
9. PIR PLUS wall panel 12
10. PIR ROOF roof panel 13
11. Basic technical information 14
12. Material and cladding coatings 15
    12.1. Material 15
    12.2. Coatings 15
13. Facing colour scheme 16
14. Facings profiling scheme 17
15. Strength issues 19
16. Thermal performance 26
17. Acoustic insulation 27
18. Fire safety 28
19. Corrosion resistance 29
20. Fasteners 30
21. Fastening roof panels lengthwise 30
22. General guidelines on assembly 32
23. Guidelines on transport 34
24. Certification documents 34

II. DETAILED HOUSING SOLUTIONS MADE OF SANDWICH PANELS WITH A POLYURETHANE CORE: PIR STANDARD, PIR PLUS WALL PANELS & PIR ROOF ROOF PANELS

1. PIR STANDARD WALL PANEL 36
    1.1. ST01.1 PIR STANDARD wall panel - joint, profilation types - MODULE 1100 36
    1.2. ST01.2 PIR STANDARD wall panel - joint, profilation types - MODULE 1000 37
    1.3. ST02 Fastening panel - vertical panel set 38
    1.4. ST03 Fastening panel - horizontal panel set 39
    1.5. ST04.1 Fastening panels to end support - horizontal panel set - Option I 40
    1.6. ST04.2 Fastening panels to end support - horizontal panel set - Option II 41
    1.7. ST05 Ground beam wall panels start - vertical panel set 42
    1.8. ST06 Ground beam wall panels start - horizontal panel set 43
    1.9. ST07 Panels start below top of ground beam - vertical panel set 44
    1.10. ST08 Panels start below top of ground beam - horizontal panel set 45
    1.11. ST09.1 Corner joint - vertical panel set - Option I 46
    1.12. ST09.2 Corner joint - vertical panel set - Option II 47
    1.13. ST09.3 Corner joint - vertical panel set - Option III 48
    1.14. ST010.1 Corner joint - horizontal panel set - Option I 49
    1.15. ST010.2 Corner joint - horizontal panel set - Option II 50
    1.16. ST010.3 Corner joint - horizontal panel set - Option III 51
    1.17. ST11 Lengthwise panel joint - vertical panel set 52
    1.18. ST12 Panels - window joint - horizontal panel set 53
    1.19. ST13 Panels - window joint - vertical panel set 54
    1.20. ST14.1 Panels - window joint - without additional construction - Y-Y 55
    1.21. ST14.2 Panels - window joint - without additional construction - X-X 56

2. PIR PLUS WALL PANEL 57
2.1. PL01 PIR PLUS wall panel - joint, profilation types - MODULE 1000 57
2.2. PL02 Fastening panel - vertical panel set 58
2.3. PL03 Fastening panel - horizontal panel set 59
2.4. PL04.1 Fastening panels to end support - horizontal panel set - Option I 60
2.5. PL04.2 Fastening panels to end support - horizontal panel set - Option II 61
2.6. PL04.3 Fastening panels to end support - horizontal panel set - Option III 62
2.7. PL05 Ground beam wall panels start - vertical panel set 63
2.8. PL06 Ground beam wall panels start - horizontal panel set 64
2.9. PL07 Panels start below top of ground beam - vertical panel set 65
2.10. PL08 Panels start below top of ground beam - horizontal panel set 66
2.11. PL09.1 Corner joint - vertical panel set - Option I 67
2.12. PL09.2 Corner joint - vertical panel set - Option II 68
2.13. PL09.3 Corner joint - vertical panel set - Option III 69
2.14. PL010.1 Corner joint - horizontal panel set - Option I 70
2.15. PL010.2 Corner joint - horizontal panel set - Option II 71
2.16. PL010.3 Corner joint - horizontal panel set - Option III 72
2.17. PL11 Lengthwise panel joint - vertical panel set 73
2.18. PL12 Panels - window joint - horizontal panel set 74
2.19. PL13 Panels - window joint - vertical panel set 75
2.20. PL14.1 Panels - window joint - without additional construction - Y-Y 76
2.21. PL14.2 Panels - window joint - without additional construction - X-X 77

4. PIR ROOF ROOF PANEL 78
4.1. R01 PIR ROOF roof panel - joint, profilation types- MODULE 1000mm 78
4.2. R02 Fastening type 79
4.3. R03 Monopith roof 80
4.4. R04.1 Gable wall 81
4.5. R04.2 Gable wall 82
4.6. R05 Roof parapet joint 83
4.7. R06 Roof ridge joint 84
4.8. R07 Internal gutter joint 85
4.9. R08.1 Roof eave joint 86
4.10. R08.2 Roof eave joint 87
4.11. R09.1 Panel lenthwise joint 88
4.12. R09.2 Panel lenthwise joint 89
4.13. R10 Internal gutter joint near parapet 90
4.14. R11.1 Roof ridge skylight-longitudal section 91
4.15. R11.2 Roof ridge skylight-cross section 92
4.16. R12 Roof skylight near ridge 93
4.17. R13.1 Roof skylight - longitudal section 94
4.18. R13.2 Roof skylight - cross section 95
4.19. R14.1 Straight lock panel joint 96
4.20. R14.2 Straight lock panel joint 97
4.21. R15.1 MW ROOF joint 98
4.22. R15.2 MW ROOF joint 99

I. TECHNICAL INFORMATION CONCERNING ENCLOSURES MADE OF BALEXMETAL SANDWICH PANELS WITH A POLYURETHANE CORE

1. GENERAL INFORMATION – ABOUT THE COMPANY

Balex Metal Sp. z o.o. is the leading manufacturer of steel construction materials in Poland. The Company's offer includes complete solutions as well as steel roofing and façade systems for housebuilding industry, commercial and agricultural construction.

The range of products is recognised by customers in Poland, Lithuania, Latvia, Estonia, Ukraine, Czech Republic, Slovakia, Germany, Denmark, Sweden and Norway. Consulting and sales services are provided through own network of regional branches, cooperating distributors and a team of professional sales advisors.

Its leading position in the market of manufacturing double-clad insulating core sandwich panels, the Balex Metal Company owes to its technologically advanced production lines purchased from the most renowned European companies, the team of employees with excellent qualifications as well as its special attention to quality.

2. BALEX METAL'S SANDWICH PANELS

Investors, architects, general contractors and construction companies performing investment projects are interested in commercially available, complete solutions. At the same time, they expect comprehensive deliveries of all necessary elements and construction materials included in the design. In order to meet these expectations, BALEX METAL included in its offer complete solutions of curtain walls and roof coverings.

The key elements of these solutions are wall and roof sandwich panels comprising two steel sheet skins bond with a structural-insulating core. The Balex Metal offer includes steel clad panels with two types of insulating core:

- sandwich panels with mineral wool core of fibres oriented perpendicularly to facing, marketed under the commercial name MW
- sandwich panels with a polyisocyanurate core designated as PIR and covered by the present catalogue

The basic type of wall sandwich panels are panels with visible fastening (designated as PIR STANDARD), which are fastened through to the supporting structure with an exposed fastener. The other type of wall sandwich panels are panels with fasteners not visible on the façade (designated as PIR PLUS). A specially designed panel lock conceals points of fastening and makes them not visible on a finished façade.

Roof sandwich panels are characterized by very deep profiling of their exterior trapezoidal facing. This is related to carrying long-term loads, including snow and own weight accounting for creep.

Besides sandwich panels, the offer also include a wide range of various elements, such as steel flashings, accesso-ries (fasteners, screws, rivets), sealing materials, windows and doors adapted for fastening to sandwich panels, rooflights as well as systems of gutters and downpipes.

Compatibility is the feature distinguishing the BALEX's wall sandwich panels.

This feature is used when designing joints of fire separation walls with exterior walls of a structure. An example of this is a vertical arrangement joint of polyurethane core panels with mineral wool core panels. This way, achieved is a vertical strip in an exterior wall made of non-flammable material of demanded fire resistance at various fire zones.

When designing a joint of a flooring and a spandrel wall, which should be executed of non-combustible materials of appropriate fire resistance, it is appropriate to combine mineral wool core panel elements with panels of other core types.

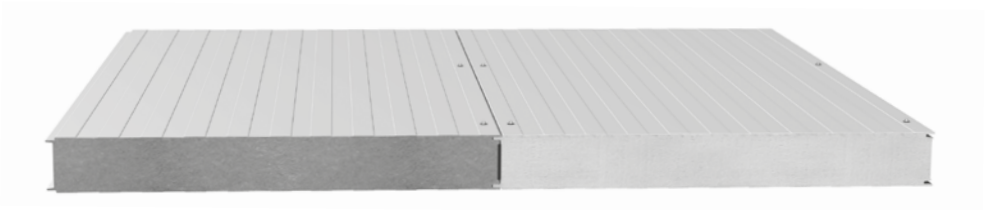


FIG. 1. A joint of the MW STANDARD and PIR STANDARD wall panels

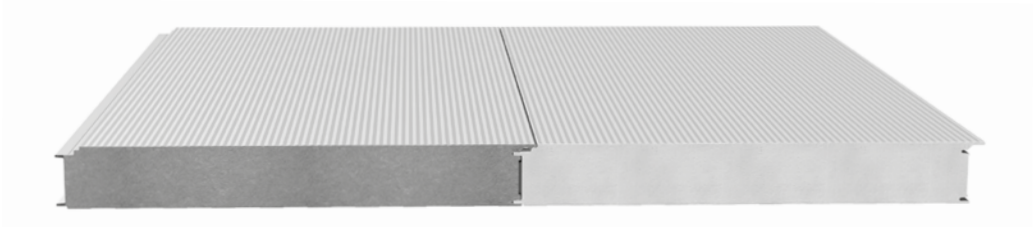


FIG. 2. A joint of the MW PLUS and PIR PLUS wall panels

3. PIR SANDWICH PANELS CONSTRUCTION

BALEX METAL company provides wide range of modern wall and roof panels with polyurethane core marked with PIR trade name.

PIR sandwich panels consist of two stainless steel facings and a construction and insulation core. The core is made of non-freon polyurethane foam, with density of 40 kg/m^3 (environmentally friendly due to the applied foaming agent) and it is responsible for transmitting shear stress, maintaining fixed distance between facing panels and ensuring high heat insulation values.

Sandwich panel's facings are made of S250GD steel metal sheet with thickness range from 0.40 mm to 0.70 mm and galvanized. Facing panels are responsible for transmitting normal stress and securing a particular facility against weather conditions. Such construction of the panel makes it very light retaining its high capacity and rigidity at the same time and enables increasing the span of supports (purlins and girders).

The variety of facing profiles along with their wide colour scheme allows architects and users to shape the elevation of particular buildings retaining proportions between their aesthetic and functional values.

4. PIR SANDWICH PANELS PRODUCTION TECHNOLOGY

The production of PIR sandwich panels with a polyurethane core was started in spring 2004. Pentane is used as the foaming agent which makes the production process environmentally friendly i.e. not damaging the ozone layer. As a result the technological process of producing sandwich panels with polyurethane core consists in injection of mixed components forming rigid polyurethane foam between two continuously moving steel strips (with previously profiled longitudinal joints and the main profile) and applying sealing strip and aluminium foil into the longitudinal panel joint at the same time. High quality and fixed repeatability of technical parameters of PIR sandwich panels has been achieved due to the application of top class materials and continuous control of production.

5. PIR PANELS TYPES

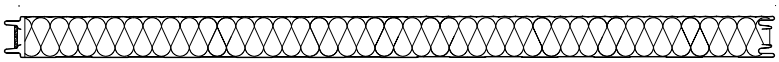
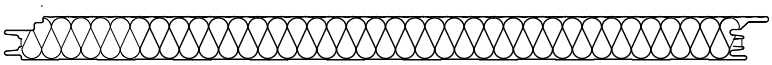
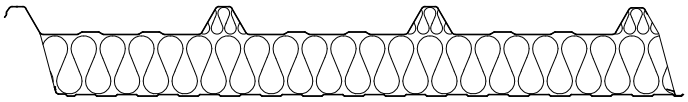
We provide five types of PIR panels:

PIR STANDARD wall panel – standard wall panel with modular width (so called ‚covering width') 1000 mm or 1100 mm enables quick assembly. Standard wall panel is fastened to the construction with the use of eyelet fasteners.

PIR PLUS wall panel – wall panel with concealed fastening, not visible from the elevation side with modular width of 1000 mm. The fastening concealed from the elevation side as well as different types of profilings: make the panels attractive in terms of both architecture and functionality.

PIR ROOF roof panel – roof panel with a modular width of 1000 mm, featuring a diagonal lock, no gaskets, a maximum fire resistance rating of RE 60, and a trapezoidal outer surface, provides high load-bearing capacity both when supporting live loads and during installation.

Table 2. Types of BALEX METAL sandwich panels

Panel type	Panel thickness [mm]	Panel shape
1	2	3
PIR STANDARD wall panel	40 50 60 80 100 120	
PIR PLUS wall panel	60 80 100 120	
PIR ROOFroof panel	40/85 60/105 80/125 100/145 120/165 150/195 160/205 180/225	

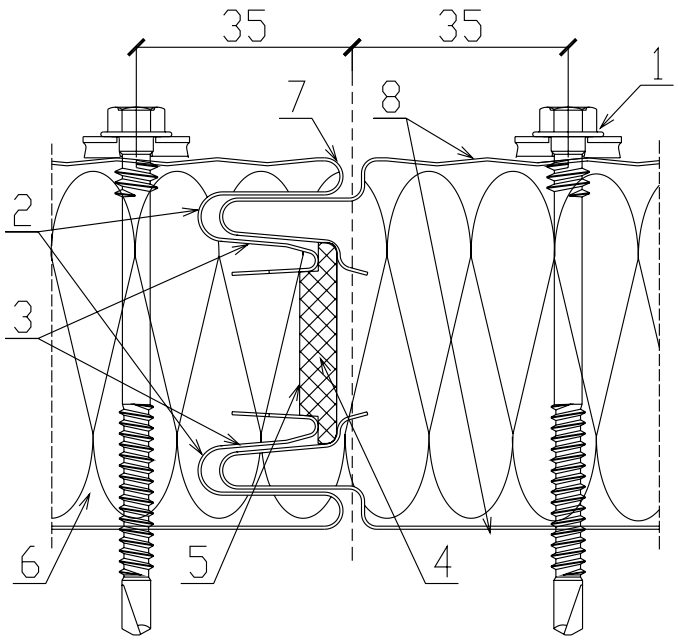
6. THE SCOPE OF PANEL APPLICATION

Sandwich panels are commonly used building material for light constructions of industrial halls, warehouses, sports and production halls, pavilions and commercial facilities, offices, social buildings, garages, workshops, administrative and public buildings, storehouses, freezers and cold stores (including food industry facilities). Wide colour palette and variety of profiling shapes make it possible to realize many construction projects. The panel's construction enables fast and easy assembly regardless of weather conditions in both vertical and horizontal arrangement. PIR STANDARD and PIR PLUS wall panels can also be applied on self-supporting suspended ceilings. The designer guided by the destination of a particular facility, operating conditions, possible influence of the internal environment as well as weather conditions decides on the kind and the arrangement of sandwich panels. Taking low heat-transfer coefficient into account PIR sandwich panels seem to be a perfect choice for heated buildings' elevations offering a possibility of complex and quick completion of the construction project. PIR panels are designed for low and normal temperatures. Fixed temperature on the panel's surface should not exceed + 60°C.

7. PIR PANELS CONTACT TYPES

A new structural solution has been implemented in PIR sandwich panels. Unique shape of longitudinal joints with optimal proportion between the thickness of feather and the depth of the groove on both claddings on the internal and external side had considerable effect on increasing the fire resistance parameters of wall panels. This solution has been applied in both PIR STANDARD and PIR PLUS panels. The appropriate joint shape guarantees good thermal performance and leaktightness to rain waters, air and water vapour penetration. In the case of PIR STANDARD roof panel the lower cladding is factory-removed at one end of the panel together with the core as standard in order to facilitate joining panels lengthwise and the assembly of gutters.

8. PIR STANDARD WALL PANEL



1) Joint fasteners

2) Unique both sided form of the panel joint in the shape of a double lock.

3) Conical inclination of the panel's internal joint surface facilitating assembly.

4) Continuous polyurethane sealing strip or polyurethane sealing strip laminated with aluminium foil and additionally covered with carbon fibre cloth applied in the production process, preventing water vapour penetration and retaining good thermal performance and increasing fire resistance.

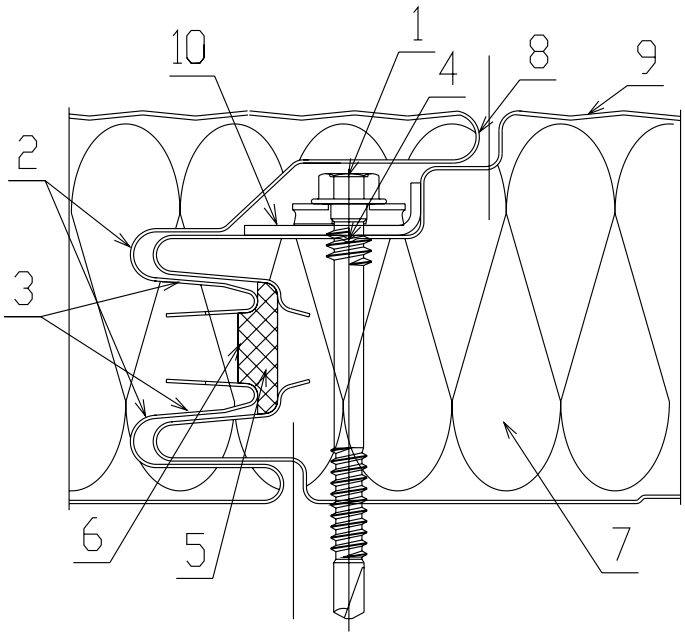
5) Aluminium foil or aluminium foil laminated with high fire resistance material preventing water vapour penetration and gas diffusion in order to maintain a fixed heat-transfer coefficient and increasing fire resistance.

6) A core made of rigid polyurethane foam produced on the basis of environmentally and ozone layer friendly technology with the lowest heat-transfer coefficient in comparison with other thermal insulation materials.

7) A technology of profiling the shape of metal claddings retaining the integrity and durability of protection coatings.

8) Wide range of external cladding profiles meeting stiff architectural requirements.

9. PIR PLUS WALL PANEL



1) Concealed fasteners masked with especially shaped tongue of the external panel's cladding.

2) Unique both sided form of the panel joint in the shape of a double lock.

3) Conical inclination of the panel's internal joint surface facilitating assembly.

4) Longitudinal groove that facilitates the process of positioning joint fasteners.

5) Continuous polyurethane sealing strip or polyurethane sealing strip laminated with aluminium foil and additionally covered with carbon fibre cloth applied in the production process, preventing water vapour penetration and retaining good thermal performance and increasing fire resistance.

6) Aluminium foil or aluminium foil laminated with high fire resistance material preventing water vapour penetration and gas diffusion in order to maintain a fixed heat-transfer coefficient and increasing fire resistance.

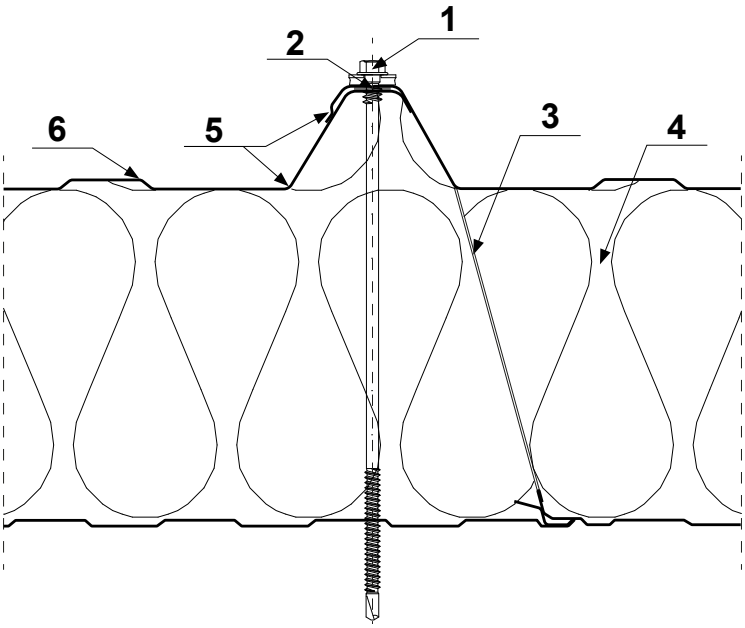
7) A core made of rigid polyurethane foam produced on the basis of environmentally and ozone layer friendly technology with the lowest heat-transfer coefficient in comparison with other thermal insulation materials.

8) A technology of profiling the shape of metal claddings retaining the integrity and durability of protection coatings.

9) Wide range of external cladding profiles meeting stiff architectural requirements.

10) Steel ŁB 25 washers in the panel lock increasing the joint's load-bearing capacity.

10. PIR ROOF ROOF PANEL



1) Joint fasteners.
2) Gasket installed during production
3) Aluminium foil preventing water vapour penetration and gas diffusion in order to maintain a fixed heat-transfer coefficient.
4) A core made of rigid polyurethane foam produced on the basis of environmentally and ozone layer friendly technology with the lowest heat-transfer coefficient in comparison with other thermal insulation materials.
5) A technology of profiling the shape of metal claddings retaining the integrity and durability of protection coatings.
6) Trapezoidal profile of the external cladding increasing the load-bearing capacity of the roof panel.

11. BASIC TECHNICAL INFORMATION

Table 3. Technical information

Panel type	Panel thickness [mm]	Cladding thickness [mm]		Panel weight [kg/m²]	Panel length L [m]	
		INT	EXT		min	max
1	2	3	4	5	6	7
PIR STANDARD wall panel	40	0,40*-0,70	0,50-0,70	10,3	2,00	18,00
	50			10,6		
	60			11,1		
	80			11,8		
	100			12,6		
	120			13,2		
PIR PLUS wall panel	60	0,40*-0,70	0,50-0,70	11,4	2,00	18,00
	80			12,1		
	100			12,9		
	120			13,8		
PIR ROOF roof panel	40/85	0,40*-0,70	0,50-0,70	10,84	2,50	16,00
	60/105			11,65		17,00
	80/125			12,45		18,00
	100/145			13,26		
	120/165			14,07		
	150/195			15,29		
	160/205			15,69		
	180/225			16,50		

** For panels with flat cladding profiling minimal steel thickness = 0.50mm*
Note: In the case of roof PIR panels thickness marking the first number refers to the core's thickness and the second one to the total thickness of the panel

12. MATERIAL AND CLADDING COATINGS

12.1. Material

STEEL S250GD + ZINC

- steel with increased parameters, galvanized on both sides and permanently secured with anti-corrosion coating
- metal sheet thickness: 0,40 - 0,70
- coated with organic and metallic coatings

INOX *(PIR STANDARD)*

12.2. Coatings

PREMIUM offer:

Polyurethane 55µm - PUR55
- polyurethane coating with polyamid of overall thickness 55 µm
- extraordinary resistance for corrosion – up to RC5
- good resistance to intensive UV radiation – RUV4
- solution for standard, demanding and agressive environments
- aesthetic and colour durability for the whole life cycle
- used for roof coverings, wall claddings, standard environments, as well as those agressive and demanding: cold, humid, of high UV radiation, industrial and polluted
- colors are presented on the company's website

Standard offer:

Polyester 25µm - PE25
- for outdoor applications - coating thickness 25 µm: resistant to changing temperature and weather conditions, good corrosion resistance
- for indoor applications coating thickness 15 µm: indoor layers of walls and roofs
- colors are presented on the company's website

Polyester 35µm Mat - PE35M
- coating thickness 35 µm
- for outdoor applications: resistant to changing temperature and weather conditions, good corrosion resistance
- perfectly suitable for roofs of commercial and industrial facilities
- colors are presented on the company's website

Polyester 35µm - PE35
- coating thickness 35 µm
- colors: GOLDEN OAK, WINCHESTER

Special orders offer:

PVC 120µm (foodsafe) - PVC120
- coating thickness 120 µm
- white foil
- special coating with increased hardness
- it can be applied in food processing facilities and cold stores, easily washable and resistant to the majority of cleaning agents.

13. FACING COLOUR SCHEME

Colour scheme according to the Balex Metal World of Colours palette

PREMIUM coating

Polyurethane 55µm - PUR55	7016, 8017, 9005, 9006, 9007, 9010

Organic coatings:

Polyester 25µm - PE25	9010, 8004, 8019, 6005, 9005, 3011, 7024, 7016, 7035, 6020, 3016, 1015, 3000, 6011, 9006, 5010, 1003, 9007, 9002, 7047, 7040, 7012, 5003, 6018, 8017
Polyester 35µm Mat - PE35M	8004, 8017, 3011, 8019, 6020, 7024, 7016, 9005
Polyester 35µm	Golden Oak, Winchester
PVC 120µm (foodsafe) - PVC120	9010

Table 4. Classification of colours by relative brightness

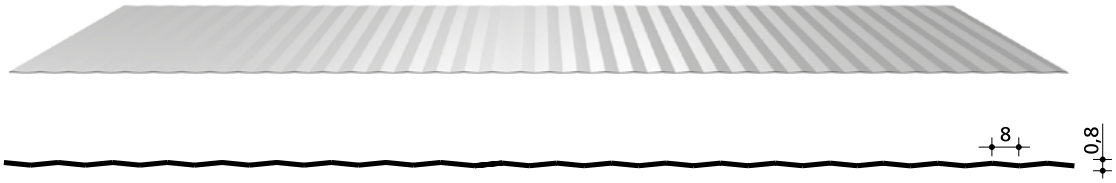
Symbol	Name	Group
9010	**white**	**very bright**
9002	**grey-white**	
7035	**light-grey**	
1015	**ivory**	
6011	**reseda-green**	**bright**
9006	**silver-metallic**	
9007	**grey-aluminium**	
1003	**signal-yellow**	
9005	**black**	**dark**
5010	**signal-blue**	
6005	**dark-green**	
6020	**fir-green**	
7024	**graphite-grey**	
7016	**graphite**	
8019	**brown-grey**	
8017	**chocolate-brown**	
8012	**red-brown**	
3016	**coral-red**	
3011	**red**	
3000	**fiery red**	

Not all colors are available for thicknesses (0.4, 0.5, 0.6, 0.7).
For more information please contact the sales representative of Balex Metal.

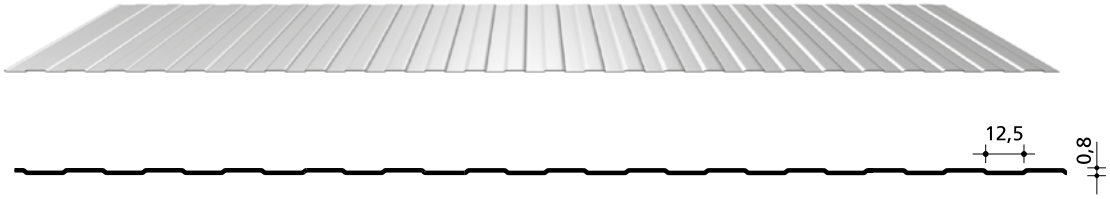
14. FACINGS PROFILING SCHEME

Sandwich panels with polyurethane core are distinguished by wide variety of available profiles, especially in the case of external elevation facing. Profile types:

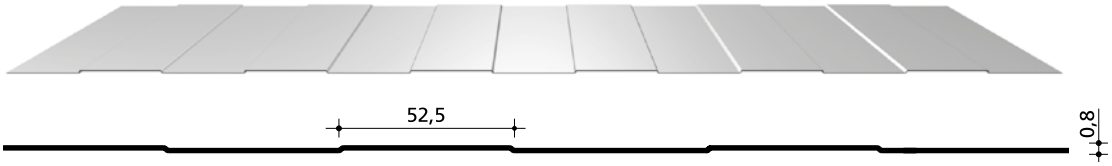
M16 = MICROPROFILED



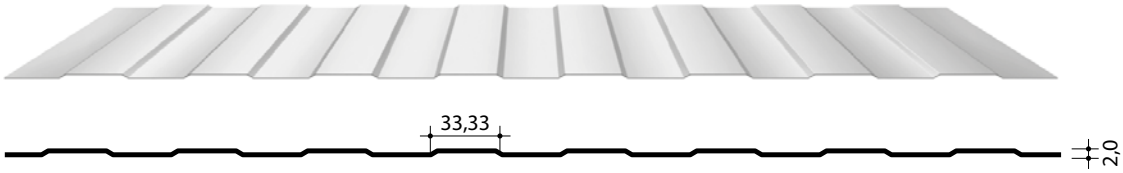
S = SOFTLINE



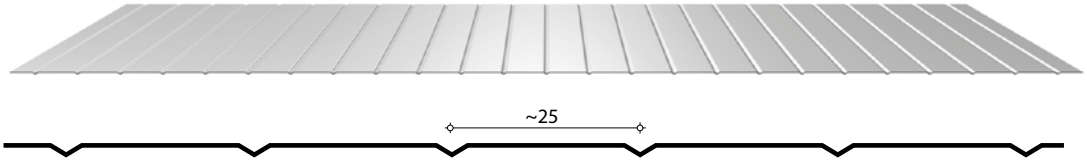
L = LINED



D = DEEP LINED



R = GROOVED



F = FLAT*

**For claddings of minimal thickness 0.50 mm*



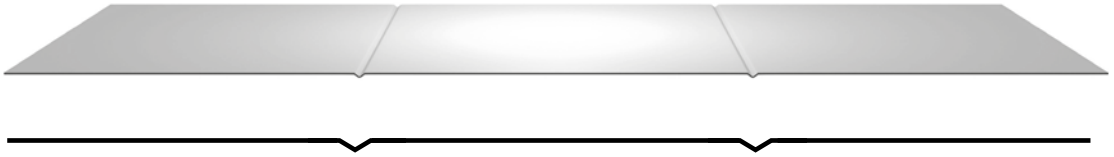
1L = CLEARLINE*

**For claddings thickness 0.50 - 0.60mm*

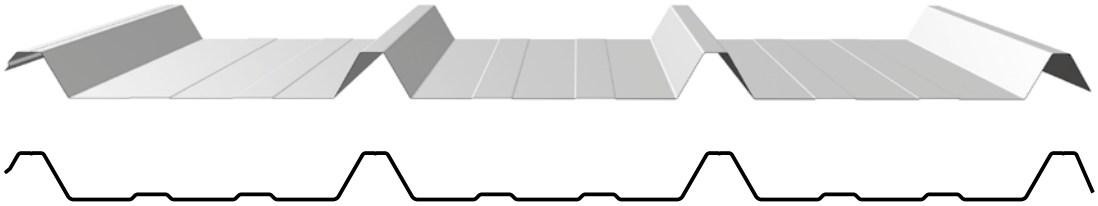


2L = DOUBLE CLEARLINE*

**For claddings thickness 0.50 - 0.60mm*



T = TRAPEZOIDAL



** 0.5 mm thickness for F/1L/2L profiling of outside cladding is available after signing the statement, which contains the acceptance of possibility of apearing visible deflection (within standard tolerance up to 0,6 mm/200 mm acc. to EN 14509).*
Stainless steel cladding not available.

The table below includes possible combinations of profile types of external and internal facings for individual panel types.

Table 5. Profile type combinations

Panel type	External facing							Internal facing		
	M16	S	L	F	1L	2L	T	L	D	F
PIR STANDARD wall panel	●		●	●	●	●		●		●
PIR PLUS 1000 wall panel	●	●	●	●	●	●		●		●
PIR ROOF roof panel							●		●	●

15. STRENGTH ISSUES

1. The following assumptions were made in the process of designing the load-bearing capacity and rigidity tables for PIR sandwich panel application:

   a) the load-bearing capacity limit is considered exceeded if real load exceeds the acceptable breaking load whose value has been calculated by means of dividing the breaking load obtained in the carried out strength tests by 2.

   b) the usability limit which means rigidity in the case of sandwich panels is considered exceeded if the deflection of wall and roof panels under short-term load exceeds 1/200 of the span spread and 1/100 of the span spread taking long-term load into consideration.

2. The scope of application of PIR panels regarding load-bearing capacity and rigidity should be in accordance with the attached tables. The values of acceptable loads indicated in the tables take the following into account:

   a) the influence of heat load caused by the difference of temperatures between the external and internal cladding (t_{int} = 25°C in summer and text t_{int}= 20°C in winter). For calculating heat loads was made an assumption of tempe-rature's difference Δt=50°C.

   b) the influence of long-term loads (in the case of roof panels)

   c) the most unfavourable load combination

   d) the increase in deflection in case of load acting towards the support, in the case of fastening panels with the use of two fasteners widthways.

3. Maximum load values provided in the table should be compare with characteristic loads:

4. Maximum loads provided in the tables have been specified for panels in **light** and **very light colours**. For checking the load capacity of dark coloured sandwich panles contact Balex Metal's design office.

5. The tables contain the following profiles:

   a) for wall plates: micro profiling on the outside and lining on the inside

   b) for roof plates: trapezoidal on the outside and lining on the inside

   In order to specify permissible loads for other profile patterns please contact the design office of Balex Metal.

6. In order to establish acceptable loads for span spreads not included in the tables you can use interpolation.

7. The tables give widths of supports for maximum loads. In other cases, the minimum width of intermediate supports is 60 mm, and edge supports 40 mm.

8. The panel support span in edge zones should be reduced appropriately in relation to the value provided in the tables.

9. Given values in the field of loads going from the support can be applied if the element to which a panel is fastened is not less than 1.50 mm thick.

10. The load capacity tables cover standard S250GD steel. For information on the load capacity of panels using stainless steel (INOX), please contact the Balex Metal Design Office - **bp@balex.eu.**

PIR STANDARD wall sandwich panel

1-span system - maximum characteristic loads for PU-PIR-W-ST sandwich panels with a polyurethane core, in claddings of 0,50/0,50 mm thickness, profile types: M16/L, 40 mm outer support width. Direction of force - TO THE SUPPORT

Core thickness	Colour group	Wall sandwich panel PIR STANDARD (PU-PIR-W-ST) Maximum loads [kN/m²] for the span [m]																												
		1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50	3,75	4,00	4,25	4,50	4,75	5,00	5,25	5,50	5,75	6,00	6,25	6,50	6,75	7,00	7,25	7,50	7,75	8,00
40	I	3,90	3,25	2,78	2,43	2,15	1,72	1,38	1,10	0,89	0,72	0,59	0,48	0,38	0,29	0,22	0,16	0,12	0,08	0,05	0,03	0,01	0,00	0,00	0,00	0,00	0,00	0,00	0,00
	II	3,90	3,25	2,78	2,43	2,15	1,72	1,38	1,10	0,89	0,72	0,59	0,48	0,38	0,29	0,22	0,16	0,12	0,08	0,05	0,03	0,01	0,00	0,00	0,00	0,00	0,00	0,00	0,00
	III	3,90	3,25	2,78	2,43	2,15	1,72	1,38	1,10	0,89	0,72	0,59	0,48	0,38	0,29	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
50	I	4,15	3,46	2,96	2,59	2,30	2,07	1,88	1,63	1,36	1,13	0,95	0,79	0,66	0,56	0,47	0,40	0,33	0,27	0,22	0,17	0,14	0,11	0,08	0,06	0,04	0,03	0,00	0,00
	II	4,15	3,46	2,96	2,59	2,30	2,07	1,88	1,63	1,36	1,13	0,95	0,79	0,66	0,56	0,47	0,40	0,33	0,27	0,22	0,17	0,14	0,11	0,08	0,06	0,04	0,03	0,00	0,00
	III	4,15	3,46	2,96	2,59	2,30	2,07	1,88	1,63	1,36	1,13	0,95	0,79	0,66	0,56	0,47	0,40	0,33	0,27	0,22	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
60	I	4,55	3,79	3,24	2,84	2,52	2,27	2,06	1,89	1,74	1,52	1,30	1,12	0,97	0,83	0,71	0,61	0,52	0,45	0,40	0,34	0,30	0,25	0,21	0,17	0,14	0,12	0,10	0,08
	II	4,55	3,79	3,24	2,84	2,52	2,27	2,06	1,89	1,74	1,52	1,30	1,12	0,97	0,83	0,71	0,61	0,52	0,45	0,40	0,34	0,30	0,25	0,21	0,17	0,14	0,12	0,10	0,08
	III	4,55	3,79	3,24	2,84	2,52	2,27	2,06	1,89	1,74	1,52	1,30	1,12	0,97	0,83	0,71	0,61	0,52	0,45	0,40	0,34	0,30	0,25	0,21	0,17	0,00	0,00	0,00	0,00
80	I	5,28	4,40	3,77	3,30	2,93	2,64	2,40	2,20	2,03	1,88	1,76	1,65	1,55	1,39	1,23	1,09	0,97	0,87	0,76	0,67	0,60	0,53	0,47	0,42	0,38	0,34	0,30	0,27
	II	5,28	4,40	3,77	3,30	2,93	2,64	2,40	2,20	2,03	1,88	1,76	1,65	1,55	1,39	1,23	1,09	0,97	0,87	0,76	0,67	0,60	0,53	0,47	0,42	0,38	0,34	0,30	0,27
	III	5,28	4,40	3,77	3,30	2,93	2,64	2,40	2,20	2,03	1,88	1,76	1,65	1,55	1,39	1,23	1,09	0,97	0,87	0,76	0,67	0,60	0,53	0,47	0,42	0,38	0,34	0,30	0,27
100	I	5,28	4,40	3,77	3,30	2,93	2,64	2,40	2,20	2,03	1,88	1,76	1,65	1,55	1,46	1,39	1,32	1,25	1,20	1,14	1,03	0,94	0,85	0,77	0,70	0,63	0,57	0,52	0,47
	II	5,28	4,40	3,77	3,30	2,93	2,64	2,40	2,20	2,03	1,88	1,76	1,65	1,55	1,46	1,39	1,32	1,25	1,20	1,14	1,03	0,94	0,85	0,77	0,70	0,63	0,57	0,52	0,47
	III	5,28	4,40	3,77	3,30	2,93	2,64	2,40	2,20	2,03	1,88	1,76	1,65	1,55	1,46	1,39	1,32	1,25	1,20	1,14	1,03	0,94	0,85	0,77	0,70	0,63	0,57	0,52	0,47
120	I	4,93	4,11	3,52	3,08	2,74	2,46	2,24	2,05	1,89	1,76	1,64	1,54	1,45	1,37	1,29	1,23	1,17	1,12	1,07	1,02	0,99	0,95	0,92	0,89	0,85	0,81	0,74	0,69
	II	4,93	4,11	3,52	3,08	2,74	2,46	2,24	2,05	1,89	1,76	1,64	1,54	1,45	1,37	1,29	1,23	1,17	1,12	1,07	1,02	0,99	0,95	0,92	0,89	0,85	0,81	0,74	0,69
	III	4,93	4,11	3,52	3,08	2,74	2,46	2,24	2,05	1,89	1,76	1,64	1,54	1,45	1,37	1,29	1,23	1,17	1,12	1,07	1,02	0,99	0,95	0,92	0,89	0,85	0,81	0,74	0,69

2-span system - maximum characteristic loads for PU-PIR-W-ST sandwich panels with a polyurethane core, in claddings of 0,50/0,50 mm thickness, profile types: M16/L, 40 mm outer support width, 60 mm inner support width. Direction of force - TO THE SUPPORT

Core thickness	Colour group	Wall sandwich panel PIR STANDARD (PU-PIR-W-ST) Maximum loads [kN/m²] for the span [m]																												
		1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50	3,75	4,00	4,25	4,50	4,75	5,00	5,25	5,50	5,75	6,00	6,25	6,50	6,75	7,00	7,25	7,50	7,75	8,00
40	I	3,43	2,86	2,45	2,14	1,90	1,71	1,56	1,43	1,28	1,08	0,94	0,81	0,71	0,62	0,55	0,50	0,45	0,40	0,37	0,33	0,31	0,28	0,26	0,24	0,22	0,21	0,00	0,00
	II	3,43	2,86	2,45	2,14	1,90	1,71	1,56	1,43	1,28	1,08	0,94	0,81	0,71	0,62	0,55	0,50	0,45	0,40	0,37	0,33	0,31	0,28	0,26	0,24	0,22	0,21	0,00	0,00
	III	3,43	2,86	2,45	2,14	1,90	1,71	1,56	1,43	1,28	1,08	0,94	0,81	0,71	0,62	0,55	0,50	0,45	0,40	0,37	0,33	0,31	0,28	0,26	0,24	0,22	0,21	0,00	0,00
50	I	3,70	3,08	2,64	2,31	2,05	1,85	1,68	1,54	1,41	1,18	1,00	0,87	0,76	0,67	0,59	0,53	0,47	0,42	0,39	0,35	0,32	0,29	0,27	0,25	0,23	0,22	0,20	0,19
	II	3,70	3,08	2,64	2,31	2,05	1,85	1,68	1,54	1,41	1,18	1,00	0,87	0,76	0,67	0,59	0,53	0,47	0,42	0,39	0,35	0,32	0,29	0,27	0,25	0,23	0,22	0,20	0,19
	III	3,70	3,08	2,64	2,31	2,05	1,85	1,68	1,54	1,41	1,18	1,00	0,87	0,76	0,67	0,59	0,53	0,47	0,42	0,39	0,35	0,32	0,29	0,27	0,25	0,23	0,22	0,20	0,19
60	I	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,45	1,20	1,01	0,88	0,76	0,66	0,57	0,50	0,44	0,39	0,35	0,32	0,29	0,26	0,24	0,22	0,20	0,19	0,17	0,16
	II	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,45	1,20	1,01	0,88	0,76	0,66	0,57	0,50	0,44	0,39	0,35	0,32	0,29	0,26	0,24	0,22	0,20	0,19	0,17	0,16
	III	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,45	1,20	1,01	0,88	0,76	0,66	0,57	0,50	0,44	0,39	0,35	0,32	0,29	0,26	0,24	0,22	0,20	0,19	0,17	0,16
80	I	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,14	1,00	0,85	0,73	0,64	0,56	0,49	0,44	0,39	0,36	0,32	0,29	0,27	0,25	0,23	0,21	0,19
	II	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,14	1,00	0,85	0,73	0,64	0,56	0,49	0,44	0,39	0,36	0,32	0,29	0,27	0,25	0,23	0,21	0,19
	III	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,14	1,00	0,85	0,73	0,64	0,56	0,49	0,44	0,39	0,36	0,32	0,29	0,27	0,25	0,23	0,21	0,19
100	I	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,23	1,15	1,00	0,86	0,75	0,65	0,57	0,50	0,45	0,40	0,36	0,33	0,30	0,27	0,25	0,23	0,21
	II	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,23	1,15	1,00	0,86	0,75	0,65	0,57	0,50	0,45	0,40	0,36	0,33	0,30	0,27	0,25	0,23	0,21
	III	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,23	1,15	1,00	0,86	0,75	0,65	0,57	0,50	0,45	0,40	0,36	0,33	0,30	0,27	0,25	0,23	0,21
120	I	3,70	3,08	2,64	2,31	2,05	1,85	1,68	1,54	1,42	1,32	1,23	1,15	1,08	1,02	0,98	0,93	0,89	0,83	0,73	0,65	0,58	0,52	0,47	0,43	0,39	0,36	0,33	0,31
	II	3,70	3,08	2,64	2,31	2,05	1,85	1,68	1,54	1,42	1,32	1,23	1,15	1,08	1,02	0,98	0,93	0,89	0,83	0,73	0,65	0,58	0,52	0,47	0,43	0,39	0,36	0,33	0,31
	III	3,70	3,08	2,64	2,31	2,05	1,85	1,68	1,54	1,42	1,32	1,23	1,15	1,08	1,02	0,98	0,93	0,89	0,83	0,73	0,65	0,58	0,52	0,47	0,43	0,39	0,36	0,33	0,31

3-span system - maximum characteristic loads for PU-PIR-W-ST sandwich panels with a polyurethane core, in claddings of 0,50/0,50 mm thickness, profile types: M16/L, 40 mm outer support width, 60 mm inner support width. Direction of force - TO THE SUPPORT

Core thickness	Colour group	Wall sandwich panel PIR STANDARD (PU-PIR-W-ST) Maximum loads [kN/m²] for the span [m]																												
		1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50	3,75	4,00	4,25	4,50	4,75	5,00	5,25	5,50	5,75	6,00	6,25	6,50	6,75	7,00	7,25	7,50	7,75	8,00
40	I	3,43	2,86	2,45	2,14	1,90	1,71	1,56	1,43	1,32	1,17	1,01	0,89	0,77	0,67	0,59	0,52	0,46	0,41	0,37	0,33	0,29	0,26	0,23	0,21	0,19	0,17	0,15	0,14
	II	3,43	2,86	2,45	2,14	1,90	1,71	1,56	1,43	1,32	1,17	1,01	0,89	0,77	0,67	0,59	0,52	0,46	0,41	0,37	0,33	0,29	0,26	0,23	0,21	0,19	0,17	0,15	0,14
	III	3,43	2,86	2,45	2,14	1,90	1,71	1,56	1,43	1,32	1,17	1,01	0,89	0,77	0,67	0,59	0,52	0,46	0,41	0,37	0,33	0,29	0,26	0,23	0,21	0,19	0,17	0,15	0,14
50	I	3,70	3,08	2,64	2,31	2,05	1,85	1,68	1,54	1,42	1,32	1,23	1,08	0,96	0,85	0,76	0,68	0,62	0,56	0,51	0,47	0,43	0,39	0,36	0,33	0,30	0,27	0,25	0,23
	II	3,70	3,08	2,64	2,31	2,05	1,85	1,68	1,54	1,42	1,32	1,23	1,08	0,96	0,85	0,76	0,68	0,62	0,56	0,51	0,47	0,43	0,39	0,36	0,33	0,30	0,27	0,25	0,23
	III	3,70	3,08	2,64	2,31	2,05	1,85	1,68	1,54	1,42	1,32	1,23	1,08	0,96	0,85	0,76	0,68	0,62	0,56	0,51	0,47	0,43	0,39	0,36	0,33	0,30	0,27	0,25	0,23
60	I	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,23	1,07	0,95	0,84	0,75	0,67	0,61	0,55	0,50	0,46	0,42	0,39	0,36	0,33	0,31	0,29	0,27	0,25
	II	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,23	1,07	0,95	0,84	0,75	0,67	0,61	0,55	0,50	0,46	0,42	0,39	0,36	0,33	0,31	0,29	0,27	0,25
	III	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,23	1,07	0,95	0,84	0,75	0,67	0,61	0,55	0,50	0,46	0,42	0,39	0,36	0,33	0,31	0,29	0,27	0,25
80	I	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,23	1,16	1,06	0,96	0,86	0,77	0,70	0,63	0,58	0,53	0,49	0,45	0,42	0,39	0,36	0,34	0,32
	II	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,23	1,16	1,06	0,96	0,86	0,77	0,70	0,63	0,58	0,53	0,49	0,45	0,42	0,39	0,36	0,34	0,32
	III	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,23	1,16	1,06	0,96	0,86	0,77	0,70	0,63	0,58	0,53	0,49	0,45	0,42	0,39	0,36	0,34	0,32
100	I	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,23	1,16	1,10	1,04	1,00	0,91	0,82	0,75	0,68	0,63	0,58	0,53	0,49	0,46	0,42	0,40	0,37
	II	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,23	1,16	1,10	1,04	1,00	0,91	0,82	0,75	0,68	0,63	0,58	0,53	0,49	0,46	0,42	0,40	0,37
	III	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,23	1,16	1,10	1,04	1,00	0,91	0,82	0,75	0,68	0,63	0,58	0,53	0,49	0,46	0,42	0,40	0,37
120	I	3,70	3,08	2,64	2,31	2,05	1,85	1,68	1,54	1,42	1,32	1,23	1,15	1,08	1,02	0,98	0,93	0,89	0,84	0,81	0,77	0,74	0,71	0,68	0,63	0,59	0,55	0,51	0,48
	II	3,70	3,08	2,64	2,31	2,05	1,85	1,68	1,54	1,42	1,32	1,23	1,15	1,08	1,02	0,98	0,93	0,89	0,84	0,81	0,77	0,74	0,71	0,68	0,63	0,59	0,55	0,51	0,48
	III	3,70	3,08	2,64	2,31	2,05	1,85	1,68	1,54	1,42	1,32	1,23	1,15	1,08	1,02	0,98	0,93	0,89	0,84	0,81	0,77	0,74	0,71	0,68	0,63	0,59	0,55	0,51	0,48

PIR STANDARD wall sandwich panel

1-span system - maximum characteristic loads for PU-PIR-W-ST sandwich panels with a polyurethane core, in claddings of 0,50/0,50 mm thickness, profile types: M16/L, 40 mm outer support width.
Direction of force - FROM THE SUPPORT

Core thickness	Colour group	Wall sandwich panel PIR STANDARD (PU-PIR-W-ST) Maximum loads [kN/m²] for the span [m]																												
		1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50	3,75	4,00	4,25	4,50	4,75	5,00	5,25	5,50	5,75	6,00	6,25	6,50	6,75	7,00	7,25	7,50	7,75	8,00
40	I	-3,90	-3,25	-2,78	-2,28	-1,80	-1,46	-1,20	-1,01	-0,87	-0,75	-0,65	-0,55	-0,46	-0,38	-0,32	-0,28	-0,24	-0,20	-0,16	-0,13	-0,10	-0,08	0,00	0,00	0,00	0,00	0,00	0,00
	II	-3,90	-3,25	-2,78	-2,28	-1,80	-1,46	-1,20	-1,01	-0,87	-0,72	-0,59	-0,48	-0,38	-0,29	-0,22	-0,16	-0,12	-0,08	-0,05	-0,03	-0,01	0,00	0,00	0,00	0,00	0,00	0,00	0,00
	III	-3,90	-3,25	-2,78	-2,28	-1,80	-1,46	-1,20	-0,94	-0,67	-0,46	-0,31	-0,20	-0,11	-0,04	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
50	I	-4,15	-3,46	-2,96	-2,59	-2,26	-1,83	-1,51	-1,27	-1,08	-0,94	-0,82	-0,72	-0,64	-0,57	-0,51	-0,46	-0,40	-0,34	-0,30	-0,26	-0,23	-0,20	-0,17	-0,15	-0,13	-0,11	-0,09	0,00
	II	-4,15	-3,46	-2,96	-2,59	-2,26	-1,83	-1,51	-1,27	-1,08	-0,94	-0,82	-0,72	-0,64	-0,56	-0,47	-0,40	-0,33	-0,27	-0,22	-0,17	-0,14	-0,11	-0,08	-0,06	-0,04	-0,03	0,00	0,00
	III	-4,15	-3,46	-2,96	-2,59	-2,26	-1,83	-1,51	-1,27	-1,08	-0,94	-0,81	-0,61	-0,46	-0,34	-0,24	-0,17	-0,11	-0,06	-0,02	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
60	I	-4,55	-3,79	-3,24	-2,84	-2,52	-2,21	-1,83	-1,53	-1,31	-1,13	-0,99	-0,87	-0,77	-0,69	-0,61	-0,55	-0,50	-0,46	-0,42	-0,38	-0,35	-0,31	-0,27	-0,24	-0,22	0,00	0,00	0,00
	II	-4,55	-3,79	-3,24	-2,84	-2,52	-2,21	-1,83	-1,53	-1,31	-1,13	-0,99	-0,87	-0,77	-0,69	-0,61	-0,55	-0,50	-0,45	-0,40	-0,34	-0,30	-0,25	-0,21	-0,17	-0,14	0,00	0,00	0,00
	III	-4,55	-3,79	-3,24	-2,84	-2,52	-2,21	-1,83	-1,53	-1,31	-1,13	-0,99	-0,87	-0,77	-0,69	-0,56	-0,44	-0,35	-0,27	-0,20	-0,15	-0,10	-0,07	-0,04	-0,01	0,00	0,00	0,00	0,00
80	I	-6,70	-5,58	-4,78	-4,19	-3,71	-3,00	-2,48	-2,08	-1,78	-1,53	-1,33	-1,17	-1,04	-0,93	-0,84	-0,75	-0,68	-0,62	-0,57	-0,52	-0,48	-0,44	-0,41	-0,38	-0,36	-0,33	-0,31	-0,27
	II	-6,70	-5,58	-4,78	-4,19	-3,71	-3,00	-2,48	-2,08	-1,78	-1,53	-1,33	-1,17	-1,04	-0,93	-0,84	-0,75	-0,68	-0,62	-0,57	-0,52	-0,48	-0,44	-0,41	-0,38	-0,36	-0,32	-0,30	-0,27
	III	-6,70	-5,58	-4,78	-4,19	-3,71	-3,00	-2,48	-2,08	-1,78	-1,53	-1,33	-1,17	-1,04	-0,93	-0,84	-0,75	-0,68	-0,62	-0,57	-0,52	-0,48	-0,41	-0,34	-0,28	-0,23	-0,19	-0,15	-0,12
100	I	-9,93	-8,27	-7,09	-5,96	-4,71	-3,82	-3,15	-2,65	-2,26	-1,94	-1,69	-1,49	-1,32	-1,17	-1,05	-0,96	-0,87	-0,79	-0,72	-0,67	-0,61	-0,57	-0,52	-0,49	-0,45	-0,42	-0,40	-0,37
	II	-9,93	-8,27	-7,09	-5,96	-4,71	-3,82	-3,15	-2,65	-2,26	-1,94	-1,69	-1,49	-1,32	-1,17	-1,05	-0,96	-0,87	-0,79	-0,72	-0,67	-0,61	-0,57	-0,52	-0,49	-0,45	-0,42	-0,40	-0,37
	III	-9,93	-8,27	-7,09	-5,96	-4,71	-3,82	-3,15	-2,65	-2,26	-1,94	-1,69	-1,49	-1,32	-1,17	-1,05	-0,96	-0,87	-0,79	-0,72	-0,67	-0,61	-0,57	-0,52	-0,49	-0,45	-0,42	-0,40	-0,37
120	I	-11,02	-9,18	-7,87	-6,88	-5,70	-4,62	-3,82	-3,21	-2,73	-2,35	-2,05	-1,80	-1,60	-1,42	-1,28	-1,15	-1,04	-0,96	-0,88	-0,81	-0,74	-0,69	-0,64	-0,59	-0,55	-0,51	-0,48	-0,45
	II	-11,02	-9,18	-7,87	-6,88	-5,70	-4,62	-3,82	-3,21	-2,73	-2,35	-2,05	-1,80	-1,60	-1,42	-1,28	-1,15	-1,04	-0,96	-0,88	-0,81	-0,74	-0,69	-0,64	-0,59	-0,55	-0,51	-0,48	-0,45
	III	-11,02	-9,18	-7,87	-6,88	-5,70	-4,62	-3,82	-3,21	-2,73	-2,35	-2,05	-1,80	-1,60	-1,42	-1,28	-1,15	-1,04	-0,96	-0,88	-0,81	-0,74	-0,69	-0,64	-0,59	-0,55	-0,51	-0,48	-0,45

2-span system - maximum characteristic loads for PU-PIR-W-ST sandwich panels with a polyurethane core, in claddings of 0,50/0,50 mm thickness, profile types: M16/L, 40 mm outer support width, 60 mm inner support width.
Direction of force - FROM THE SUPPORT

Core thickness	Colour group	Wall sandwich panel PIR STANDARD (PU-PIR-W-ST) Maximum loads [kN/m²] for the span [m]																												
		1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50	3,75	4,00	4,25	4,50	4,75	5,00	5,25	5,50	5,75	6,00	6,25	6,50	6,75	7,00	7,25	7,50	7,75	8,00
40	I	-3,90	-3,24	-2,78	-2,28	-1,80	-1,46	-1,20	-1,01	-0,87	-0,75	-0,65	-0,57	-0,51	-0,45	-0,41	-0,37	-0,33	-0,30	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
	II	-3,90	-3,24	-2,78	-2,28	-1,80	-1,46	-1,20	-1,01	-0,87	-0,75	-0,65	-0,57	-0,51	-0,45	-0,41	-0,37	-0,33	-0,30	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
	III	-3,90	-3,24	-2,78	-2,28	-1,80	-1,46	-1,20	-1,01	-0,87	-0,75	-0,65	-0,57	-0,51	-0,45	-0,41	-0,37	-0,33	-0,30	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
50	I	-4,15	-3,45	-2,96	-2,59	-2,26	-1,83	-1,51	-1,27	-1,08	-0,94	-0,82	-0,72	-0,64	-0,57	-0,51	-0,46	-0,42	-0,38	-0,35	-0,32	-0,29	-0,27	-0,25	-0,23	-0,22	-0,20	-0,19	-0,18
	II	-4,15	-3,45	-2,96	-2,59	-2,26	-1,83	-1,51	-1,27	-1,08	-0,94	-0,82	-0,72	-0,64	-0,57	-0,51	-0,46	-0,42	-0,38	-0,35	-0,32	-0,29	-0,27	-0,25	-0,23	-0,22	-0,20	-0,19	-0,18
	III	-4,15	-3,45	-2,96	-2,59	-2,26	-1,83	-1,51	-1,27	-1,08	-0,94	-0,82	-0,72	-0,64	-0,57	-0,51	-0,46	-0,42	-0,38	-0,35	-0,32	-0,29	-0,27	-0,25	-0,23	-0,22	-0,20	-0,19	-0,18
60	I	-4,54	-3,78	-3,24	-2,84	-2,52	-2,21	-1,83	-1,53	-1,31	-1,13	-0,99	-0,87	-0,77	-0,69	-0,61	-0,55	-0,50	-0,46	-0,42	-0,39	-0,36	-0,33	-0,30	-0,28	-0,26	-0,24	-0,23	-0,19
	II	-4,54	-3,78	-3,24	-2,84	-2,52	-2,21	-1,83	-1,53	-1,31	-1,13	-0,99	-0,87	-0,77	-0,69	-0,61	-0,55	-0,50	-0,46	-0,42	-0,39	-0,36	-0,33	-0,30	-0,28	-0,26	-0,24	-0,23	-0,19
	III	-4,54	-3,78	-3,24	-2,84	-2,52	-2,21	-1,83	-1,53	-1,31	-1,13	-0,99	-0,87	-0,77	-0,69	-0,61	-0,55	-0,50	-0,46	-0,42	-0,39	-0,36	-0,33	-0,30	-0,28	-0,26	-0,24	-0,23	-0,19
80	I	-6,69	-5,58	-4,78	-4,18	-3,71	-3,00	-2,48	-2,08	-1,78	-1,53	-1,33	-1,17	-1,04	-0,93	-0,84	-0,75	-0,68	-0,62	-0,57	-0,52	-0,48	-0,44	-0,41	-0,38	-0,36	-0,33	-0,31	-0,27
	II	-6,69	-5,58	-4,78	-4,18	-3,71	-3,00	-2,48	-2,08	-1,78	-1,53	-1,33	-1,17	-1,04	-0,93	-0,84	-0,75	-0,68	-0,62	-0,57	-0,52	-0,48	-0,44	-0,41	-0,38	-0,36	-0,33	-0,31	-0,27
	III	-6,69	-5,58	-4,78	-4,18	-3,71	-3,00	-2,48	-2,08	-1,78	-1,53	-1,33	-1,17	-1,04	-0,93	-0,84	-0,75	-0,68	-0,62	-0,57	-0,52	-0,48	-0,44	-0,41	-0,38	-0,36	-0,33	-0,31	-0,27
100	I	-9,91	-8,26	-7,08	-5,96	-4,71	-3,82	-3,15	-2,65	-2,26	-1,94	-1,69	-1,49	-1,32	-1,17	-1,05	-0,96	-0,87	-0,79	-0,72	-0,67	-0,61	-0,57	-0,52	-0,49	-0,45	-0,42	-0,40	-0,37
	II	-9,91	-8,26	-7,08	-5,96	-4,71	-3,82	-3,15	-2,65	-2,26	-1,94	-1,69	-1,49	-1,32	-1,17	-1,05	-0,96	-0,87	-0,79	-0,72	-0,67	-0,61	-0,57	-0,52	-0,49	-0,45	-0,42	-0,40	-0,37
	III	-9,91	-8,26	-7,08	-5,96	-4,71	-3,82	-3,15	-2,65	-2,26	-1,94	-1,69	-1,49	-1,32	-1,17	-1,05	-0,96	-0,87	-0,79	-0,72	-0,67	-0,61	-0,57	-0,52	-0,49	-0,45	-0,42	-0,40	-0,37
120	I	-11,01	-9,17	-7,86	-6,88	-5,70	-4,62	-3,82	-3,21	-2,73	-2,35	-2,05	-1,80	-1,60	-1,42	-1,28	-1,15	-1,04	-0,96	-0,88	-0,81	-0,74	-0,69	-0,64	-0,59	-0,55	-0,51	-0,48	-0,45
	II	-11,01	-9,17	-7,86	-6,88	-5,70	-4,62	-3,82	-3,21	-2,73	-2,35	-2,05	-1,80	-1,60	-1,42	-1,28	-1,15	-1,04	-0,96	-0,88	-0,81	-0,74	-0,69	-0,64	-0,59	-0,55	-0,51	-0,48	-0,45
	III	-11,01	-9,17	-7,86	-6,88	-5,70	-4,62	-3,82	-3,21	-2,73	-2,35	-2,05	-1,80	-1,60	-1,42	-1,28	-1,15	-1,04	-0,96	-0,88	-0,81	-0,74	-0,69	-0,64	-0,59	-0,55	-0,51	-0,48	-0,45

3-span system - maximum characteristic loads for PU-PIR-W-ST sandwich panels with a polyurethane core, in claddings of 0,50/0,50 mm thickness, profile types: M16/L, 40 mm outer support width, 60 mm inner support width.
Direction of force - FROM THE SUPPORT

Core thickness	Colour group	Wall sandwich panel PIR STANDARD (PU-PIR-W-ST) Maximum loads [kN/m²] for the span [m]																												
		1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50	3,75	4,00	4,25	4,50	4,75	5,00	5,25	5,50	5,75	6,00	6,25	6,50	6,75	7,00	7,25	7,50	7,75	8,00
40	I	-3,89	-3,24	-2,78	-2,28	-1,80	-1,46	-1,20	-1,01	-0,87	-0,75	-0,65	-0,57	-0,51	-0,45	-0,41	-0,37	-0,33	-0,30	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
	II	-3,89	-3,24	-2,78	-2,28	-1,80	-1,46	-1,20	-1,01	-0,87	-0,75	-0,65	-0,57	-0,51	-0,45	-0,41	-0,37	-0,33	-0,30	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
	III	-3,89	-3,24	-2,78	-2,28	-1,80	-1,46	-1,20	-1,01	-0,87	-0,75	-0,65	-0,57	-0,51	-0,45	-0,41	-0,37	-0,33	-0,30	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
50	I	-4,14	-3,45	-2,96	-2,59	-2,26	-1,83	-1,51	-1,27	-1,08	-0,94	-0,82	-0,72	-0,64	-0,57	-0,51	-0,46	-0,42	-0,38	-0,35	-0,32	-0,29	-0,27	-0,25	-0,23	-0,22	-0,20	-0,19	-0,18
	II	-4,14	-3,45	-2,96	-2,59	-2,26	-1,83	-1,51	-1,27	-1,08	-0,94	-0,82	-0,72	-0,64	-0,57	-0,51	-0,46	-0,42	-0,38	-0,35	-0,32	-0,29	-0,27	-0,25	-0,23	-0,22	-0,20	-0,19	-0,18
	III	-4,14	-3,45	-2,96	-2,59	-2,26	-1,83	-1,51	-1,27	-1,08	-0,94	-0,82	-0,72	-0,64	-0,57	-0,51	-0,46	-0,42	-0,38	-0,35	-0,32	-0,29	-0,27	-0,25	-0,23	-0,22	-0,20	-0,19	-0,18
60	I	-4,54	-3,78	-3,24	-2,84	-2,52	-2,21	-1,83	-1,53	-1,31	-1,13	-0,99	-0,87	-0,77	-0,69	-0,61	-0,55	-0,50	-0,46	-0,42	-0,39	-0,36	-0,33	-0,30	-0,28	-0,26	-0,24	-0,23	-0,19
	II	-4,54	-3,78	-3,24	-2,84	-2,52	-2,21	-1,83	-1,53	-1,31	-1,13	-0,99	-0,87	-0,77	-0,69	-0,61	-0,55	-0,50	-0,46	-0,42	-0,39	-0,36	-0,33	-0,30	-0,28	-0,26	-0,24	-0,23	-0,19
	III	-4,54	-3,78	-3,24	-2,84	-2,52	-2,21	-1,83	-1,53	-1,31	-1,13	-0,99	-0,87	-0,77	-0,69	-0,61	-0,55	-0,50	-0,46	-0,42	-0,39	-0,36	-0,33	-0,30	-0,28	-0,26	-0,24	-0,23	-0,19
80	I	-6,69	-5,58	-4,78	-4,18	-3,71	-3,00	-2,48	-2,08	-1,78	-1,53	-1,33	-1,17	-1,04	-0,93	-0,84	-0,75	-0,68	-0,62	-0,57	-0,52	-0,48	-0,44	-0,41	-0,38	-0,36	0,00	-0,31	-0,27
	II	-6,69	-5,58	-4,78	-4,18	-3,71	-3,00	-2,48	-2,08	-1,78	-1,53	-1,33	-1,17	-1,04	-0,93	-0,84	-0,75	-0,68	-0,62	-0,57	-0,52	-0,48	-0,44	-0,41	-0,38	-0,36	0,00	-0,31	-0,27
	III	-6,69	-5,58	-4,78	-4,18	-3,71	-3,00	-2,48	-2,08	-1,78	-1,53	-1,33	-1,17	-1,04	-0,93	-0,84	-0,75	-0,68	-0,62	-0,57	-0,52	-0,48	-0,44	-0,41	-0,38	-0,36	0,00	-0,31	-0,27
100	I	-9,91	-8,26	-7,08	-5,96	-4,71	-3,82	-3,15	-2,65	-2,26	-1,94	-1,69	-1,49	-1,32	-1,17	-1,05	-0,96	-0,87	-0,79	-0,72	-0,67	-0,61	-0,57	-0,52	-0,49	-0,45	-0,33	-0,40	-0,37
	II	-9,91	-8,26	-7,08	-5,96	-4,71	-3,82	-3,15	-2,65	-2,26	-1,94	-1,69	-1,49	-1,32	-1,17	-1,05	-0,96	-0,87	-0,79	-0,72	-0,67	-0,61	-0,57	-0,52	-0,49	-0,45	-0,33	-0,40	-0,37
	III	-9,91	-8,26	-7,08	-5,96	-4,71	-3,82	-3,15	-2,65	-2,26	-1,94	-1,69	-1,49	-1,32	-1,17	-1,05	-0,96	-0,87	-0,79	-0,72	-0,67	-0,61	-0,57	-0,52	-0,49	-0,45	-0,33	-0,40	-0,37
120	I	-11,00	-9,17	-7,86	-6,88	-5,70	-4,62	-3,82	-3,21	-2,73	-2,35	-2,05	-1,80	-1,60	-1,42	-1,28	-1,15	-1,04	-0,96	-0,88	-0,81	-0,74	-0,69	-0,64	-0,59	-0,55	-0,51	-0,48	-0,45
	II	-11,00	-9,17	-7,86	-6,88	-5,70	-4,62	-3,82	-3,21	-2,73	-2,35	-2,05	-1,80	-1,60	-1,42	-1,28	-1,15	-1,04	-0,96	-0,88	-0,81	-0,74	-0,69	-0,64	-0,59	-0,55	-0,51	-0,48	-0,45
	III	-11,00	-9,17	-7,86	-6,88	-5,70	-4,62	-3,82	-3,21	-2,73	-2,35	-2,05	-1,80	-1,60	-1,42	-1,28	-1,15	-1,04	-0,96	-0,88	-0,81	-0,74	-0,69	-0,64	-0,59	-0,55	-0,51	-0,48	-0,45

PIR PLUS wall sandwich panel

1-span system - maximum characteristic loads for PU-PIR-W-PLUS sandwich panels with a polyurethane core, in claddings of 0,50/0,50 mm thickness, profile types: M16/L, 40 mm outer support width.
Direction of force - TO THE SUPPORT

Core thickness	Colour group	Wall sandwich panel PIR PLUS (PU-PIR-W-PLUS) Maximum loads [kN/m²] for the span [m]																												
		1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50	3,75	4,00	4,25	4,50	4,75	5,00	5,25	5,50	5,75	6,00	6,25	6,50	6,75	7,00	7,25	7,50	7,75	8,00
60	I	4,55	3,79	3,24	2,84	2,52	2,27	2,06	1,89	1,74	1,52	1,30	1,12	0,97	0,83	0,71	0,61	0,52	0,45	0,40	0,34	0,30	0,25	0,21	0,17	0,14	0,12	0,10	0,08
	II	4,55	3,79	3,24	2,84	2,52	2,27	2,06	1,89	1,74	1,52	1,30	1,12	0,97	0,83	0,71	0,61	0,52	0,45	0,40	0,34	0,30	0,25	0,21	0,17	0,14	0,12	0,10	0,08
	III	4,55	3,79	3,24	2,84	2,52	2,27	2,06	1,89	1,74	1,52	1,30	1,12	0,97	0,83	0,71	0,61	0,52	0,45	0,40	0,34	0,30	0,25	0,21	0,17	0,00	0,00	0,00	0,00
80	I	5,28	4,40	3,77	3,30	2,93	2,64	2,40	2,20	2,03	1,88	1,76	1,65	1,55	1,39	1,23	1,09	0,97	0,87	0,76	0,67	0,60	0,53	0,47	0,42	0,38	0,34	0,30	0,27
	II	5,28	4,40	3,77	3,30	2,93	2,64	2,40	2,20	2,03	1,88	1,76	1,65	1,55	1,39	1,23	1,09	0,97	0,87	0,76	0,67	0,60	0,53	0,47	0,42	0,38	0,34	0,30	0,27
	III	5,28	4,40	3,77	3,30	2,93	2,64	2,40	2,20	2,03	1,88	1,76	1,65	1,55	1,39	1,23	1,09	0,97	0,87	0,76	0,67	0,60	0,53	0,47	0,42	0,38	0,34	0,30	0,27
100	I	5,28	4,40	3,77	3,30	2,93	2,64	2,40	2,20	2,03	1,88	1,76	1,65	1,55	1,46	1,39	1,32	1,25	1,20	1,14	1,03	0,94	0,85	0,77	0,70	0,63	0,57	0,52	0,47
	II	5,28	4,40	3,77	3,30	2,93	2,64	2,40	2,20	2,03	1,88	1,76	1,65	1,55	1,46	1,39	1,32	1,25	1,20	1,14	1,03	0,94	0,85	0,77	0,70	0,63	0,57	0,52	0,47
	III	5,28	4,40	3,77	3,30	2,93	2,64	2,40	2,20	2,03	1,88	1,76	1,65	1,55	1,46	1,39	1,32	1,25	1,20	1,14	1,03	0,94	0,85	0,77	0,70	0,63	0,57	0,52	0,47
120	I	4,93	4,11	3,52	3,08	2,74	2,46	2,24	2,05	1,89	1,76	1,64	1,54	1,45	1,37	1,29	1,23	1,17	1,12	1,07	1,02	0,99	0,95	0,92	0,89	0,85	0,81	0,74	0,69
	II	4,93	4,11	3,52	3,08	2,74	2,46	2,24	2,05	1,89	1,76	1,64	1,54	1,45	1,37	1,29	1,23	1,17	1,12	1,07	1,02	0,99	0,95	0,92	0,89	0,85	0,81	0,74	0,69
	III	4,93	4,11	3,52	3,08	2,74	2,46	2,24	2,05	1,89	1,76	1,64	1,54	1,45	1,37	1,29	1,23	1,17	1,12	1,07	1,02	0,99	0,95	0,92	0,89	0,85	0,81	0,74	0,69

2-span system - maximum characteristic loads for PU-PIR-W-PLUS sandwich panels with a polyurethane core, in claddings of 0,50/0,50 mm thickness, profile types: M16/L, 40 mm outer support width, 60 mm inner support width.
Direction of force - TO THE SUPPORT

Core thickness	Colour group	Wall sandwich panel PIR PLUS (PU-PIR-W-PLUS) Maximum loads [kN/m²] for the span [m]																												
		1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50	3,75	4,00	4,25	4,50	4,75	5,00	5,25	5,50	5,75	6,00	6,25	6,50	6,75	7,00	7,25	7,50	7,75	8,00
60	I	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,45	1,20	1,01	0,88	0,76	0,66	0,57	0,50	0,44	0,39	0,35	0,32	0,29	0,26	0,24	0,22	0,20	0,19	0,17	0,16
	II	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,45	1,20	1,01	0,88	0,76	0,66	0,57	0,50	0,44	0,39	0,35	0,32	0,29	0,26	0,24	0,22	0,20	0,19	0,17	0,16
	III	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,45	1,20	1,01	0,88	0,76	0,66	0,57	0,50	0,44	0,39	0,35	0,32	0,29	0,26	0,24	0,22	0,20	0,19	0,17	0,16
80	I	5,28	4,40	3,77	3,30	2,93	2,64	2,40	2,20	2,03	1,88	1,76	1,65	1,55	1,39	1,23	1,09	0,97	0,87	0,76	0,67	0,60	0,53	0,47	0,42	0,38	0,34	0,30	0,27
	II	5,28	4,40	3,77	3,30	2,93	2,64	2,40	2,20	2,03	1,88	1,76	1,65	1,55	1,39	1,23	1,09	0,97	0,87	0,76	0,67	0,60	0,53	0,47	0,42	0,38	0,34	0,30	0,27
	III	5,28	4,40	3,77	3,30	2,93	2,64	2,40	2,20	2,03	1,88	1,76	1,65	1,55	1,39	1,23	1,09	0,97	0,87	0,76	0,67	0,60	0,53	0,47	0,42	0,38	0,34	0,30	0,27
100	I	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,23	1,15	1,00	0,86	0,75	0,65	0,57	0,50	0,45	0,40	0,36	0,33	0,30	0,27	0,25	0,23	0,21
	II	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,23	1,15	1,00	0,86	0,75	0,65	0,57	0,50	0,45	0,40	0,36	0,33	0,30	0,27	0,25	0,23	0,21
	III	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,23	1,15	1,00	0,86	0,75	0,65	0,57	0,50	0,45	0,40	0,36	0,33	0,30	0,27	0,25	0,23	0,21
120	I	3,70	3,08	2,64	2,31	2,05	1,85	1,68	1,54	1,42	1,32	1,23	1,15	1,08	1,02	0,98	0,93	0,89	0,83	0,73	0,65	0,58	0,52	0,47	0,43	0,39	0,36	0,33	0,31
	II	3,70	3,08	2,64	2,31	2,05	1,85	1,68	1,54	1,42	1,32	1,23	1,15	1,08	1,02	0,98	0,93	0,89	0,83	0,73	0,65	0,58	0,52	0,47	0,43	0,39	0,36	0,33	0,31
	III	3,70	3,08	2,64	2,31	2,05	1,85	1,68	1,54	1,42	1,32	1,23	1,15	1,08	1,02	0,98	0,93	0,89	0,83	0,73	0,65	0,58	0,52	0,47	0,43	0,39	0,36	0,33	0,31

3-span system - maximum characteristic loads for PU-PIR-W-PLUS sandwich panels with a polyurethane core, in claddings of 0,50/0,50 mm thickness, profile types: M16/L, 40 mm outer support width, 60 mm inner support width.
Direction of force - TO THE SUPPORT

Core thickness	Colour group	Wall sandwich panel PIR PLUS (PU-PIR-W-PLUS) Maximum loads [kN/m²] for the span [m]																												
		1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50	3,75	4,00	4,25	4,50	4,75	5,00	5,25	5,50	5,75	6,00	6,25	6,50	6,75	7,00	7,25	7,50	7,75	8,00
60	I	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,23	1,07	0,95	0,84	0,75	0,67	0,61	0,55	0,50	0,46	0,42	0,39	0,36	0,33	0,31	0,29	0,27	0,25
	II	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,23	1,07	0,95	0,84	0,75	0,67	0,61	0,55	0,50	0,46	0,42	0,39	0,36	0,33	0,31	0,29	0,27	0,25
	III	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,23	1,07	0,95	0,84	0,75	0,67	0,61	0,55	0,50	0,46	0,42	0,39	0,36	0,33	0,31	0,29	0,27	0,25
80	I	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,23	1,16	1,06	0,96	0,86	0,77	0,70	0,63	0,58	0,53	0,49	0,45	0,42	0,39	0,36	0,34	0,32
	II	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,23	1,16	1,06	0,96	0,86	0,77	0,70	0,63	0,58	0,53	0,49	0,45	0,42	0,39	0,36	0,34	0,32
	III	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,23	1,16	1,06	0,96	0,86	0,77	0,70	0,63	0,58	0,53	0,49	0,45	0,42	0,39	0,36	0,34	0,32
100	I	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,23	1,16	1,10	1,04	1,00	0,91	0,82	0,75	0,68	0,63	0,58	0,53	0,49	0,46	0,42	0,40	0,37
	II	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,23	1,16	1,10	1,04	1,00	0,91	0,82	0,75	0,68	0,63	0,58	0,53	0,49	0,46	0,42	0,40	0,37
	III	3,96	3,30	2,83	2,47	2,20	1,98	1,80	1,65	1,52	1,41	1,32	1,23	1,16	1,10	1,04	1,00	0,91	0,82	0,75	0,68	0,63	0,58	0,53	0,49	0,46	0,42	0,40	0,37
120	I	3,70	3,08	2,64	2,31	2,05	1,85	1,68	1,54	1,42	1,32	1,23	1,15	1,08	1,02	0,98	0,93	0,89	0,84	0,81	0,77	0,74	0,71	0,68	0,63	0,59	0,55	0,51	0,48
	II	3,70	3,08	2,64	2,31	2,05	1,85	1,68	1,54	1,42	1,32	1,23	1,15	1,08	1,02	0,98	0,93	0,89	0,84	0,81	0,77	0,74	0,71	0,68	0,63	0,59	0,55	0,51	0,48
	III	3,70	3,08	2,64	2,31	2,05	1,85	1,68	1,54	1,42	1,32	1,23	1,15	1,08	1,02	0,98	0,93	0,89	0,84	0,81	0,77	0,74	0,71	0,68	0,63	0,59	0,55	0,51	0,48

PIR PLUS wall sandwich panel

1-span system - maximum characteristic loads for PU-PIR-W-PLUS sandwich panels with a polyurethane core, in claddings of 0,50/0,50 mm thickness, profile types: M16/L, 40 mm outer support width.
Direction of force - FROM THE SUPPORT

Core thickness	Colour group	Wall sandwich panel PIR PLUS (PU-PIR-W-PLUS) Maximum loads [kN/m²] for the span [m]																												
		1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50	3,75	4,00	4,25	4,50	4,75	5,00	5,25	5,50	5,75	6,00	6,25	6,50	6,75	7,00	7,25	7,50	7,75	8,00
60	I	-4,55	-3,79	-3,24	-2,84	-2,52	-2,21	-1,83	-1,53	-1,31	-1,13	-0,99	-0,87	-0,77	-0,69	-0,61	-0,55	-0,50	-0,46	-0,42	-0,38	-0,35	-0,31	-0,27	-0,24	-0,22	0,00	0,00	0,00
	II	-4,55	-3,79	-3,24	-2,84	-2,52	-2,21	-1,83	-1,53	-1,31	-1,13	-0,99	-0,87	-0,77	-0,69	-0,61	-0,55	-0,50	-0,45	-0,40	-0,34	-0,30	-0,25	-0,21	-0,17	-0,14	0,00	0,00	0,00
	III	-4,55	-3,79	-3,24	-2,84	-2,52	-2,21	-1,83	-1,53	-1,31	-1,13	-0,99	-0,87	-0,77	-0,69	-0,56	-0,44	-0,35	-0,27	-0,20	-0,15	-0,10	-0,07	-0,04	-0,01	0,00	0,00	0,00	0,00
80	I	-6,70	-5,58	-4,78	-4,19	-3,71	-3,00	-2,48	-2,08	-1,78	-1,53	-1,33	-1,17	-1,04	-0,93	-0,84	-0,75	-0,68	-0,62	-0,57	-0,52	-0,48	-0,44	-0,41	-0,38	-0,36	-0,33	-0,31	-0,27
	II	-6,70	-5,58	-4,78	-4,19	-3,71	-3,00	-2,48	-2,08	-1,78	-1,53	-1,33	-1,17	-1,04	-0,93	-0,84	-0,75	-0,68	-0,62	-0,57	-0,52	-0,48	-0,44	-0,41	-0,38	-0,36	-0,32	-0,30	-0,27
	III	-6,70	-5,58	-4,78	-4,19	-3,71	-3,00	-2,48	-2,08	-1,78	-1,53	-1,33	-1,17	-1,04	-0,93	-0,84	-0,75	-0,68	-0,62	-0,57	-0,52	-0,48	-0,41	-0,34	-0,28	-0,23	-0,19	-0,15	-0,12
100	I	-9,93	-8,27	-7,09	-5,96	-4,71	-3,82	-3,15	-2,65	-2,26	-1,94	-1,69	-1,49	-1,32	-1,17	-1,05	-0,96	-0,87	-0,79	-0,72	-0,67	-0,61	-0,57	-0,52	-0,49	-0,45	-0,42	-0,40	-0,37
	II	-9,93	-8,27	-7,09	-5,96	-4,71	-3,82	-3,15	-2,65	-2,26	-1,94	-1,69	-1,49	-1,32	-1,17	-1,05	-0,96	-0,87	-0,79	-0,72	-0,67	-0,61	-0,57	-0,52	-0,49	-0,45	-0,42	-0,40	-0,37
	III	-9,93	-8,27	-7,09	-5,96	-4,71	-3,82	-3,15	-2,65	-2,26	-1,94	-1,69	-1,49	-1,32	-1,17	-1,05	-0,96	-0,87	-0,79	-0,72	-0,67	-0,61	-0,57	-0,52	-0,49	-0,45	-0,42	-0,40	-0,37
120	I	-11,02	-9,18	-7,87	-6,88	-5,70	-4,62	-3,82	-3,21	-2,73	-2,35	-2,05	-1,80	-1,60	-1,42	-1,28	-1,15	-1,04	-0,96	-0,88	-0,81	-0,74	-0,69	-0,64	-0,59	-0,55	-0,51	-0,48	-0,45
	II	-11,02	-9,18	-7,87	-6,88	-5,70	-4,62	-3,82	-3,21	-2,73	-2,35	-2,05	-1,80	-1,60	-1,42	-1,28	-1,15	-1,04	-0,96	-0,88	-0,81	-0,74	-0,69	-0,64	-0,59	-0,55	-0,51	-0,48	-0,45
	III	-11,02	-9,18	-7,87	-6,88	-5,70	-4,62	-3,82	-3,21	-2,73	-2,35	-2,05	-1,80	-1,60	-1,42	-1,28	-1,15	-1,04	-0,96	-0,88	-0,81	-0,74	-0,69	-0,64	-0,59	-0,55	-0,51	-0,48	-0,45

2-span system - maximum characteristic loads for PU-PIR-W-PLUS sandwich panels with a polyurethane core, in claddings of 0,50/0,50 mm thickness, profile types: M16/L, 40 mm outer support width, 60 mm inner support width.
Direction of force - FROM THE SUPPORT

Core thickness	Colour group	Wall sandwich panel PIR PLUS (PU-PIR-W-PLUS) Maximum loads [kN/m²] for the span [m]																												
		1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50	3,75	4,00	4,25	4,50	4,75	5,00	5,25	5,50	5,75	6,00	6,25	6,50	6,75	7,00	7,25	7,50	7,75	8,00
60	I	-4,54	-3,78	-3,24	-2,84	-2,52	-2,21	-1,83	-1,53	-1,31	-1,13	-0,99	-0,87	-0,77	-0,69	-0,61	-0,55	-0,50	-0,46	-0,42	-0,39	-0,36	-0,33	-0,30	-0,28	-0,26	-0,24	-0,23	-0,19
	II	-4,54	-3,78	-3,24	-2,84	-2,52	-2,21	-1,83	-1,53	-1,31	-1,13	-0,99	-0,87	-0,77	-0,69	-0,61	-0,55	-0,50	-0,46	-0,42	-0,39	-0,36	-0,33	-0,30	-0,28	-0,26	-0,24	-0,23	-0,19
	III	-4,54	-3,78	-3,24	-2,84	-2,52	-2,21	-1,83	-1,53	-1,31	-1,13	-0,99	-0,87	-0,77	-0,69	-0,61	-0,55	-0,50	-0,46	-0,42	-0,39	-0,36	-0,33	-0,30	-0,28	-0,26	-0,24	-0,23	-0,19
80	I	-6,69	-5,58	-4,78	-4,18	-3,71	-3,00	-2,48	-2,08	-1,78	-1,53	-1,33	-1,17	-1,04	-0,93	-0,84	-0,75	-0,68	-0,62	-0,57	-0,52	-0,48	-0,44	-0,41	-0,38	-0,36	-0,33	-0,31	-0,27
	II	-6,69	-5,58	-4,78	-4,18	-3,71	-3,00	-2,48	-2,08	-1,78	-1,53	-1,33	-1,17	-1,04	-0,93	-0,84	-0,75	-0,68	-0,62	-0,57	-0,52	-0,48	-0,44	-0,41	-0,38	-0,36	-0,33	-0,31	-0,27
	III	-6,69	-5,58	-4,78	-4,18	-3,71	-3,00	-2,48	-2,08	-1,78	-1,53	-1,33	-1,17	-1,04	-0,93	-0,84	-0,75	-0,68	-0,62	-0,57	-0,52	-0,48	-0,44	-0,41	-0,38	-0,36	-0,33	-0,31	-0,27
100	I	-9,91	-8,26	-7,08	-5,96	-4,71	-3,82	-3,15	-2,65	-2,26	-1,94	-1,69	-1,49	-1,32	-1,17	-1,05	-0,96	-0,87	-0,79	-0,72	-0,67	-0,61	-0,57	-0,52	-0,49	-0,45	-0,42	-0,40	-0,37
	II	-9,91	-8,26	-7,08	-5,96	-4,71	-3,82	-3,15	-2,65	-2,26	-1,94	-1,69	-1,49	-1,32	-1,17	-1,05	-0,96	-0,87	-0,79	-0,72	-0,67	-0,61	-0,57	-0,52	-0,49	-0,45	-0,42	-0,40	-0,37
	III	-9,91	-8,26	-7,08	-5,96	-4,71	-3,82	-3,15	-2,65	-2,26	-1,94	-1,69	-1,49	-1,32	-1,17	-1,05	-0,96	-0,87	-0,79	-0,72	-0,67	-0,61	-0,57	-0,52	-0,49	-0,45	-0,42	-0,40	-0,37
120	I	-11,01	-9,17	-7,86	-6,88	-5,70	-4,62	-3,82	-3,21	-2,73	-2,35	-2,05	-1,80	-1,60	-1,42	-1,28	-1,15	-1,04	-0,96	-0,88	-0,81	-0,74	-0,69	-0,64	-0,59	-0,55	-0,51	-0,48	-0,45
	II	-11,01	-9,17	-7,86	-6,88	-5,70	-4,62	-3,82	-3,21	-2,73	-2,35	-2,05	-1,80	-1,60	-1,42	-1,28	-1,15	-1,04	-0,96	-0,88	-0,81	-0,74	-0,69	-0,64	-0,59	-0,55	-0,51	-0,48	-0,45
	III	-11,01	-9,17	-7,86	-6,88	-5,70	-4,62	-3,82	-3,21	-2,73	-2,35	-2,05	-1,80	-1,60	-1,42	-1,28	-1,15	-1,04	-0,96	-0,88	-0,81	-0,74	-0,69	-0,64	-0,59	-0,55	-0,51	-0,48	-0,45

3-span system - maximum characteristic loads for PU-PIR-W-PLUS sandwich panels with a polyurethane core, in claddings of 0,50/0,50 mm thickness, profile types: M16/L, 40 mm outer support width, 60 mm inner support width.
Direction of force - FROM THE SUPPORT

Core thickness	Colour group	Wall sandwich panel PIR PLUS (PU-PIR-W-PLUS) Maximum loads [kN/m²] for the span [m]																												
		1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50	3,75	4,00	4,25	4,50	4,75	5,00	5,25	5,50	5,75	6,00	6,25	6,50	6,75	7,00	7,25	7,50	7,75	8,00
60	I	-4,54	-3,78	-3,24	-2,84	-2,52	-2,21	-1,83	-1,53	-1,31	-1,13	-0,99	-0,87	-0,77	-0,69	-0,61	-0,55	-0,50	-0,46	-0,42	-0,39	-0,36	-0,33	-0,30	-0,28	-0,26	-0,24	-0,23	-0,19
	II	-4,54	-3,78	-3,24	-2,84	-2,52	-2,21	-1,83	-1,53	-1,31	-1,13	-0,99	-0,87	-0,77	-0,69	-0,61	-0,55	-0,50	-0,46	-0,42	-0,39	-0,36	-0,33	-0,30	-0,28	-0,26	-0,24	-0,23	-0,19
	III	-4,54	-3,78	-3,24	-2,84	-2,52	-2,21	-1,83	-1,53	-1,31	-1,13	-0,99	-0,87	-0,77	-0,69	-0,61	-0,55	-0,50	-0,46	-0,42	-0,39	-0,36	-0,33	-0,30	-0,28	-0,26	-0,24	-0,23	-0,19
80	I	-6,69	-5,58	-4,78	-4,18	-3,71	-3,00	-2,48	-2,08	-1,78	-1,53	-1,33	-1,17	-1,04	-0,93	-0,84	-0,75	-0,68	-0,62	-0,57	-0,52	-0,48	-0,44	-0,41	-0,38	-0,36	0,00	-0,31	-0,27
	II	-6,69	-5,58	-4,78	-4,18	-3,71	-3,00	-2,48	-2,08	-1,78	-1,53	-1,33	-1,17	-1,04	-0,93	-0,84	-0,75	-0,68	-0,62	-0,57	-0,52	-0,48	-0,44	-0,41	-0,38	-0,36	0,00	-0,31	-0,27
	III	-6,69	-5,58	-4,78	-4,18	-3,71	-3,00	-2,48	-2,08	-1,78	-1,53	-1,33	-1,17	-1,04	-0,93	-0,84	-0,75	-0,68	-0,62	-0,57	-0,52	-0,48	-0,44	-0,41	-0,38	-0,36	0,00	-0,31	-0,27
100	I	-9,91	-8,26	-7,08	-5,96	-4,71	-3,82	-3,15	-2,65	-2,26	-1,94	-1,69	-1,49	-1,32	-1,17	-1,05	-0,96	-0,87	-0,79	-0,72	-0,67	-0,61	-0,57	-0,52	-0,49	-0,45	-0,42	-0,40	-0,37
	II	-9,91	-8,26	-7,08	-5,96	-4,71	-3,82	-3,15	-2,65	-2,26	-1,94	-1,69	-1,49	-1,32	-1,17	-1,05	-0,96	-0,87	-0,79	-0,72	-0,67	-0,61	-0,57	-0,52	-0,49	-0,45	-0,42	-0,40	-0,37
	III	-9,91	-8,26	-7,08	-5,96	-4,71	-3,82	-3,15	-2,65	-2,26	-1,94	-1,69	-1,49	-1,32	-1,17	-1,05	-0,96	-0,87	-0,79	-0,72	-0,67	-0,61	-0,57	-0,52	-0,49	-0,45	-0,42	-0,40	-0,37
120	I	-11,00	-9,17	-7,86	-6,88	-5,70	-4,62	-3,82	-3,21	-2,73	-2,35	-2,05	-1,80	-1,60	-1,42	-1,28	-1,15	-1,04	-0,96	-0,88	-0,81	-0,74	-0,69	-0,64	-0,59	-0,55	-0,51	-0,48	-0,45
	II	-11,00	-9,17	-7,86	-6,88	-5,70	-4,62	-3,82	-3,21	-2,73	-2,35	-2,05	-1,80	-1,60	-1,42	-1,28	-1,15	-1,04	-0,96	-0,88	-0,81	-0,74	-0,69	-0,64	-0,59	-0,55	-0,51	-0,48	-0,45
	III	-11,00	-9,17	-7,86	-6,88	-5,70	-4,62	-3,82	-3,21	-2,73	-2,35	-2,05	-1,80	-1,60	-1,42	-1,28	-1,15	-1,04	-0,96	-0,88	-0,81	-0,74	-0,69	-0,64	-0,59	-0,55	-0,51	-0,48	-0,45

PIR ROOF roof sandwich panel

1-span system - maximum characteristic loads for PIR ROOF roof sandwich panel with a polyurethane core, in claddings of 0,50/0,40 mm and profile types: T/D; 40mm outer support width.
Direction of force - TO THE SUPPORT

Core thickness [mm]	Colours group	Roof sandwich panel PIR ROOF																						
		Characteristic maximum load [kN/m²] at the given span [m]																						
		0,50	0,75	1,00	1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50	3,75	4,00	4,25	4,50	4,75	5,00	5,25	5,50	5,75	6,00
40/85	I	9,21	6,11	3,99	2,72	2,01	1,57	1,26	1,05	0,90	0,77	0,66	0,57	0,50	0,44	0,33	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
	II	9,21	6,11	3,99	2,72	2,01	1,57	1,26	1,05	0,90	0,77	0,66	0,57	0,50	0,44	0,33	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
	III	9,21	6,11	3,99	2,72	2,01	1,57	1,26	1,05	0,90	0,77	0,66	0,57	0,50	0,44	0,33	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
60/105	I	10,05	6,67	4,27	2,99	2,28	1,84	1,53	1,30	1,13	1,00	0,89	0,80	0,71	0,64	0,58	0,51	0,41	0,33	0,26	0,00	0,00	0,00	0,00
	II	10,05	6,67	4,27	2,99	2,28	1,84	1,53	1,30	1,13	1,00	0,89	0,80	0,71	0,64	0,58	0,51	0,41	0,33	0,26	0,00	0,00	0,00	0,00
	III	10,05	6,67	4,27	2,99	2,28	1,84	1,53	1,30	1,13	1,00	0,89	0,80	0,71	0,64	0,58	0,51	0,41	0,33	0,26	0,00	0,00	0,00	0,00
80/125	I	10,04	6,66	4,80	3,51	2,78	2,30	1,97	1,73	1,53	1,36	1,23	1,11	1,00	0,93	0,84	0,77	0,70	0,63	0,52	0,43	0,36	0,29	0,24
	II	10,04	6,66	4,80	3,51	2,78	2,30	1,97	1,73	1,53	1,36	1,23	1,11	1,00	0,93	0,84	0,77	0,70	0,63	0,52	0,43	0,36	0,29	0,24
	III	10,04	6,66	4,80	3,51	2,78	2,30	1,97	1,73	1,53	1,36	1,23	1,11	1,00	0,93	0,84	0,77	0,70	0,63	0,52	0,43	0,36	0,29	0,24
100/145	I	10,88	7,22	5,22	3,94	3,20	2,72	2,37	2,11	1,89	1,71	1,55	1,41	1,29	1,18	1,08	1,00	0,92	0,85	0,78	0,71	0,61	0,52	0,44
	II	10,88	7,22	5,22	3,94	3,20	2,72	2,37	2,11	1,89	1,71	1,55	1,41	1,29	1,18	1,08	1,00	0,92	0,85	0,78	0,71	0,61	0,52	0,44
	III	10,88	7,22	5,22	3,94	3,20	2,72	2,37	2,11	1,89	1,71	1,55	1,41	1,29	1,18	1,08	1,00	0,92	0,85	0,78	0,71	0,61	0,52	0,44
120/165	I	10,87	7,21	5,23	4,00	3,30	2,84	2,52	2,26	2,05	1,87	1,71	1,57	1,45	1,33	1,23	1,13	1,05	0,98	0,91	0,84	0,78	0,72	0,65
	II	10,87	7,21	5,23	4,00	3,30	2,84	2,52	2,26	2,05	1,87	1,71	1,57	1,45	1,33	1,23	1,13	1,05	0,98	0,91	0,84	0,78	0,72	0,65
	III	10,87	7,21	5,23	4,00	3,30	2,84	2,52	2,26	2,05	1,87	1,71	1,57	1,45	1,33	1,23	1,13	1,05	0,98	0,91	0,84	0,78	0,72	0,65
150/195	I	10,86	7,20	5,17	4,04	3,39	2,97	2,62	2,31	2,07	1,87	1,70	1,56	1,44	1,33	1,24	1,16	1,09	1,02	0,98	0,92	0,88	0,83	0,79
	II	10,86	7,20	5,17	4,04	3,39	2,97	2,62	2,31	2,07	1,87	1,70	1,56	1,44	1,33	1,24	1,16	1,09	1,02	0,98	0,92	0,88	0,83	0,79
	III	10,86	7,20	5,17	4,04	3,39	2,97	2,62	2,31	2,07	1,87	1,70	1,56	1,44	1,33	1,24	1,16	1,09	1,02	0,98	0,92	0,88	0,83	0,79
160/205	I	10,86	7,19	5,08	3,98	3,36	2,95	2,61	2,31	2,06	1,86	1,70	1,55	1,43	1,33	1,24	1,16	1,08	1,02	0,97	0,92	0,87	0,83	0,79
	II	10,86	7,19	5,08	3,98	3,36	2,95	2,61	2,31	2,06	1,86	1,70	1,55	1,43	1,33	1,24	1,16	1,08	1,02	0,97	0,92	0,87	0,83	0,79
	III	10,86	7,19	5,08	3,98	3,36	2,95	2,61	2,31	2,06	1,86	1,70	1,55	1,43	1,33	1,24	1,16	1,08	1,02	0,97	0,92	0,87	0,83	0,79
180/225	I	10,85	7,02	4,85	3,82	3,24	2,86	2,58	2,30	2,05	1,85	1,69	1,55	1,43	1,32	1,23	1,15	1,08	1,01	0,97	0,91	0,87	0,82	0,78
	II	10,85	7,02	4,85	3,82	3,24	2,86	2,58	2,30	2,05	1,85	1,69	1,55	1,43	1,32	1,23	1,15	1,08	1,01	0,97	0,91	0,87	0,82	0,78
	III	10,85	7,02	4,85	3,82	3,24	2,86	2,58	2,30	2,05	1,85	1,69	1,55	1,43	1,32	1,23	1,15	1,08	1,01	0,97	0,91	0,87	0,82	0,78

2-span system - maximum characteristic loads for PIR ROOF roof sandwich panel with a polyurethane core, in claddings of 0,50/0,40 mm and profile types: T/D; 40mm outer support width, 60mm inner support width.
Direction of force - TO THE SUPPORT

Core thickness [mm]	Colours group	Roof sandwich panel PIR ROOF																						
		Characteristic maximum load [kN/m²] at the given span [m]																						
		0,50	0,75	1,00	1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50	3,75	4,00	4,25	4,50	4,75	5,00	5,25	5,50	5,75	6,00
40/85	I	6,89	4,56	3,40	2,70	2,01	1,57	1,26	1,05	0,90	0,77	0,66	0,57	0,50	0,44	0,39	0,34	0,30	0,27	0,24	0,21	0,19	0,17	0,15
	II	6,89	4,56	3,40	2,70	2,01	1,57	1,26	1,05	0,90	0,77	0,66	0,57	0,50	0,44	0,39	0,34	0,30	0,27	0,24	0,21	0,19	0,17	0,15
	III	6,89	4,56	3,40	2,70	2,01	1,57	1,26	1,05	0,90	0,77	0,66	0,57	0,50	0,44	0,39	0,34	0,30	0,27	0,24	0,21	0,19	0,17	0,15
60/105	I	7,51	4,98	3,71	2,95	2,28	1,84	1,53	1,25	1,03	0,87	0,74	0,64	0,56	0,49	0,43	0,38	0,34	0,30	0,27	0,24	0,22	0,20	0,18
	II	7,51	4,98	3,71	2,95	2,28	1,84	1,53	1,25	1,03	0,87	0,74	0,64	0,56	0,49	0,43	0,38	0,34	0,30	0,27	0,24	0,22	0,20	0,18
	III	7,51	4,98	3,71	2,95	2,28	1,84	1,53	1,25	1,03	0,87	0,74	0,64	0,56	0,49	0,43	0,38	0,34	0,30	0,27	0,24	0,22	0,20	0,18
80/125	I	7,50	4,97	3,70	2,94	2,43	2,07	1,70	1,38	1,14	0,98	0,84	0,73	0,64	0,57	0,50	0,45	0,40	0,36	0,33	0,30	0,27	0,24	0,22
	II	7,50	4,97	3,70	2,94	2,43	2,07	1,70	1,38	1,14	0,98	0,84	0,73	0,64	0,57	0,50	0,45	0,40	0,36	0,33	0,30	0,27	0,24	0,22
	III	7,50	4,97	3,70	2,94	2,43	2,07	1,70	1,38	1,14	0,98	0,84	0,73	0,64	0,57	0,50	0,45	0,40	0,36	0,33	0,30	0,27	0,24	0,22
100/145	I	8,13	5,39	4,01	3,18	2,63	2,24	1,84	1,50	1,26	1,07	0,94	0,82	0,72	0,64	0,57	0,51	0,46	0,42	0,38	0,34	0,31	0,29	0,26
	II	8,13	5,39	4,01	3,18	2,63	2,24	1,84	1,50	1,26	1,07	0,94	0,82	0,72	0,64	0,57	0,51	0,46	0,42	0,38	0,34	0,31	0,29	0,26
	III	8,13	5,39	4,01	3,18	2,63	2,24	1,84	1,50	1,26	1,07	0,94	0,82	0,72	0,64	0,57	0,51	0,46	0,42	0,38	0,34	0,31	0,29	0,26
120/165	I	8,12	5,38	4,00	3,18	2,63	2,23	1,93	1,58	1,32	1,13	0,99	0,87	0,76	0,68	0,61	0,55	0,49	0,45	0,41	0,37	0,34	0,31	0,28
	II	8,12	5,38	4,00	3,18	2,63	2,23	1,93	1,58	1,32	1,13	0,99	0,87	0,76	0,68	0,61	0,55	0,49	0,45	0,41	0,37	0,34	0,31	0,28
	III	8,12	5,38	4,00	3,18	2,63	2,23	1,93	1,58	1,32	1,13	0,99	0,87	0,76	0,68	0,61	0,55	0,49	0,45	0,41	0,37	0,34	0,31	0,28
150/195	I	8,11	5,37	3,99	3,17	2,62	2,22	1,93	1,69	1,42	1,21	1,05	0,94	0,83	0,74	0,66	0,59	0,54	0,49	0,44	0,41	0,37	0,34	0,31
	II	8,11	5,37	3,99	3,17	2,62	2,22	1,93	1,69	1,42	1,21	1,05	0,94	0,83	0,74	0,66	0,59	0,54	0,49	0,44	0,41	0,37	0,34	0,31
	III	8,11	5,37	3,99	3,17	2,62	2,22	1,93	1,69	1,42	1,21	1,05	0,94	0,83	0,74	0,66	0,59	0,54	0,49	0,44	0,41	0,37	0,34	0,31
160/205	I	8,11	5,36	3,99	3,16	2,61	2,22	1,92	1,70	1,44	1,23	1,07	0,95	0,84	0,75	0,67	0,60	0,55	0,50	0,45	0,41	0,38	0,35	0,32
	II	8,11	5,36	3,99	3,16	2,61	2,22	1,92	1,70	1,44	1,23	1,07	0,95	0,84	0,75	0,67	0,60	0,55	0,50	0,45	0,41	0,38	0,35	0,32
	III	8,11	5,36	3,99	3,16	2,61	2,22	1,92	1,70	1,44	1,23	1,07	0,95	0,84	0,75	0,67	0,60	0,55	0,50	0,45	0,41	0,38	0,35	0,32
180/225	I	8,10	5,36	3,98	3,16	2,61	2,21	1,92	1,69	1,47	1,26	1,09	0,97	0,86	0,77	0,69	0,62	0,56	0,51	0,46	0,42	0,39	0,35	0,32
	II	8,10	5,36	3,98	3,16	2,61	2,21	1,92	1,69	1,47	1,26	1,09	0,97	0,86	0,77	0,69	0,62	0,56	0,51	0,46	0,42	0,39	0,35	0,32
	III	8,10	5,36	3,98	3,16	2,61	2,21	1,92	1,69	1,47	1,26	1,09	0,97	0,86	0,77	0,69	0,62	0,56	0,51	0,46	0,42	0,39	0,35	0,32

3-span system - maximum characteristic loads for PIR ROOF roof sandwich panel with a polyurethane core, in claddings of 0,50/0,40 mm and profile types: T/D; 40mm outer support width, 60mm inner support width.
Direction of force - TO THE SUPPORT

Core thickness [mm]	Colours group	Roof sandwich panel PIR ROOF																						
		Characteristic maximum load [kN/m²] at the given span [m]																						
		0,50	0,75	1,00	1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50	3,75	4,00	4,25	4,50	4,75	5,00	5,25	5,50	5,75	6,00
40/85	I	6,89	4,56	3,40	2,70	2,01	1,57	1,26	1,05	0,90	0,77	0,66	0,57	0,50	0,44	0,39	0,34	0,30	0,27	0,24	0,21	0,19	0,17	0,15
	II	6,89	4,56	3,40	2,70	2,01	1,57	1,26	1,05	0,90	0,77	0,66	0,57	0,50	0,44	0,39	0,34	0,30	0,27	0,24	0,21	0,19	0,17	0,15
	III	6,89	4,56	3,40	2,70	2,01	1,57	1,26	1,05	0,90	0,77	0,66	0,57	0,50	0,44	0,39	0,34	0,30	0,27	0,24	0,21	0,19	0,17	0,15
60/105	I	7,51	4,98	3,71	2,95	2,28	1,84	1,53	1,30	1,13	1,00	0,89	0,80	0,71	0,63	0,56	0,50	0,45	0,40	0,37	0,33	0,30	0,28	0,25
	II	7,51	4,98	3,71	2,95	2,28	1,84	1,53	1,30	1,13	1,00	0,89	0,80	0,71	0,63	0,56	0,50	0,45	0,40	0,37	0,33	0,30	0,28	0,25
	III	7,51	4,98	3,71	2,95	2,28	1,84	1,53	1,30	1,13	1,00	0,89	0,80	0,71	0,63	0,56	0,50	0,45	0,40	0,37	0,33	0,30	0,28	0,25
80/125	I	7,50	4,97	3,70	2,94	2,43	2,07	1,79	1,58	1,41	1,19	1,03	0,91	0,81	0,72	0,64	0,58	0,52	0,47	0,43	0,39	0,36	0,33	0,30
	II	7,50	4,97	3,70	2,94	2,43	2,07	1,79	1,58	1,41	1,19	1,03	0,91	0,81	0,72	0,64	0,58	0,52	0,47	0,43	0,39	0,36	0,33	0,30
	III	7,50	4,97	3,70	2,94	2,43	2,07	1,79	1,58	1,41	1,19	1,03	0,91	0,81	0,72	0,64	0,58	0,52	0,47	0,43	0,39	0,36	0,33	0,30
100/145	I	8,13	5,39	4,01	3,18	2,63	2,24	1,95	1,72	1,51	1,29	1,12	1,00	0,89	0,79	0,71	0,64	0,58	0,53	0,49	0,45	0,41	0,38	0,35
	II	8,13	5,39	4,01	3,18	2,63	2,24	1,95	1,72	1,51	1,29	1,12	1,00	0,89	0,79	0,71	0,64	0,58	0,53	0,49	0,45	0,41	0,38	0,35
	III	8,13	5,39	4,01	3,18	2,63	2,24	1,95	1,72	1,51	1,29	1,12	1,00	0,89	0,79	0,71	0,64	0,58	0,53	0,49	0,45	0,41	0,38	0,35
120/165	I	8,12	5,38	4,00	3,18	2,63	2,23	1,94	1,71	1,53	1,34	1,17	1,03	0,93	0,83	0,75	0,68	0,62	0,56	0,52	0,47	0,44	0,40	0,37
	II	8,12	5,38	4,00	3,18	2,63	2,23	1,94	1,71	1,53	1,34	1,17	1,03	0,93	0,83	0,75	0,68	0,62	0,56	0,52	0,47	0,44	0,40	0,37
	III	8,12	5,38	4,00	3,18	2,63	2,23	1,94	1,71	1,53	1,34	1,17	1,03	0,93	0,83	0,75	0,68	0,62	0,56	0,52	0,47	0,44	0,40	0,37
150/195	I	8,11	5,37	3,99	3,17	2,62	2,22	1,93	1,70	1,52	1,37	1,23	1,09	0,98	0,88	0,80	0,72	0,66	0,60	0,55	0,51	0,47	0,43	0,40
	II	8,11	5,37	3,99	3,17	2,62	2,22	1,93	1,70	1,52	1,37	1,23	1,09	0,98	0,88	0,80	0,72	0,66	0,60	0,55	0,51	0,47	0,43	0,40
	III	8,11	5,37	3,99	3,17	2,62	2,22	1,93	1,70	1,52	1,37	1,23	1,09	0,98	0,88	0,80	0,72	0,66	0,60	0,55	0,51	0,47	0,43	0,40
160/205	I	8,11	5,36	3,99	3,16	2,61	2,22	1,92	1,70	1,51	1,36	1,24	1,10	0,99	0,89	0,80	0,73	0,66	0,61	0,56	0,51	0,47	0,44	0,41
	II	8,11	5,36	3,99	3,16	2,61	2,22	1,92	1,70	1,51	1,36	1,24	1,10	0,99	0,89	0,80	0,73	0,66	0,61	0,56	0,51	0,47	0,44	0,41
	III	8,11	5,36	3,99	3,16	2,61	2,22	1,92	1,70	1,51	1,36	1,24	1,10	0,99	0,89	0,80	0,73	0,66	0,61	0,56	0,51	0,47	0,44	0,41
180/225	I	8,10	5,36	3,98	3,16	2,61	2,21	1,92	1,69	1,50	1,35	1,23	1,11	1,00	0,90	0,81	0,74	0,67	0,61	0,56	0,52	0,48	0,44	0,41
	II	8,10	5,36	3,98	3,16	2,61	2,21	1,92	1,69	1,50	1,35	1,23	1,11	1,00	0,90	0,81	0,74	0,67	0,61	0,56	0,52	0,48	0,44	0,41
	III	8,10	5,36	3,98	3,16	2,61	2,21	1,92	1,69	1,50	1,35	1,23	1,11	1,00	0,90	0,81	0,74	0,67	0,61	0,56	0,52	0,48	0,44	0,41

PIR ROOF roof sandwich panel

1-span system - maximum characteristic loads for PIR ROOF roof sandwich panel with a polyurethane core, in claddings of 0,50/0,40 mm and profile types: T/D; 40mm outer support width.
Direction of force - FROM THE SUPPORT

Core thickness [mm]	Colours group	Roof sandwich panel PIR ROOF																						
		Characteristic maximum load [kN/m²] at the given span [m]																						
		0,50	0,75	1,00	1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50	3,75	4,00	4,25	4,50	4,75	5,00	5,25	5,50	5,75	6,00
40/85	I	-16,39	-7,87	-4,83	-3,38	-2,55	-2,02	-1,65	-1,38	-1,18	-1,02	-0,91	-0,80	-0,72	-0,64	-0,58	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
	II	-16,39	-7,87	-4,83	-3,38	-2,55	-2,02	-1,65	-1,38	-1,18	-1,02	-0,91	-0,80	-0,72	-0,64	-0,52	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
	III	-16,39	-7,87	-4,83	-3,38	-2,55	-2,02	-1,65	-1,38	-1,18	-1,02	-0,87	-0,60	-0,40	-0,26	-0,16	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
60/105	I	-16,94	-8,42	-5,38	-3,92	-3,08	-2,53	-2,15	-1,85	-1,62	-1,43	-1,27	-1,14	-1,03	-0,95	-0,86	-0,79	-0,73	-0,64	-0,57	0,00	0,00	0,00	0,00
	II	-16,94	-8,42	-5,38	-3,92	-3,08	-2,53	-2,15	-1,85	-1,62	-1,43	-1,27	-1,14	-1,03	-0,95	-0,86	-0,74	-0,64	-0,56	-0,50	0,00	0,00	0,00	0,00
	III	-16,93	-8,42	-5,38	-3,92	-3,08	-2,53	-2,15	-1,85	-1,62	-1,43	-1,27	-1,14	-0,99	-0,76	-0,58	-0,44	-0,34	-0,25	-0,19	0,00	0,00	0,00	0,00
80/125	I	-17,93	-9,39	-6,32	-4,82	-3,93	-3,33	-2,88	-2,53	-2,24	-2,00	-1,79	-1,62	-1,47	-1,33	-1,22	-1,11	-1,02	-0,94	-0,85	-0,77	-0,69	-0,63	-0,57
	II	-17,93	-9,39	-6,32	-4,82	-3,93	-3,33	-2,88	-2,53	-2,24	-2,00	-1,79	-1,62	-1,47	-1,33	-1,22	-1,11	-0,99	-0,87	-0,77	-0,69	-0,61	-0,55	-0,50
	III	-17,93	-9,39	-6,32	-4,82	-3,93	-3,33	-2,88	-2,53	-2,24	-2,00	-1,79	-1,62	-1,47	-1,30	-1,11	-0,94	-0,76	-0,62	-0,50	-0,41	-0,34	-0,27	-0,23
100/145	I	-18,95	-10,38	-7,28	-5,75	-4,81	-4,16	-3,65	-3,25	-2,90	-2,61	-2,35	-2,13	-1,93	-1,76	-1,61	-1,48	-1,36	-1,25	-1,13	-1,02	-0,94	-0,85	-0,78
	II	-18,95	-10,38	-7,28	-5,75	-4,81	-4,16	-3,65	-3,25	-2,90	-2,61	-2,35	-2,13	-1,93	-1,76	-1,61	-1,48	-1,36	-1,22	-1,08	-0,98	-0,88	-0,79	-0,72
	III	-18,95	-10,38	-7,28	-5,75	-4,81	-4,16	-3,65	-3,25	-2,90	-2,61	-2,35	-2,13	-1,93	-1,76	-1,59	-1,38	-1,20	-1,05	-0,93	-0,78	-0,66	-0,56	-0,48
120/165	I	-19,54	-10,98	-7,89	-6,35	-5,41	-4,75	-4,23	-3,80	-3,43	-3,11	-2,82	-2,57	-2,35	-2,15	-1,97	-1,82	-1,68	-1,52	-1,38	-1,26	-1,15	-1,05	-0,97
	II	-19,54	-10,98	-7,89	-6,35	-5,41	-4,75	-4,23	-3,80	-3,43	-3,11	-2,82	-2,57	-2,35	-2,15	-1,97	-1,82	-1,68	-1,52	-1,38	-1,25	-1,13	-1,02	-0,94
	III	-19,54	-10,98	-7,89	-6,35	-5,41	-4,75	-4,23	-3,80	-3,43	-3,11	-2,82	-2,57	-2,35	-2,15	-1,97	-1,77	-1,56	-1,38	-1,22	-1,09	-0,99	-0,89	-0,79
150/195	I	-20,39	-11,85	-8,76	-7,22	-6,28	-5,60	-5,06	-4,60	-4,20	-3,84	-3,52	-3,23	-2,97	-2,73	-2,52	-2,33	-2,10	-1,89	-1,71	-1,56	-1,42	-1,31	-1,21
	II	-20,39	-11,85	-8,76	-7,22	-6,28	-5,60	-5,06	-4,60	-4,20	-3,84	-3,52	-3,23	-2,97	-2,73	-2,52	-2,33	-2,10	-1,89	-1,71	-1,56	-1,42	-1,31	-1,21
	III	-20,39	-11,85	-8,76	-7,22	-6,28	-5,60	-5,06	-4,60	-4,20	-3,84	-3,52	-3,23	-2,97	-2,73	-2,52	-2,33	-2,10	-1,89	-1,70	-1,53	-1,38	-1,25	-1,13
160/205	I	-20,56	-12,02	-8,94	-7,41	-6,47	-5,80	-5,26	-4,81	-4,40	-4,04	-3,71	-3,42	-3,15	-2,90	-2,68	-2,48	-2,22	-2,00	-1,81	-1,65	-1,51	-1,39	-1,28
	II	-20,56	-12,02	-8,94	-7,41	-6,47	-5,80	-5,26	-4,81	-4,40	-4,04	-3,71	-3,42	-3,15	-2,90	-2,68	-2,48	-2,22	-2,00	-1,81	-1,65	-1,51	-1,39	-1,28
	III	-20,56	-12,02	-8,94	-7,41	-6,47	-5,80	-5,26	-4,81	-4,40	-4,04	-3,71	-3,42	-3,15	-2,90	-2,68	-2,48	-2,22	-2,00	-1,81	-1,65	-1,50	-1,36	-1,24
180/225	I	-20,78	-12,25	-9,19	-7,68	-6,77	-6,11	-5,59	-5,14	-4,74	-4,38	-4,05	-3,74	-3,46	-3,21	-2,95	-2,69	-2,45	-2,23	-2,02	-1,84	-1,68	-1,54	-1,42
	II	-20,78	-12,25	-9,19	-7,68	-6,77	-6,11	-5,59	-5,14	-4,74	-4,38	-4,05	-3,74	-3,46	-3,21	-2,95	-2,69	-2,45	-2,23	-2,02	-1,84	-1,68	-1,54	-1,42
	III	-20,78	-12,25	-9,19	-7,68	-6,77	-6,11	-5,59	-5,14	-4,74	-4,38	-4,05	-3,74	-3,46	-3,21	-2,95	-2,69	-2,45	-2,23	-2,02	-1,84	-1,68	-1,54	-1,42

2-span system - maximum characteristic loads for PIR ROOF roof sandwich panel with a polyurethane core, in claddings of 0,50/0,40 mm and profile types: T/D; 40mm outer support width, 60mm inner support width.
Direction of force - FROM THE SUPPORT

Core thickness [mm]	Colours group	Roof sandwich panel PIR ROOF																						
		Characteristic maximum load [kN/m²] at the given span [m]																						
		0,50	0,75	1,00	1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50	3,75	4,00	4,25	4,50	4,75	5,00	5,25	5,50	5,75	6,00
40/85	I	-16,38	-7,87	-4,83	-3,38	-2,55	-2,02	-1,65	-1,38	-1,18	-1,02	-0,91	-0,80	-0,72	-0,64	-0,58	-0,53	-0,48	-0,44	-0,41	-0,38	-0,35	-0,33	-0,31
	II	-16,38	-7,87	-4,83	-3,38	-2,55	-2,02	-1,65	-1,38	-1,18	-1,02	-0,91	-0,80	-0,72	-0,64	-0,58	-0,53	-0,48	-0,44	-0,41	-0,38	-0,35	-0,33	-0,31
	III	-16,38	-7,87	-4,83	-3,38	-2,55	-1,96	-1,55	-1,27	-1,07	-0,94	-0,82	-0,73	-0,65	-0,59	-0,54	-0,50	-0,46	-0,43	-0,41	-0,38	-0,35	-0,33	-0,31
60/105	I	-16,94	-8,42	-5,38	-3,92	-3,08	-2,50	-2,02	-1,68	-1,43	-1,24	-1,10	-0,99	-0,89	-0,81	-0,75	-0,69	-0,64	-0,60	-0,56	-0,53	-0,50	-0,47	-0,45
	II	-16,94	-8,42	-5,38	-3,92	-3,08	-2,37	-1,90	-1,57	-1,34	-1,16	-1,02	-0,92	-0,83	-0,76	-0,69	-0,64	-0,60	-0,56	-0,52	-0,49	-0,45	-0,44	-0,42
	III	-16,93	-8,42	-5,38	-3,92	-2,85	-2,16	-1,72	-1,41	-1,19	-1,03	-0,92	-0,82	-0,74	-0,67	-0,62	-0,57	-0,53	-0,50	-0,47	-0,44	-0,42	-0,40	-0,38
80/125	I	-17,94	-9,39	-6,32	-4,79	-3,55	-2,79	-2,27	-1,90	-1,63	-1,42	-1,26	-1,13	-1,02	-0,95	-0,87	-0,80	-0,75	-0,70	-0,66	-0,62	-0,59	-0,56	-0,53
	II	-17,94	-9,39	-6,32	-4,60	-3,39	-2,64	-2,13	-1,78	-1,52	-1,33	-1,17	-1,05	-0,96	-0,88	-0,81	-0,75	-0,70	-0,65	-0,61	-0,58	-0,55	-0,52	-0,50
	III	-17,93	-9,39	-6,32	-4,33	-3,14	-2,41	-1,93	-1,60	-1,36	-1,18	-1,04	-0,94	-0,85	-0,78	-0,72	-0,66	-0,62	-0,58	-0,55	-0,52	-0,49	-0,47	-0,45
100/145	I	-18,95	-10,38	-7,23	-5,03	-3,78	-2,99	-2,45	-2,07	-1,78	-1,56	-1,38	-1,24	-1,13	-1,03	-0,96	-0,89	-0,83	-0,78	-0,73	-0,69	-0,65	-0,62	-0,59
	II	-18,95	-10,38	-7,02	-4,83	-3,60	-2,82	-2,30	-1,93	-1,66	-1,45	-1,28	-1,15	-1,04	-0,97	-0,89	-0,83	-0,77	-0,72	-0,68	-0,64	-0,61	-0,58	-0,55
	III	-18,95	-10,38	-6,71	-4,54	-3,33	-2,58	-2,08	-1,73	-1,47	-1,28	-1,13	-1,01	-0,93	-0,85	-0,79	-0,73	-0,68	-0,64	-0,60	-0,57	-0,54	-0,52	-0,49
120/165	I	-19,54	-10,98	-7,09	-4,97	-3,76	-2,99	-2,46	-2,08	-1,80	-1,58	-1,40	-1,26	-1,15	-1,05	-0,98	-0,91	-0,85	-0,80	-0,75	-0,71	-0,67	-0,64	-0,61
	II	-19,54	-10,98	-6,90	-4,79	-3,59	-2,83	-2,32	-1,95	-1,67	-1,46	-1,30	-1,17	-1,06	-0,98	-0,91	-0,84	-0,79	-0,74	-0,70	-0,66	-0,63	-0,60	-0,57
	III	-19,53	-10,98	-6,61	-4,51	-3,33	-2,59	-2,10	-1,74	-1,49	-1,29	-1,14	-1,02	-0,94	-0,86	-0,80	-0,74	-0,69	-0,65	-0,61	-0,58	-0,55	-0,53	-0,50
150/195	I	-20,39	-10,91	-6,81	-4,83	-3,68	-2,95	-2,44	-2,07	-1,80	-1,58	-1,41	-1,27	-1,16	-1,06	-0,99	-0,92	-0,86	-0,81	-0,76	-0,72	-0,69	-0,66	-0,63
	II	-20,39	-10,73	-6,64	-4,66	-3,52	-2,80	-2,30	-1,94	-1,67	-1,46	-1,30	-1,17	-1,06	-0,99	-0,91	-0,85	-0,79	-0,74	-0,70	-0,67	-0,63	-0,60	-0,58
	III	-20,39	-10,45	-6,37	-4,40	-3,28	-2,56	-2,08	-1,73	-1,48	-1,29	-1,14	-1,02	-0,94	-0,86	-0,79	-0,73	-0,69	-0,65	-0,61	-0,58	-0,55	-0,53	-0,51
160/205	I	-20,55	-10,66	-6,68	-4,75	-3,63	-2,91	-2,41	-2,05	-1,78	-1,57	-1,40	-1,26	-1,15	-1,05	-0,99	-0,92	-0,86	-0,81	-0,76	-0,72	-0,68	-0,65	-0,62
	II	-20,55	-10,49	-6,51	-4,58	-3,47	-2,76	-2,27	-1,92	-1,65	-1,45	-1,29	-1,16	-1,05	-0,98	-0,90	-0,84	-0,79	-0,74	-0,70	-0,66	-0,63	-0,60	-0,57
	III	-20,56	-10,22	-6,25	-4,33	-3,23	-2,53	-2,06	-1,72	-1,46	-1,27	-1,12	-1,16	-0,93	-0,85	-0,78	-0,73	-0,68	-0,64	-0,60	-0,57	-0,55	-0,52	-0,50
180/225	I	-20,55	-10,13	-6,37	-4,54	-3,49	-2,81	-2,33	-1,99	-1,73	-1,52	-1,36	-1,23	-1,12	-1,03	-0,96	-0,90	-0,84	-0,79	-0,74	-0,71	-0,67	-0,64	-0,61
	II	-20,39	-9,97	-6,22	-4,39	-3,34	-2,66	-2,20	-1,86	-1,60	-1,41	-1,25	-1,13	-1,02	-0,95	-0,88	-0,82	-0,77	-0,72	-0,68	-0,65	-0,61	-0,59	-0,56
	III	-20,16	-9,73	-5,98	-4,16	-3,12	-2,45	-2,00	-1,67	-1,42	-1,24	-1,09	-0,99	-0,90	-0,82	-0,76	-0,70	-0,65	-0,61	-0,58	-0,55	-0,53	-0,51	-0,49

3-span system - maximum characteristic loads for PIR ROOF roof sandwich panel with a polyurethane core, in claddings of 0,50/0,40 mm and profile types: T/D; 40mm outer support width, 60mm inner support width.
Direction of force - FROM THE SUPPORT

Core thickness [mm]	Colours group	Roof sandwich panel PIR ROOF																						
		Characteristic maximum load [kN/m²] at the given span [m]																						
		0,50	0,75	1,00	1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50	3,75	4,00	4,25	4,50	4,75	5,00	5,25	5,50	5,75	6,00
40/85	I	-16,38	-7,87	-4,83	-3,38	-2,55	-2,02	-1,65	-1,38	-1,18	-1,02	-0,91	-0,80	-0,72	-0,64	-0,58	-0,53	-0,48	-0,44	-0,41	-0,38	-0,35	-0,33	-0,31
	II	-16,38	-7,87	-4,83	-3,38	-2,55	-2,02	-1,65	-1,38	-1,18	-1,02	-0,91	-0,80	-0,72	-0,64	-0,58	-0,53	-0,48	-0,44	-0,41	-0,38	-0,35	-0,33	-0,31
	III	-16,38	-7,87	-4,83	-3,38	-2,55	-2,02	-1,65	-1,38	-1,18	-1,02	-0,91	-0,80	-0,72	-0,64	-0,58	-0,53	-0,48	-0,44	-0,41	-0,38	-0,35	-0,33	-0,31
60/105	I	-16,94	-8,42	-5,38	-3,92	-3,08	-2,53	-2,15	-1,85	-1,62	-1,43	-1,27	-1,14	-1,03	-0,95	-0,86	-0,79	-0,73	-0,67	-0,62	-0,58	-0,54	-0,50	-0,47
	II	-16,93	-8,42	-5,38	-3,92	-3,08	-2,53	-2,15	-1,85	-1,62	-1,42	-1,26	-1,13	-1,02	-0,94	-0,86	-0,79	-0,73	-0,67	-0,62	-0,58	-0,54	-0,50	-0,47
	III	-16,93	-8,42	-5,38	-3,92	-3,08	-2,53	-2,14	-1,78	-1,52	-1,32	-1,17	-1,05	-0,96	-0,87	-0,80	-0,74	-0,69	-0,65	-0,61	-0,57	-0,54	-0,50	-0,47
80/125	I	-17,93	-9,39	-6,32	-4,82	-3,93	-3,24	-2,65	-2,23	-1,92	-1,68	-1,50	-1,35	-1,22	-1,12	-1,03	-0,96	-0,90	-0,84	-0,79	-0,74	-0,70	-0,67	-0,63
	II	-17,94	-9,39	-6,32	-4,82	-3,93	-3,11	-2,53	-2,13	-1,83	-1,60	-1,43	-1,28	-1,16	-1,07	-0,99	-0,92	-0,86	-0,80	-0,75	-0,71	-0,67	-0,64	-0,61
	III	-17,94	-9,39	-6,32	-4,82	-3,75	-2,91	-2,36	-1,98	-1,70	-1,49	-1,32	-1,19	-1,08	-1,07	-0,92	-0,86	-0,80	-0,75	-0,71	-0,67	-0,63	-0,60	-0,57
100/145	I	-18,95	-10,38	-7,28	-5,75	-4,33	-3,42	-2,81	-2,38	-2,06	-1,82	-1,62	-1,46	-1,33	-1,22	-1,13	-1,05	-0,99	-0,93	-0,87	-0,82	-0,78	-0,74	-0,70
	II	-18,95	-10,38	-7,28	-5,60	-4,16	-3,28	-2,69	-2,27	-1,97	-1,73	-1,54	-1,39	-1,27	-1,16	-1,08	-1,05	-0,94	-0,89	-0,83	-0,79	-0,75	-0,71	-0,68
	III	-18,95	-10,38	-7,28	-5,31	-3,91	-3,06	-2,50	-2,11	-1,82	-1,60	-1,43	-1,29	-1,17	-1,08	-1,00	-0,94	-0,88	-0,82	-0,78	-0,73	-0,70	-0,66	-0,63
120/165	I	-19,54	-10,98	-7,89	-5,67	-4,26	-3,38	-2,79	-2,37	-2,06	-1,82	-1,62	-1,47	-1,34	-1,23	-1,14	-1,06	-0,94	-0,89	-0,84	-0,80	-0,76	-0,73	
	II	-19,54	-10,98	-7,89	-5,49	-4,10	-3,24	-2,66	-2,26	-1,96	-1,72	-1,54	-1,40	-1,27	-1,17	-1,09	-1,01	-0,96	-0,90	-0,85	-0,80	-0,76	-0,73	-0,69
	III	-19,53	-10,98	-7,69	-5,21	-3,85	-3,02	-2,47	-2,09	-1,80	-1,59	-1,42	-1,28	-1,17	-1,08	-1,00	-0,94	-0,88	-0,83	-0,79	-0,75	-0,71	-0,68	-0,65
150/195	I	-20,38	-11,85	-7,82	-5,45	-4,12	-3,29	-2,72	-2,32	-2,02	-1,79	-1,61	-1,46	-1,33	-1,23	-1,14	-1,06	-0,93	-0,91	-0,90	-0,85	-0,81	-0,77	-0,74
	II	-20,39	-11,84	-7,63	-5,28	-3,96	-3,14	-2,59	-2,20	-1,91	-1,69	-1,52	-1,38	-1,26	-1,16	-1,08	-1,01	-0,96	-0,90	-0,85	-0,81	-0,77	-0,73	-0,70
	III	-20,39	-11,84	-7,35	-5,01	-3,72	-2,93	-2,40	-2,03	-1,75	-1,55	-1,38	-1,25	-1,15	-1,06	-1,00	-0,93	-0,88	-0,83	-0,78	-0,74	-0,71	-0,68	-0,65
160/205	I	-20,55	-12,01	-7,66	-5,34	-4,04	-3,23	-2,68	-2,28	-1,99	-1,76	-1,58	-1,44	-1,31	-1,21	-1,13	-1,05	-0,92	-0,91	-0,89	-0,84	-0,80	-0,77	-0,73
	II	-20,55	-12,01	-7,47	-5,17	-3,89	-3,09	-2,55	-2,17	-1,88	-1,67	-1,49	-1,35	-1,24	-1,14	-1,06	-1,00	-0,94	-0,89	-0,84	-0,80	-0,76	-0,73	-0,70
	III	-20,56	-12,01	-7,20	-4,92	-3,65	-2,87	-2,35	-1,99	-1,72	-1,52	-1,36	-1,23	-1,13	-1,04	-0,98	-0,92	-0,86	-0,81	-0,77	-0,73	-0,70	-0,67	-0,64
180/225	I	-20,77	-11,89	-7,29	-5,10	-3,87	-3,09	-2,57	-2,19	-1,91	-1,69	-1,52	-1,38	-1,27	-1,17	-1,09	-1,02	-0,97	-0,91	-0,86	-0,82	-0,78	-0,75	-0,72
	II	-20,78	-11,71	-7,12	-4,94	-3,72	-2,96	-2,44	-2,08	-1,81	-1,60	-1,43	-1,30	-1,19	-1,10	-1,02	-0,97	-0,91	-0,86	-0,82	-0,78	-0,74	-0,71	-0,68
	III	-20,78	-11,45	-6,87	-4,71	-3,50	-2,75	-2,26	-1,91	-1,65	-1,45	-1,30	-1,18	-1,08	-1,00	-0,94	-0,88	-0,83	-0,78	-0,74	-0,71	-0,68	-0,65	-0,62

16. THERMAL PERFORMANCE

PIR STANDARD, PIR PLUS wall sandwich panels and PIR ROOF roof sandwich panels are distinguished by good thermal performance parameters. Tests and calculations carried out in the Building Research Institute in the Department of Thermal Physics in Warsaw aimed at determining the thermal conductivity coefficient of polyurethane foam forming the panel's insulating core and the partition heat-transfer coefficient proved high the quality as well as high replicability of PIR panel parameters achieved due to the application of top class materials and continuous control of all production stages.

The design thermal conductivity coefficient (used for designing purposes and corresponding to the conditions of material application) reaches λ_D at least 0.022 W/mK in relation to the average partition's temperature of 10°C.

Calculation of values of heat-transfer coefficients U_c was carried out taking linear thermal bridges occurring in the panels' joints as well as spot thermal bridges occurring in the place of fastening panels to the load-bearing structure with the use of fasteners into consideration.

The values of heat-transfer coefficients U_c of a partition made of PIR wall and roof panels are provided in the table below.

Tabela 16. The values of U_c partition's heat-transfer coefficient

Panel type	Core thickness [mm]	U_c [W/m²K]
PIR STANDARD wall panel	40	0,64
	50	0,51
	60	0,39
	80	0,29
	100	0,22
	120	0,19
PIR PLUS wall panel	60	0,47
	80	0,32
	100	0,25
	120	0,19
PIR ROOF roof panel	40/85	0,60
	60/105	0,40
	80/125	0,29
	100/145	0,22
	120/165	0,19
	150/195	0,15
	160/205	0,14
	180/225	0,12

17. ACOUSTIC INSULATION

The acoustic parameters presented here are based on calculations made using the INSUL program, which has a tolerance of +/-3 dB. Calculation reports containing graphs showing sound insulation values for individual frequency ranges are available from the Balex Metal design office (bp@balex.eu).

Panel type	Thickness	Airborne sound insulation R_w [dB]	Indicator C [dB]	Indicator C_{tr} [dB]	Airborne sound insulation for medium and high frequeces R_{A1} [dB]	Airborne sound insulation for low and medium frequeces R_{A2} [dB]
PIR STANDARD wall panel	40	27	-1	-3	26	24
	50	27	0	-3	27	24
	60	27	0	-3	27	24
	80	28	-1	-3	27	25
	100	29	-1	-4	28	25
	120	29	-1	-3	28	26
PIR PLUS wall panel	60	27	0	-3	27	24
	80	28	-1	-3	27	25
	100	29	-1	-4	28	25
	120	29	-1	-3	28	26

18. FIRE SAFETY

As building components of determined fire resistance class, PIR sandwich panels with polyurethane core must meet the fire safety and suppression requirements set out in the law.
External walls and roofs of buildings are subject to fire safety classifications in the following fields:

- fire resistance
- fire spreading level

PIR sandwich panels are classified as a FR product – fire resisting – in accordance with the standards: PN-90/B-02867 and PN-EN 02872:1996.

Table 17. Fire classification for wall sandwich panels PIR STANDARD & PIR PLUS

PIR STANDARD wall sandwich panel						
Core thickness [mm]	**40**	**50**	**60**	**80**	**100**	**120**
Fire resistance (horizontal orientation)	-				**up to EI 15 (4m)**	**up to EI 20 (4m)**
Fire resistance (vertical orietnation)	-			**up to EI 15 (3m)**		**do EI 20 (3m)/ do EI 15 (4m)**
Reaction to fire with standard gasket	**B-s2,d0**				**B-s1,d0**	
Reaction to fire with carbon gasket	-		**B-s1,d0**			
External fire performance	**NRO (not spreading fire)**					

PIR PLUS wall sandwich panel				
Core thickness [mm]	**60**	**80**	**100**	**120**
Fire resistance (horizontal orientation)	-	**up to EI 15 (3m)**		
Fire resistance (vertical orietnation)	-		**up to EI 15 (4m)**	
Reaction to fire	**B-s2,d0**			
External fire performance	**NRO (not spreading fire)**			

The B-s2,d0 classification allows for end applications of PIR panels as both ceilings and enclosure walls in accordance with technical conditions specified for buildings and their placement and as for "non-ignitable, non-dripping and non-falling off when exposed to fire" products and "fire retardant" construction elements in accordance with the polish law.

Table 18. Fire classification for roof sandwich panels PIR ROOF

PIR ROOF roof sanwich panel								
Core thickness [mm]	**40**	**60**	**80**	**100**	**120**	**150**	**160**	**180**
Fire resistance	-			**up to RE 60 / up to REI 15**				
Reaction to fire	**B-s1,d0**							
Roof external fire performance	**Broof (t1), Broof (t2), Broof (t3)**							

Fire classifications remain valid for roofs with a slope of up to 15 degrees

19. CORROSION RESISTANCE

Corrosion classes and examples of typical environments according to the PN-EN ISO 12944-2 standard.

C1 corrosion class

- indoors – heated buildings with a clean atmosphere e.g. offices, shops, schools and hotels.

C2 corrosion class

- outdoors – slightly polluted atmosphere, mainly countryside
- indoors – not heated buildings with a possibility of condensation e.g. warehouses, sports halls.

C3 corrosion class

- outdoors – urban and industrial atmospheres, medium pollution with sulphur oxide (IV), coastal area with low salinit
- indoors – production rooms with high humidity and certain level of air pollution e.g. food industrial facilities, laundries, breweries, dairies.

C4 corrosion class

- outdoors – industrial areas, coastal area with medium salinity
- indoors – chemical plants, swimming pools, shipyards and yacht yards.

C5 corrosion class

- outdoors - industrial areas with high humidity and aggressive atmosphere
- indoors - buildings or areas with almost permanent condensation and with high pollution

The table below shows the relevant corrosion resistance categories depending on the coating used.

Table 19. Corrosion resistance classes.

Coating	PE15	PE25	PE35M	PE35	PUR 50/55	PVC 120 (foodsafe)	INOX
Metallic coating	min Z225 or equal (alloy coatings)						-
Corrosion resistance class (external) RC	–	up to RC3	up to RC4	up to RC4	up to RC5*	–	up to RC5*
Corrosion resistance class (internal) AC	CPI2	up to AC3	up to AC4*	up to AC4*	up to AC5*	up to CPI5*	up to AC5*

* Confirmation of the corrosion resistance class RC/AC by the steel suplier after analyzing the enviromental questionnarie only (via the Quality Departament)

** Transport, assembly, cleaning, maintenance in accordance with Balex Metal recommendations

20 FASTENERS

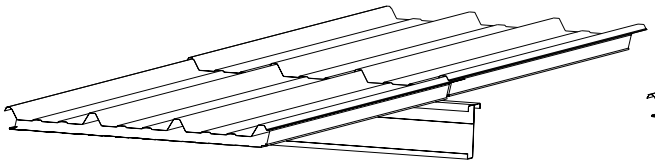
The PIR sandwich panels are fastened to supporting steel structures with self-drilling fasteners. This way, the need for pre-drilling a pilot is eliminated. Moreover, the self-drilling fasteners improve the reliability of fastening and limit the number of tools used. In the case of self-drilling fasteners, the drilling bit is always new, because the fasteners are intended for one-time application - this improves the durability of fastening.

The self-drilling fasteners are intended for fastening sandwich panels to steel structures of maximum wall thickness of 12 mm. The fasteners are made of hardened carbon steel with surface corrosion protection. All fasteners are equipped with washers with cured EPDM. Application of EPDM improves the durability and tightness of joints.

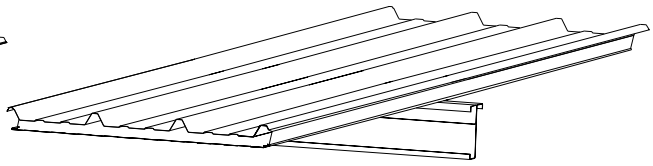
Used fasteners:
- a fastener of drilling ability up to 6 mm, intended for cold-formed steel bases
- a fastener of drilling ability up to 12 mm, intended for hot-rolled steel bases
- a fastener of drilling ability up to 16 mm, intended for hot-rolled steel bases
- A fastener for wood and concrete – intended for direct assembly in a wooden or concrete base. Before assembly, it is recommended to pre-drill the sandwich panel facing with a 5 mm drill. In the case of concrete base, it is necessary to pre-drill a hole of 5 mm diameter with a concrete drill.
- Self-drilling fastener – intended for steel bases of thickness over 12 mm. Before the assembly it is necessary to pre-drill the sandwich panel and in the steel structure with a 5.8 mm drill.

21. FASTENING ROOF PANELS LENGTHWISE

Recommended roof pitch for PIR ROOF panels is:

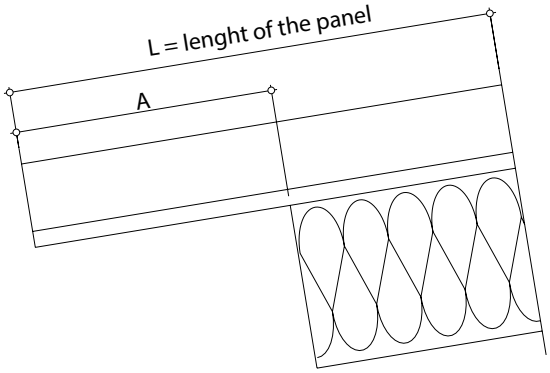


>7 % - for panels joined lengthwise or with roof skylights



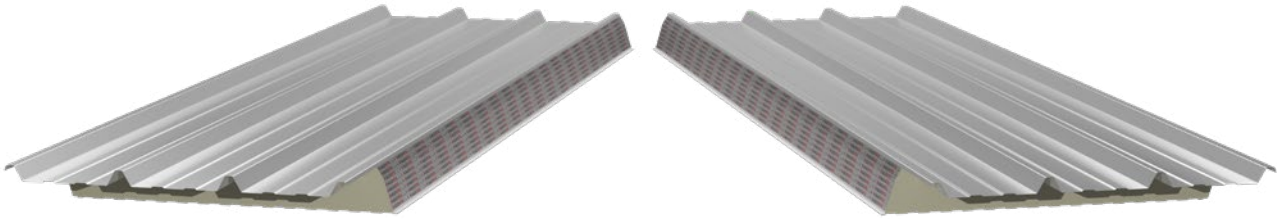
> 5% - for continuous panels without roof skylights

PIR ROOF roof panels have endings formed at the production stage that facilitate gutter assembly next to the eaves or joining panels lengthwise.



A - overlap:
- standard 50 mm next to the eaves
- standard 200 mm next to the overlap
- max. 300 mm next to the overlap
- min. 10mm without any overlap

PIR ROOF roof panel is produced as a RIGHT option as a standard. LEFT option is produced on the customers demand.



LEFT panel **RIGHT panel**

Panel side overlaps

Roof sandwich panels have factory-formed end sections during the manufacturing process, known as panel side overlaps, which facilitate installation at the eaves on the edge of the roof or the longitudinal joining of panels.

RIGHT OVERLAP - direction of roof panel installation: **from the right side**



overlap wave on the right side

LEFT OVERLAP - direction of roof panel installation: **from the left side**



overlap wave on the left side

22. GENERAL GUIDELINES ON ASSEMBLY

Before starting assembly it is recommended that one should verify the carrying construction in terms of precision of performance and its accordance with the design.

PIR sandwich panels are protected against contamination and damaging by protective film, with which the cladding is covered during the production process. After unpacking the set of panels, the protective film must be removed to avoid its permanent binding with protective varnish of lining sheets.

In the case of symmetrical sandwich panels in order to avoid mistakes during assembly aluminium foil placed in the longitudinal panel joint in the production process is additionally marked, apart from the BALEX METAL company's logo, with an arrow with the inscription ‚external side' – fig. 1

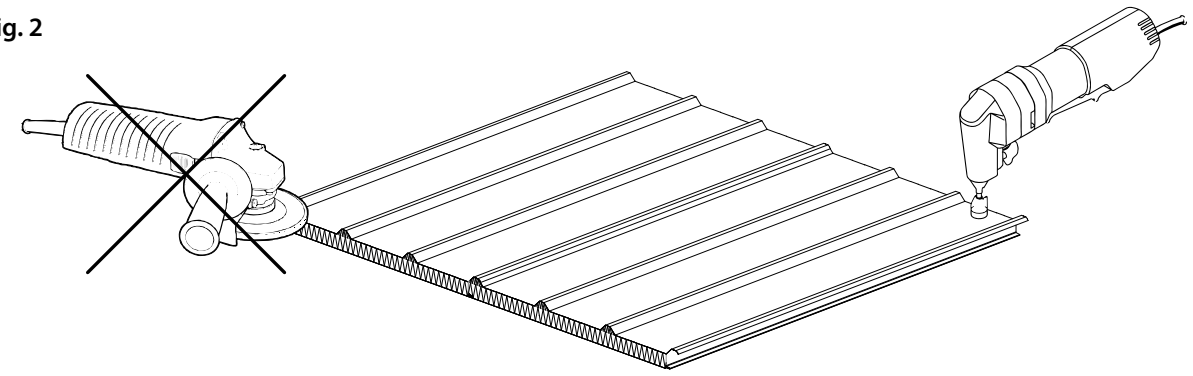
Fig. 1



In order to prevent colour coating from being damaged it is recommended to cut panels and flashings on stands covered with soft material e.g. felt, polystyrene foam.

Use circular saws with fine-toothed blades for cutting the panels, and manual scissors for sheet cutting. It is forbidden to use angle grinders and other tools heating up claddings in the process of cutting – it can lead to damaging anti-corrosion coating – fig. 2

Fig. 2



It is recommended to fasten sandwich panels with the use of appropriate fasteners depending on the type of supporting structure and the panel core thickness. The types of fastening elements and all markings are provided in the FASTENER chapter.

It is recommended to use hardened galvanized carbon steel self-drilling fasteners for fastening sandwich panels to a steel construction that is less than 12 mm thick. All fasteners are equipped with a washer with a cured EPDM which ensures many years of operation retaining the flexibility of the sealing element.

In the case of steel base (thickness >12 mm) or wooden base the application of special self-tapping fasteners with appropriately formed screw thread profile is recommended.

In the case of concrete base the application of special fasteners with universal sleeve anchor element or self-tapping fasteners with appropriately formed screw thread.

For screwing fasteners the application of electrical tools is recommended. Screwdrivers should be equipped with an appropriate head for driving long fasteners and a depth limiter fig.3 That guarantees appropriate assembly i.e. maintaining perpendicular location of a particular fastener in relation to a panel, minimised risk of damaging panels' surface and tightness of the fastening fig. 4. It is possible to use universal screwdrivers with standard short heads. However tools of this type should be equipped with a fastener depth limiter. Optimal parameters of electric tools for sandwich panel assembly are provided in the list below.

- power: 600 - 750 W
- rotation: 1500 - 2000 rpm
- torque: 600 - 700 Ncm

Fig. 3

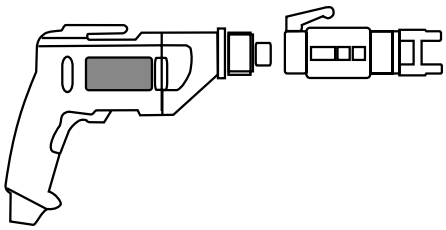
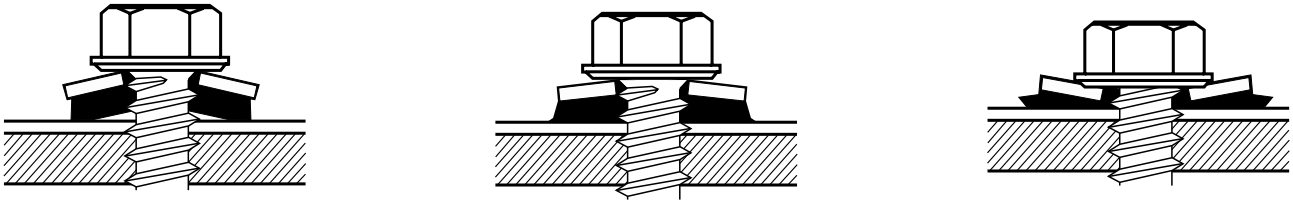
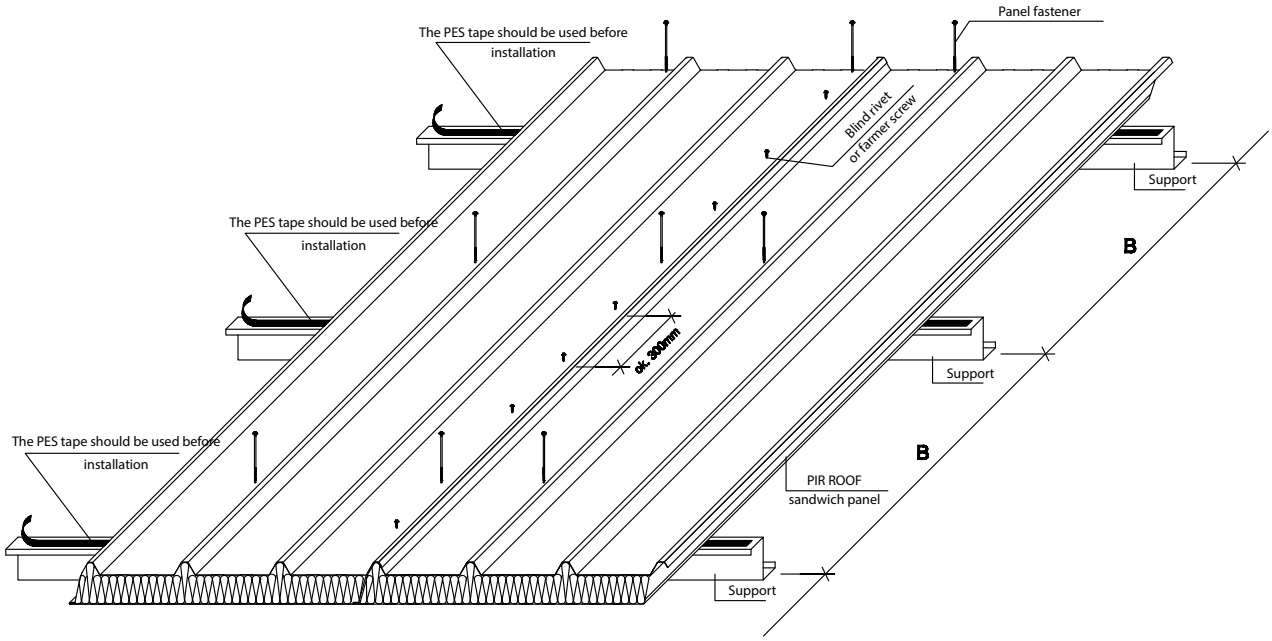


Fig. 4



too weak (not tightened) **appropriate** **too tight**

Fastening PIR ROOF panels to the roof construction.



Note !
During installation of panels longer than 8 m, the assembly crew should consist of at least 2 persons

After finishing cutting and drilling you should carefully remove all steel waste and filings that can cause discolouration of the surface of a cladding. The whole enclosure should be sealed with the use of appropriate tapes and sealing foams. All damages to varnish of screening metal sheet panels caused during assembly should be secured with touch up varnish.

23. GUIDELINES ON TRANSPORT

Recommended means of transport and its technical conditions:

Lorries with an open load-carrying body or an open trailer enabling loading long panels (up to 13.60 running metres) from both sides of a car constitute the basic means of transport for sandwich panels.

The following technical conditions are recommended for vehicles for transporting sandwich panels:

- load-carrying body with canvas cover ("CURTAIN" type)
- load-carrying body longer than transported panels (a package of panels should be placed on the platform in its entire length)
- transport belts holding the load should be placed on a package of panels on each support (belt tension can deform panels)

Unloading, transport:

During loading and unloading extreme caution should be kept, one should avoid point supports as it may lead to damaging the facing of the lowest panel. In order to avoid this problem you should distribute the load over a greater surface area. You should also pay attention not to drag one panel along another one not to scratch it.

Panel storage:

Sandwich panels should be stored on ground beams, not less than 250 mm over the ground. It is allowed to store two packages placed one on another at the most. It is recommended to store panels in airy rooms, at normal temperature, far away from acids, lye, salts and other corrosive substances. Storing uncovered panels is unacceptable. In case of storing panels under canvas cover for a short period of time (a fortnight at the most) you should ensure free air flow. If storing period is longer than 2 weeks panels should be placed in a well ventilated room and left uncovered as well as provided with free access to air in order for all layers to be ventilated. Not observing these recommendations can lead to coating decolouration, so called "white rust", permanent core damage as well as loss of guarantee.

Slight repairs and maintenance:

All damage to coating caused during transport or assembly should be covered with touch up paint. Maintenance of sandwich panels consists in performing regular inspections and securing potential damages. During inspection close attention should be paid to uncovered edges and joints.

Remarks concerning usage:

Wall sandwich panels with dark claddings are characterized by high heat absorption capacity which can cause local deformation of claddings' surface in the period of high air temperatures. As a result allow for panels' thermal motion and apply panels of limited length. This effect does not have any influence on the properties of sandwich panels but the producer warns that clients purchase wall panels in those colours on their own responsibility and do not have rights to claim against the producer due to this fact. The appearance of local deformations in roof panels practically does not appear.

Pursuant to the EN 14509 standard, it is assumed that sheets in dark colours heat to the temperature of 80°C. Therefore, Balex Metal shall not be responsible for any damages caused by high temperature, which may result in changing the position of the cladding in some places. Dark colours are defined in Item E.33 of the EN14509 standard.

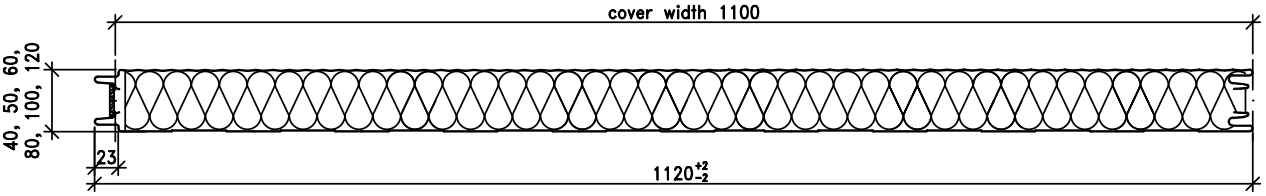
24. CERTIFICATION DOCUMENTS

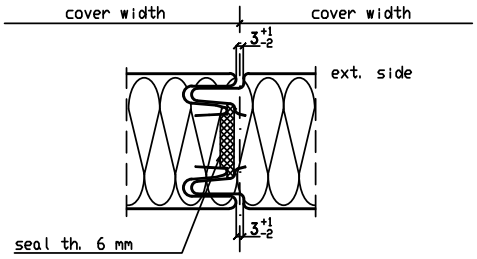
The current certification documents are available at **www.balex.eu**

II. DETAILED HOUSING SOLUTIONS MADE OF SANDWICH PANELS WITH A POLYURETHANE CORE: PIR STANDARD & PIR PLUS WALL SANDWICH PANELS, PIR ROOF ROOF SANDWICH PANELS

EN-2026-08-24

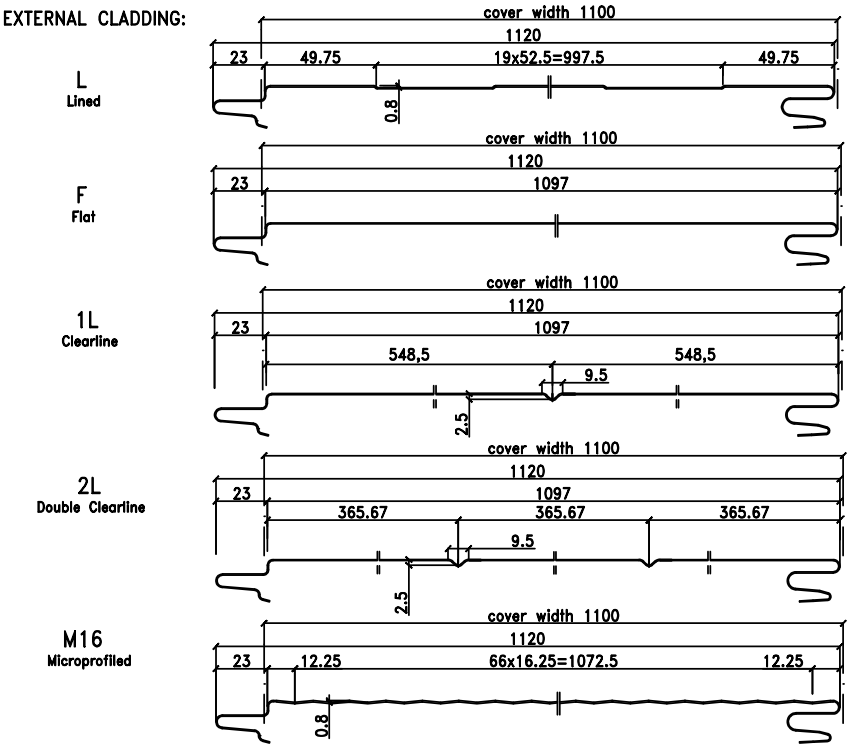
1. PIR STANDARD wall panel
1.1.
ST01/1 PIR STANDARD wall panel - joint, profilation types - MODULE 1100



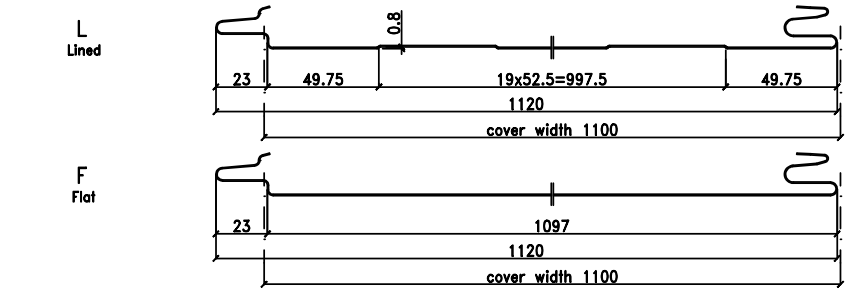


PROFILATION TYPES COVER WIDTH 1100mm

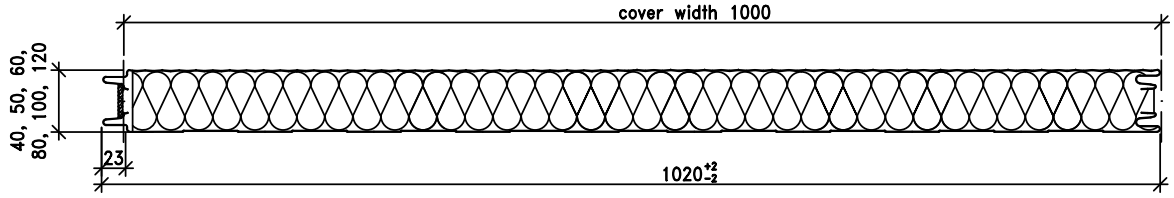
EXTERNAL CLADDING:

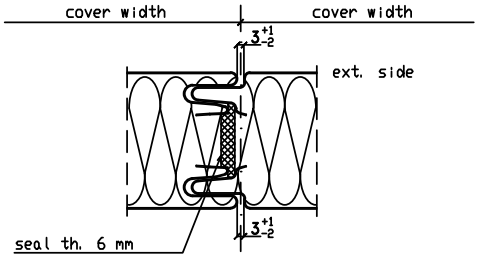


INTERNAL CLADDING:



1.2.
ST01/2 PIR STANDARD wall panel - joint, profilation types - MODULE 1000

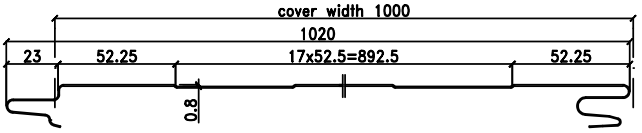




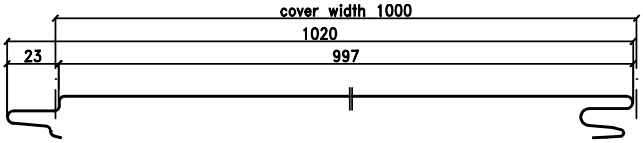
PROFILATION TYPES COVER WIDTH 1000mm

EXTERNAL CLADDING:

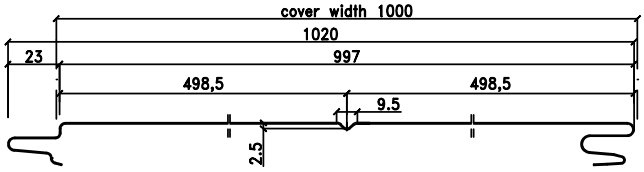
L
Lined



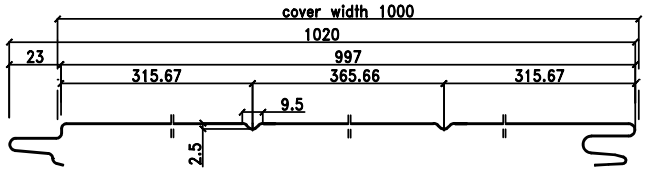
F
Flat



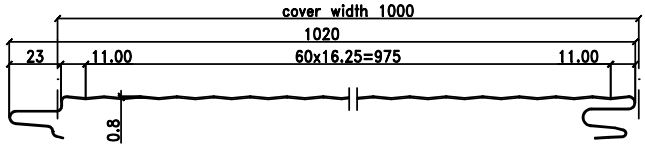
1L
Clearline



2L
Double Clearline

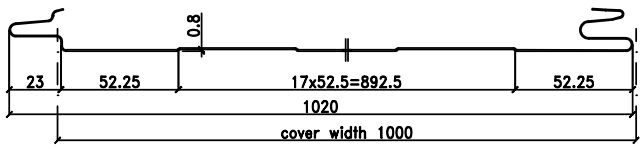


M16
Microprofiled

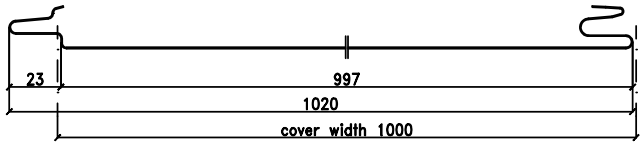


INTERNAL CLADDING:

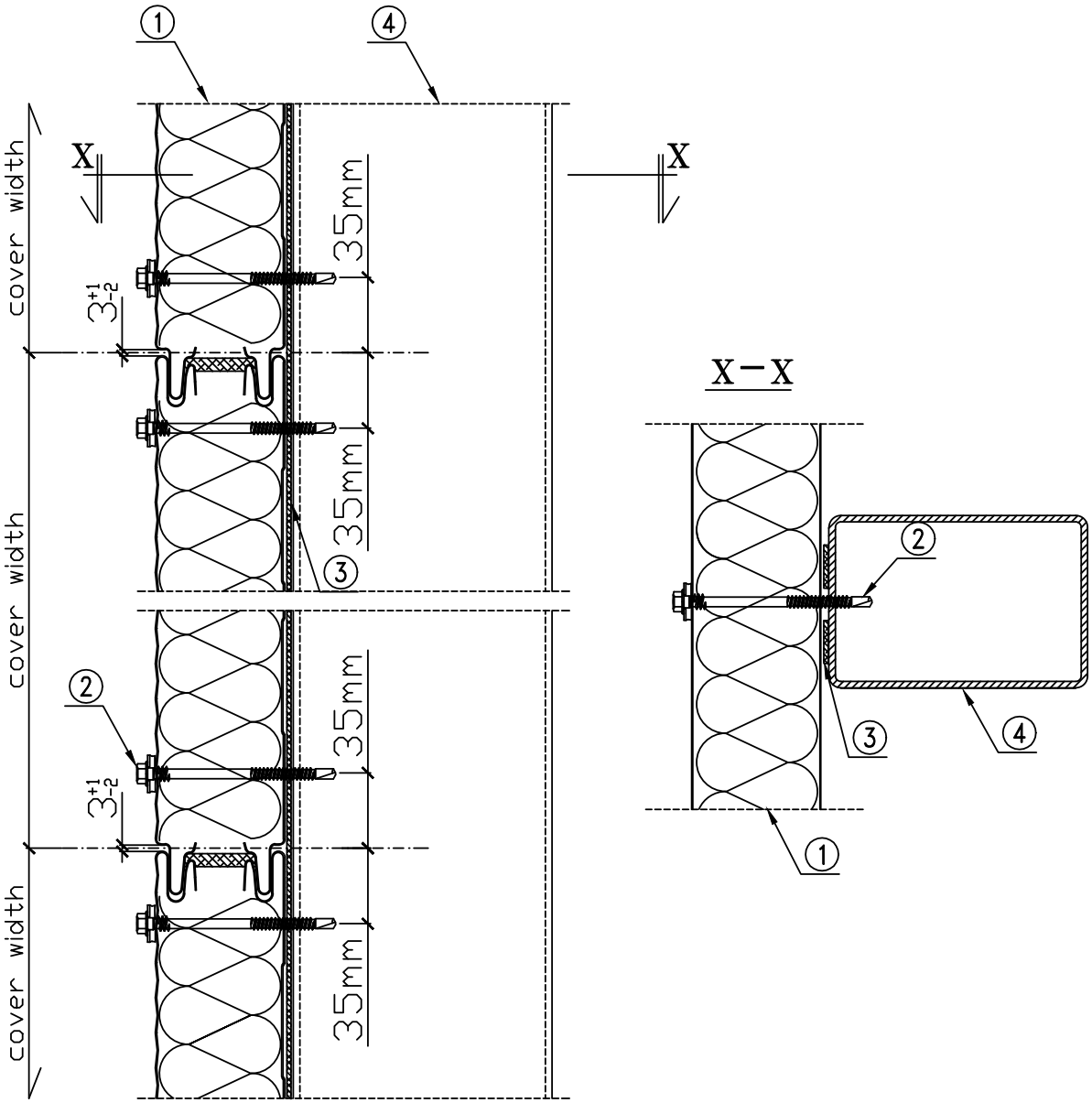
L
Lined



F
Flat

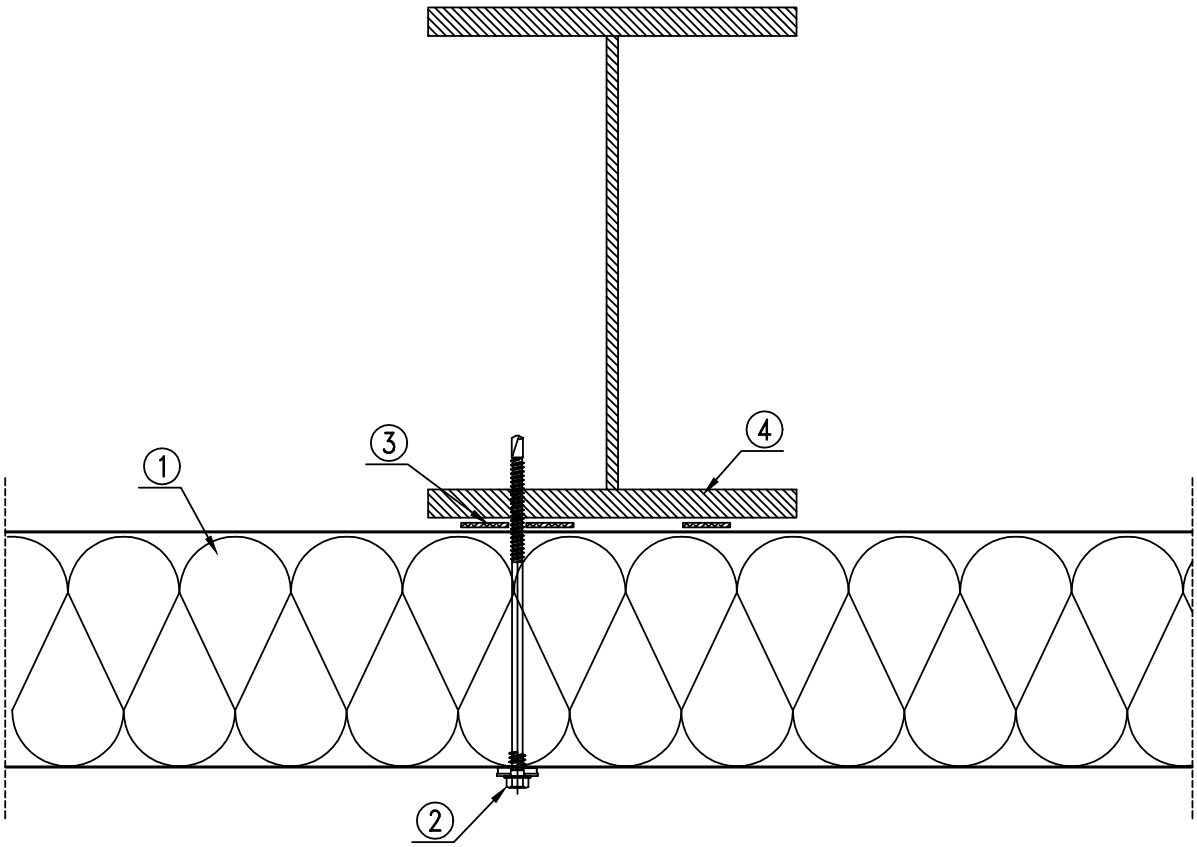


1.3.
ST02 Fastening panel - vertical panel set



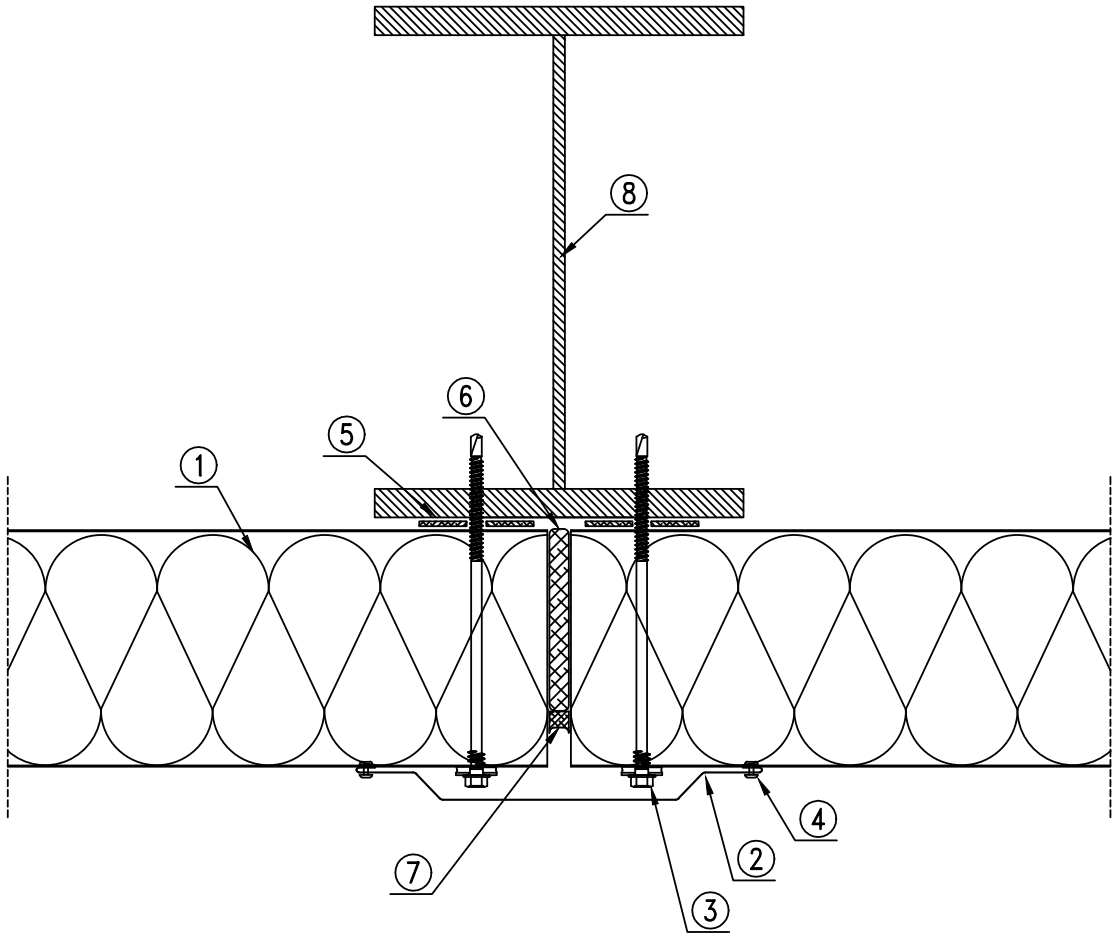
1. Wall sandwich panel PIR Standard (PU-PIR-W-ST)
2. Sandwich panel fastener
3. Self-adhesive sealing tape PES
4. Load bearing profile acc. to construction design

1.4.
ST03 Fastening panel - horizontal panel set



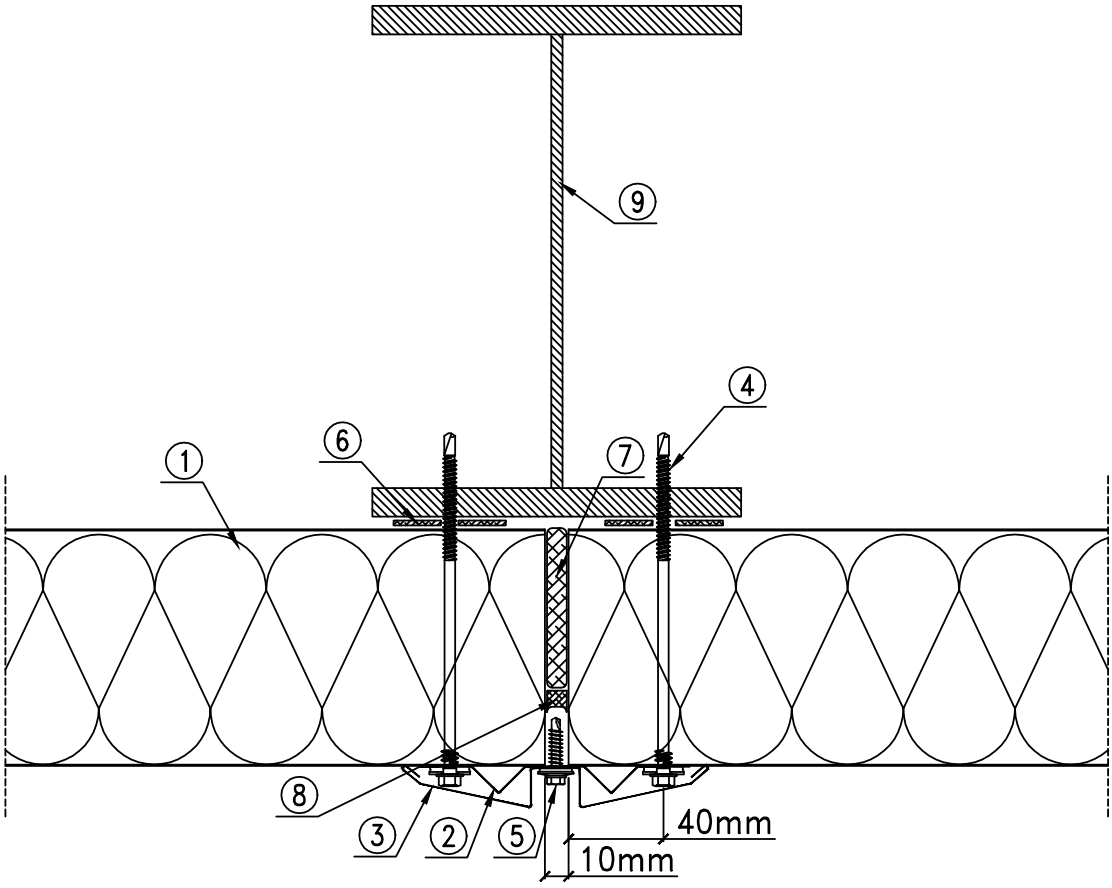
1. Wall sandwich panel PIR Standard (PU-PIR-W-ST)
2. Sandwich panel fastener
3. Self-adhesive sealing tape PES
4. Load bearing column acc. to construction design

1.5.
ST04/1 Fastening panels to end support - horizontal panel set - Option I



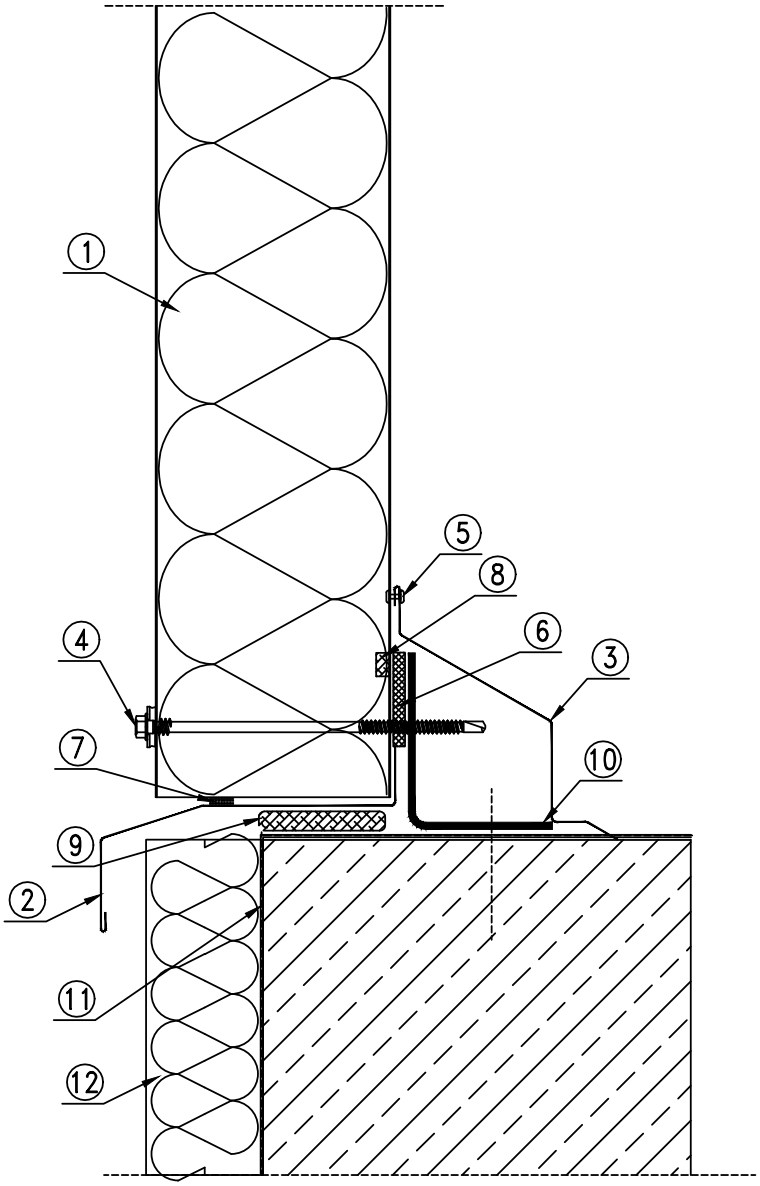
1. Wall sandwich panel PIR Standard (PU-PIR-W-ST)
2. OBR 105 contact area masking frame - straight flashing
3. Sandwich panel fastener
4. Self-drilling fastener or rivet every 300mm
5. PES self-adhesive sealing tape
6. Polyurethane gasket or mounting foam
7. PURS impregnated potyurethane gasket
8.Load bearing column acc. to construction design

1.6.
ST04/2 Fastening panels to end support - horizontal panel set - Option II



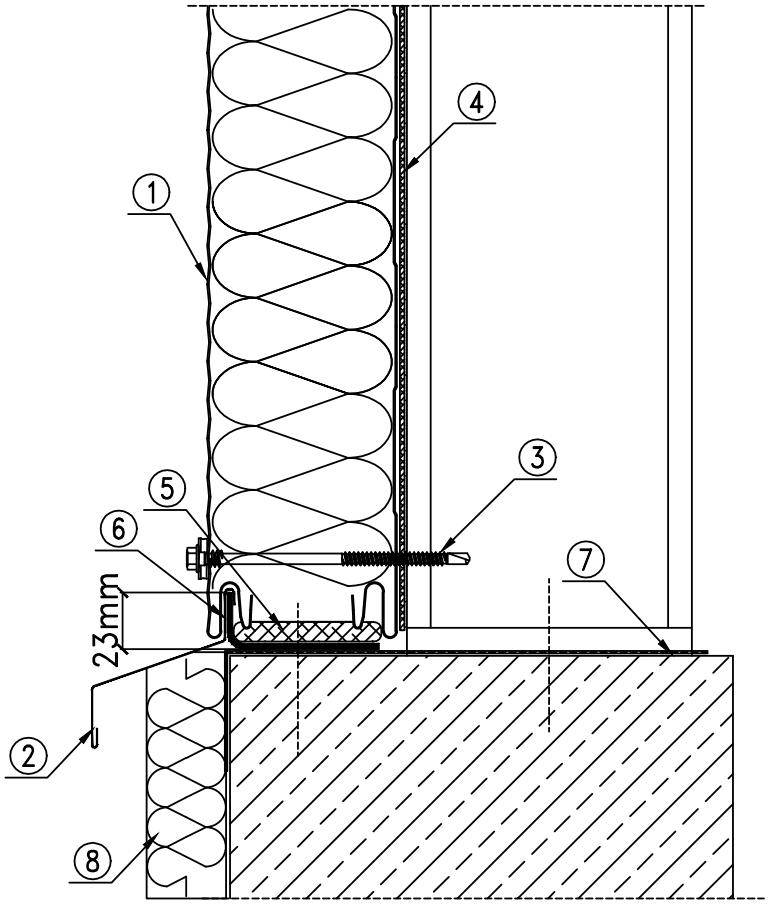
1. Wall sandwich panel PIR Standard (PU-PIR-W-ST)
2. OBR110 contact area masking ground beam flashing
3. OBR111 contact area masking frame - decorative element flashing
4. Sandwich panel fastener
5. Self-drilling fastener or rivet every 300mm
6. PES self-adhesive sealing tape
7. Polyurethane gasket or mounting foam
8. PURS self-adhesive sealing tape
9. Load bearing column acc. to construction design

1.7.
ST05 Ground beam wall panels start - vertical panel set



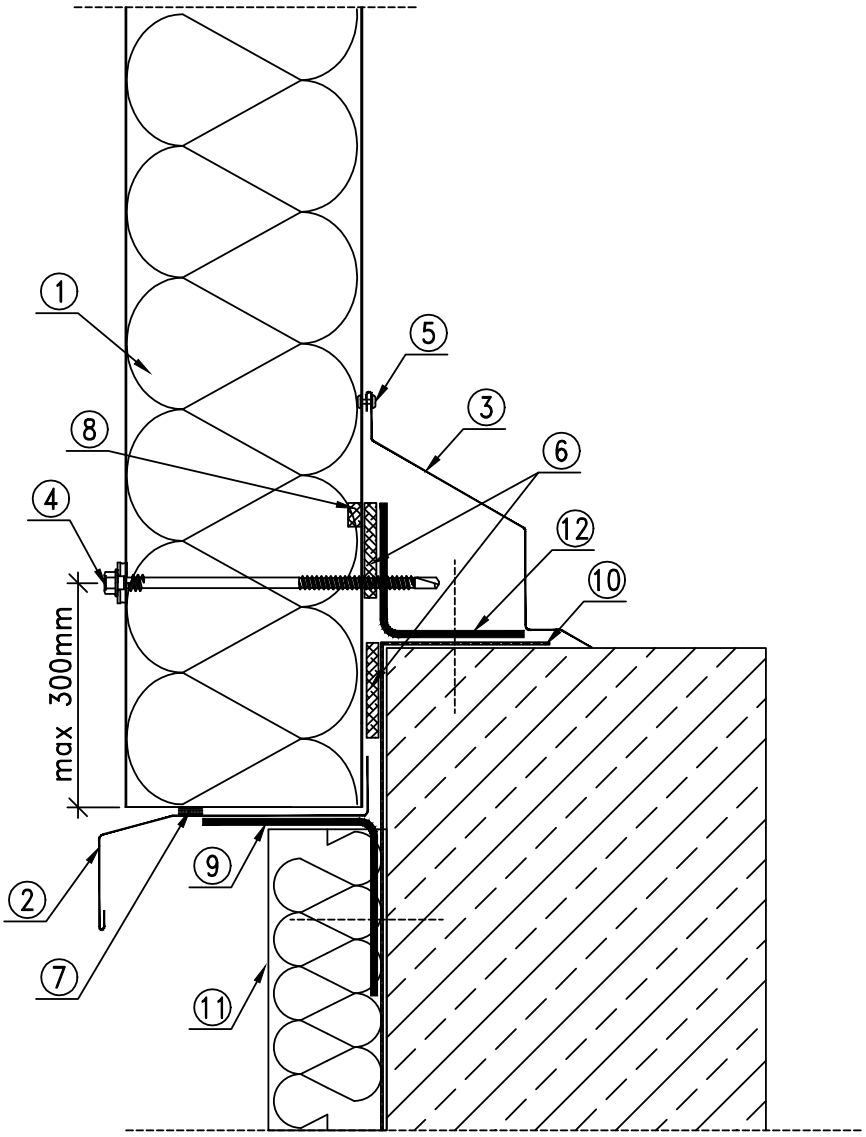
1. Wall sandwich panel PIR Standard (PU-PIR-W-ST)
2. OBR100 standard drip cap flashing
3. OBR101 Internal masking frame flashing
4. Sandwich panel fastener
5. Self-drilling fastener or rivet every 300mm
6. PUS self-adhesive sealing tape
7. Butyl sealing tape
8. Butyl mass
9. PURS impregnated potyurethane gasket
10. Load bearing profile acc. to construction design
11. Hydro barrier acc. to architecture design
12. Thermal insulation + plastering acc. to architecture design

1.8.
ST06 Ground beam wall panels start - horizontal panel set



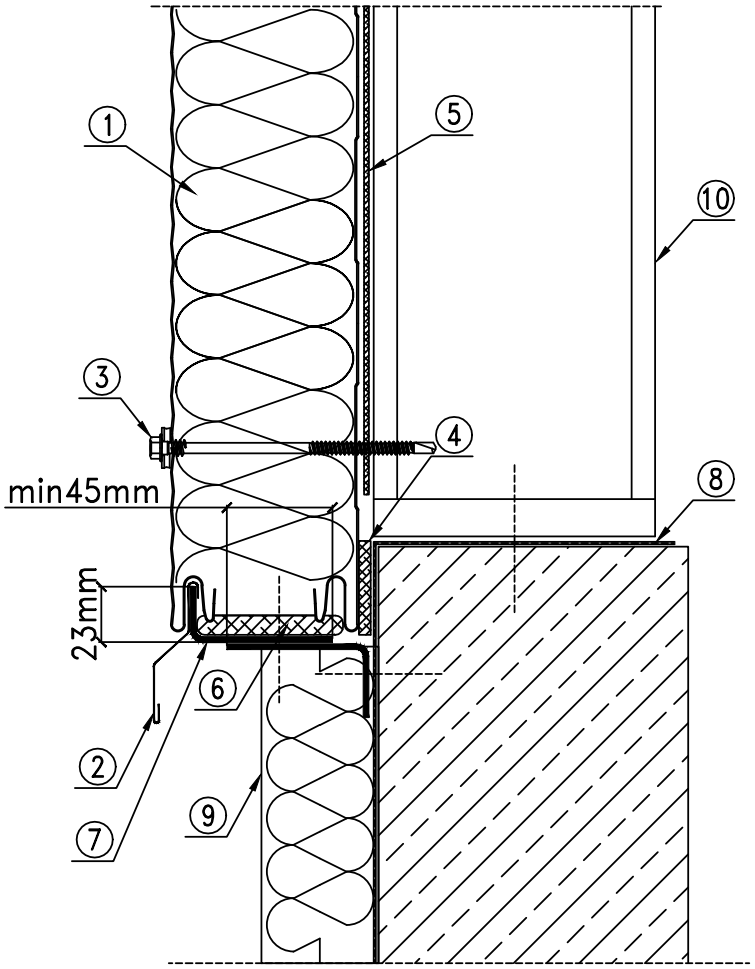
1. Wall sandwich panel PIR Standard (PU-PIR-W-ST)
2. Individual flashing
3. Sandwich panel fastener
4. PES self-adhesive tape
5. PURS impregnated potyurethane gasket
6. Load bearing profile acc. to construction design
7. Hydro barrier acc. to architecture design
8. Thermal insulation + plastering acc. to architecture design

1.9.
ST07 Panels start below top of ground beam - vertical panel set



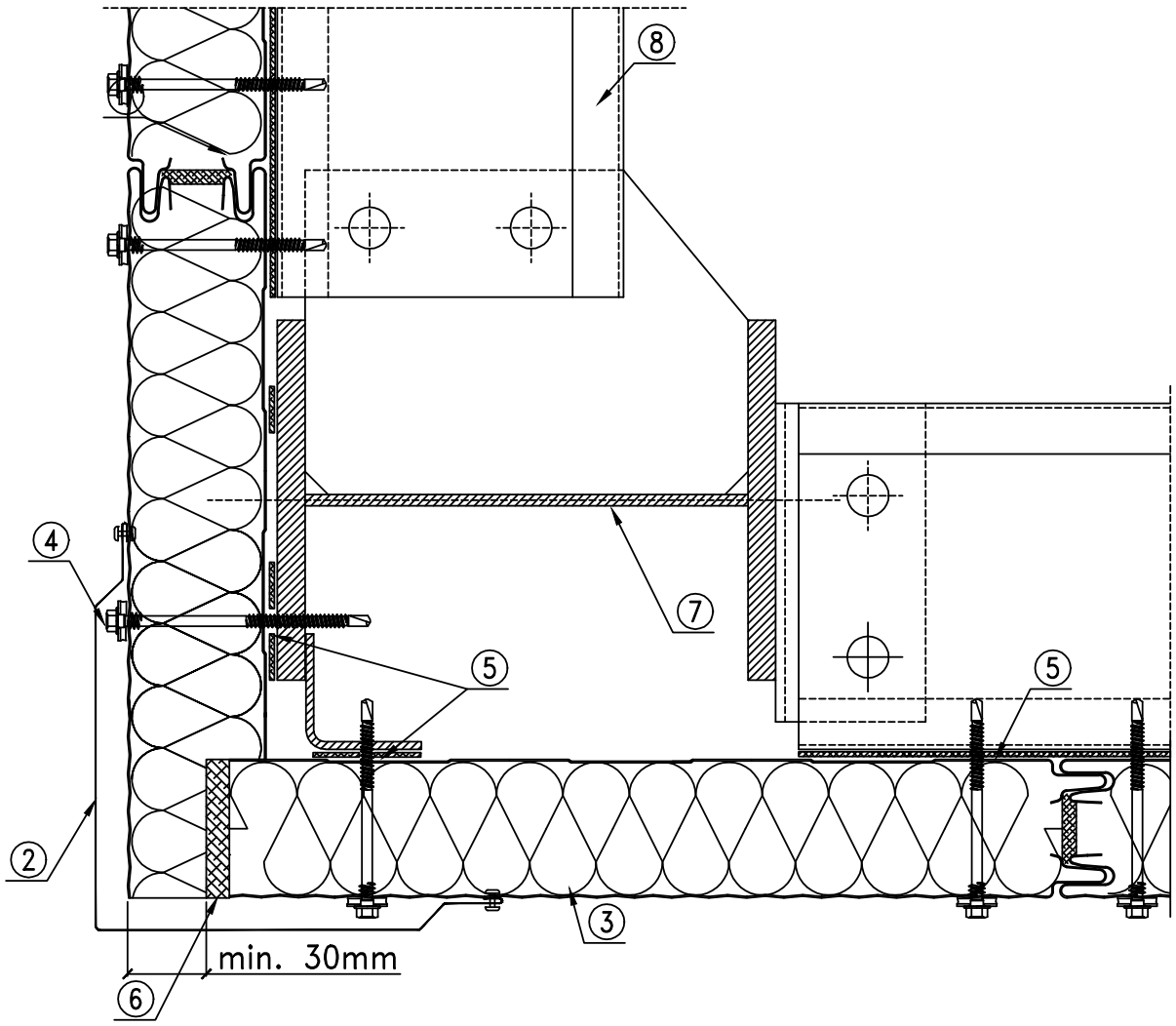
1. Wall sandwich panel PIR Standard (PU-PIR-W-ST)
2. OBR100 standard drip cap flashing
3. OBR101 Internal masking frame flashing
4. Sandwich panel fastener
5. Self-drilling fastener or rivet every 300mm
6. PUS self-adhesive sealing tape
7. Butyl sealing tape
8. Butyl mass
9. Steel profile acc. to construction design
10. Hydro barrier acc. to architecture design
11. Thermal insulation + plastering acc. to architecture design
12. Load bearing profile acc. to construction design

1.10.
ST08 Panels start below top of ground beam - horizontal panel set



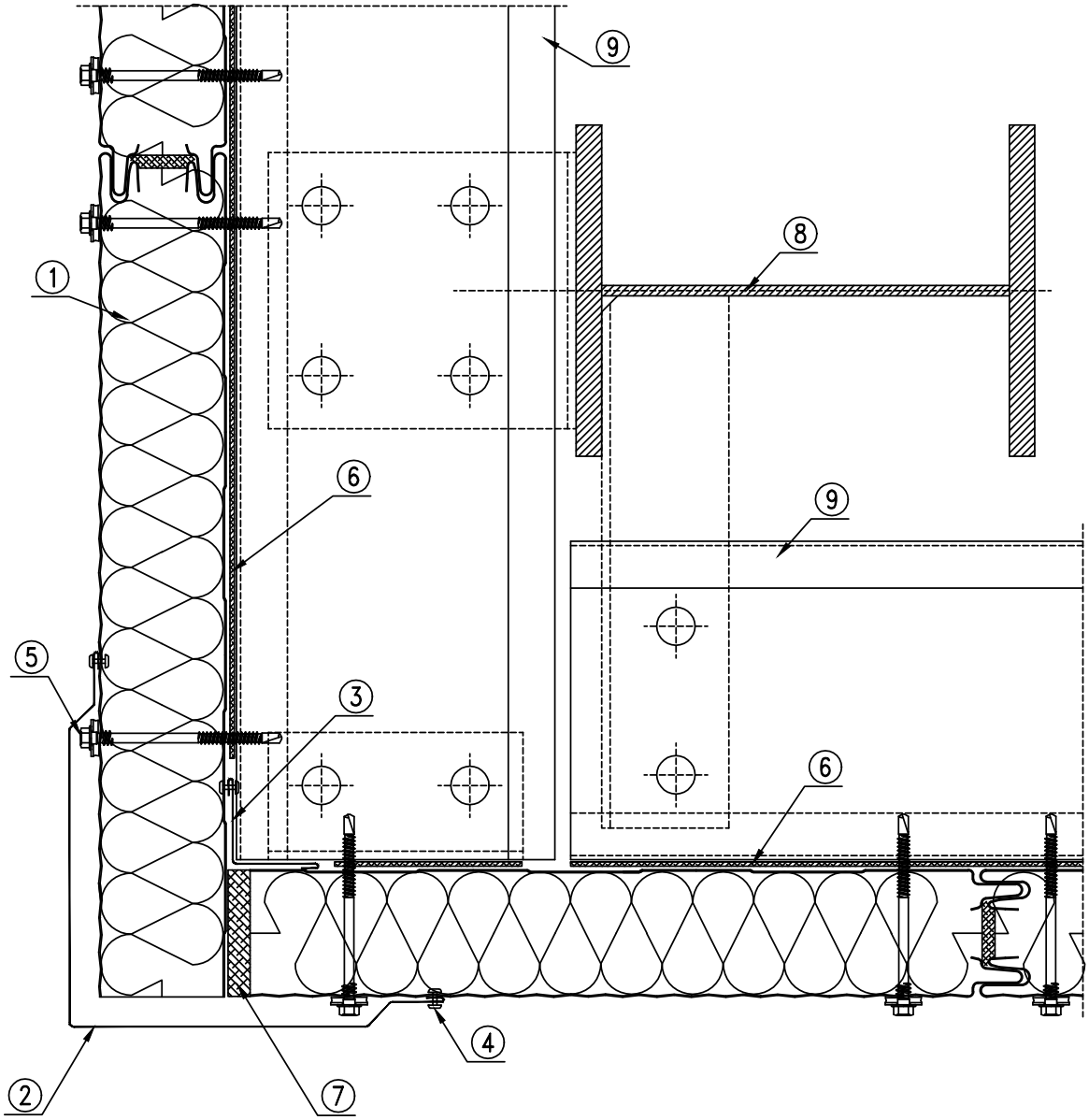
1. Wall sandwich panel PIR Standard (PU-PIR-W-ST)
2. OBR102 Tongue-groove drip cap flashing
3. Sandwich panel fastener
4. PUS self-adhesive sealing tape
5. PES self-adhesive tape
6. PURS impregnated potyurethane gasket
7. Angle profiles screwed together acc. to construction design
8. Hydro barrier acc. to architecture design
9. Thermal insulation + plastering acc. to architecture design
10. Load bearing column acc. to construction design

1.11.
ST09/1 Corner joint - vertical panel set - Option I



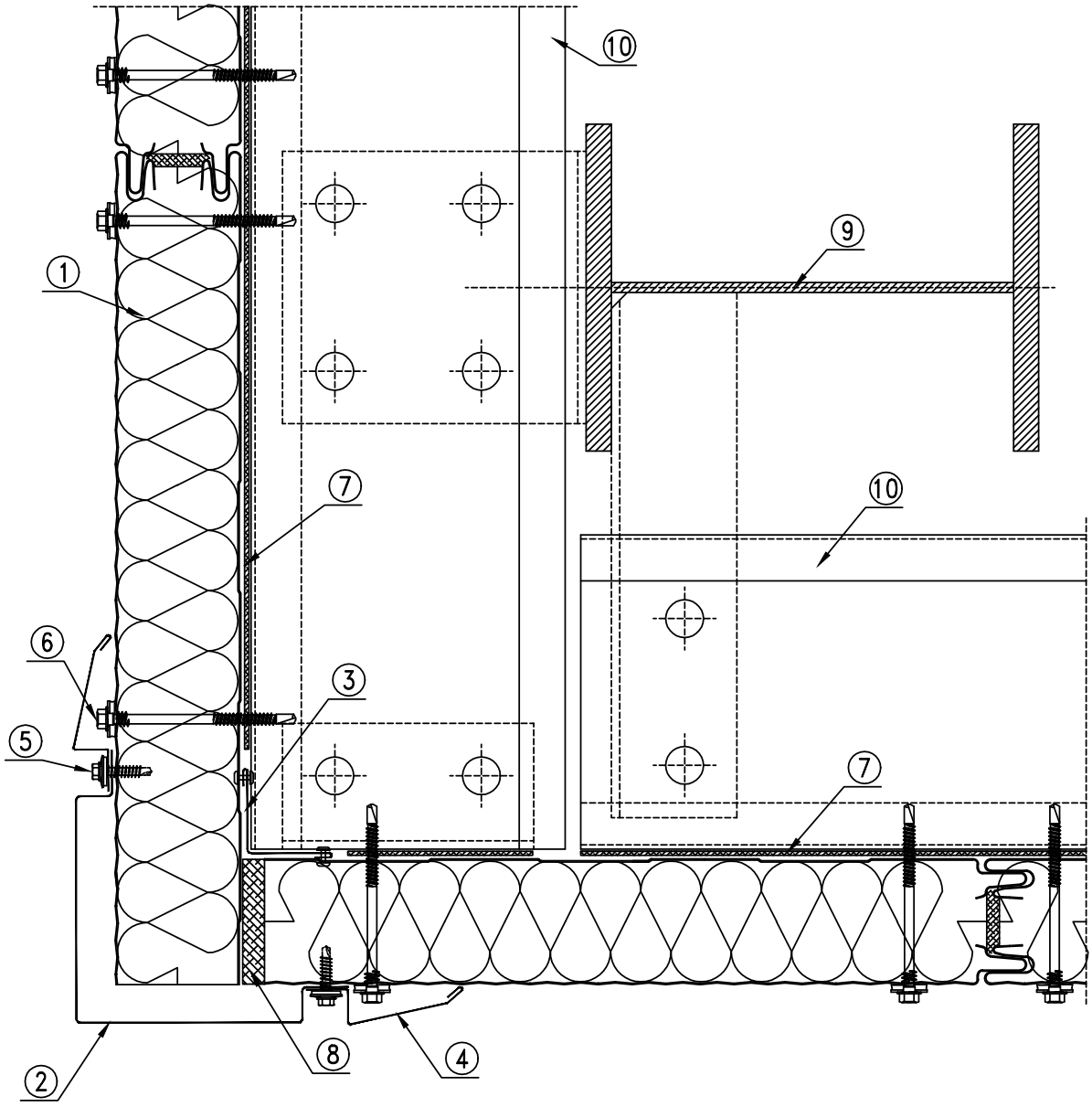
1. Wall sandwich panel PIR Standard (PU-PIR-W-ST)
2. OBR103 External corner with a hidden joint flashing
3. Self-drilling fastener or rivet every 300mm
4. Sandwich panel fastener
5. PES self-adhesive tape
6. Polyurethane gasket or mounting foam
7. Load bearing column + angle profile
acc. to construction design
8. Load bearing beam acc. to construction design

1.12.
ST09/2 Corner joint - vertical panel set - Option II



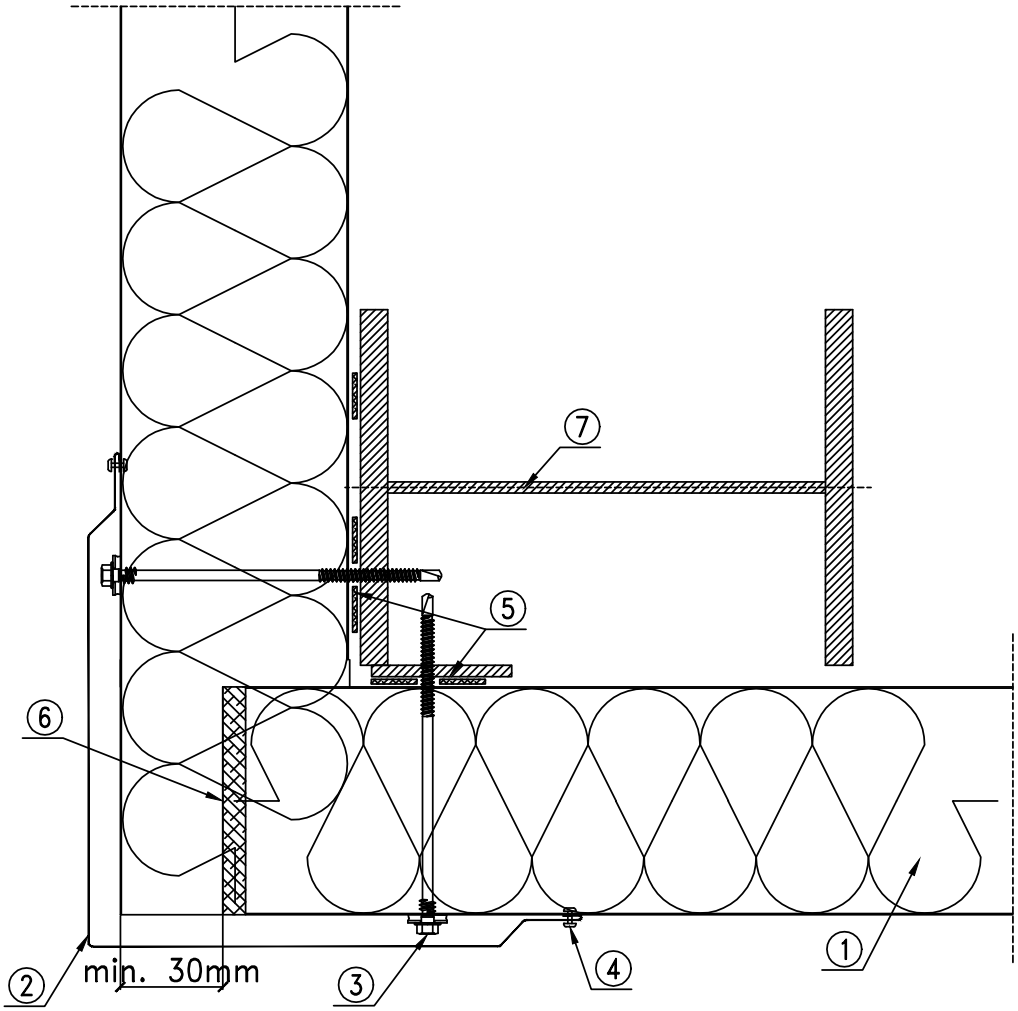
1. Wall sandwich panel PIR Standard (PU-PIR-W-ST)
2. OBR103 External corner with a hidden joint flashin
3. OBR104 Internal strip flashing
4. Self-drilling fastener or rivet every 300mm
5. Sandwich panel fastener
6. PES self-adhesive tape
7. Polyurethane gasket or mounting foam
8. Load bearing column + angle profile
acc. to construction design
9.Load bearing beam acc. to construction design

1.13.
ST09/3 Corner joint - vertical panel set - Option III



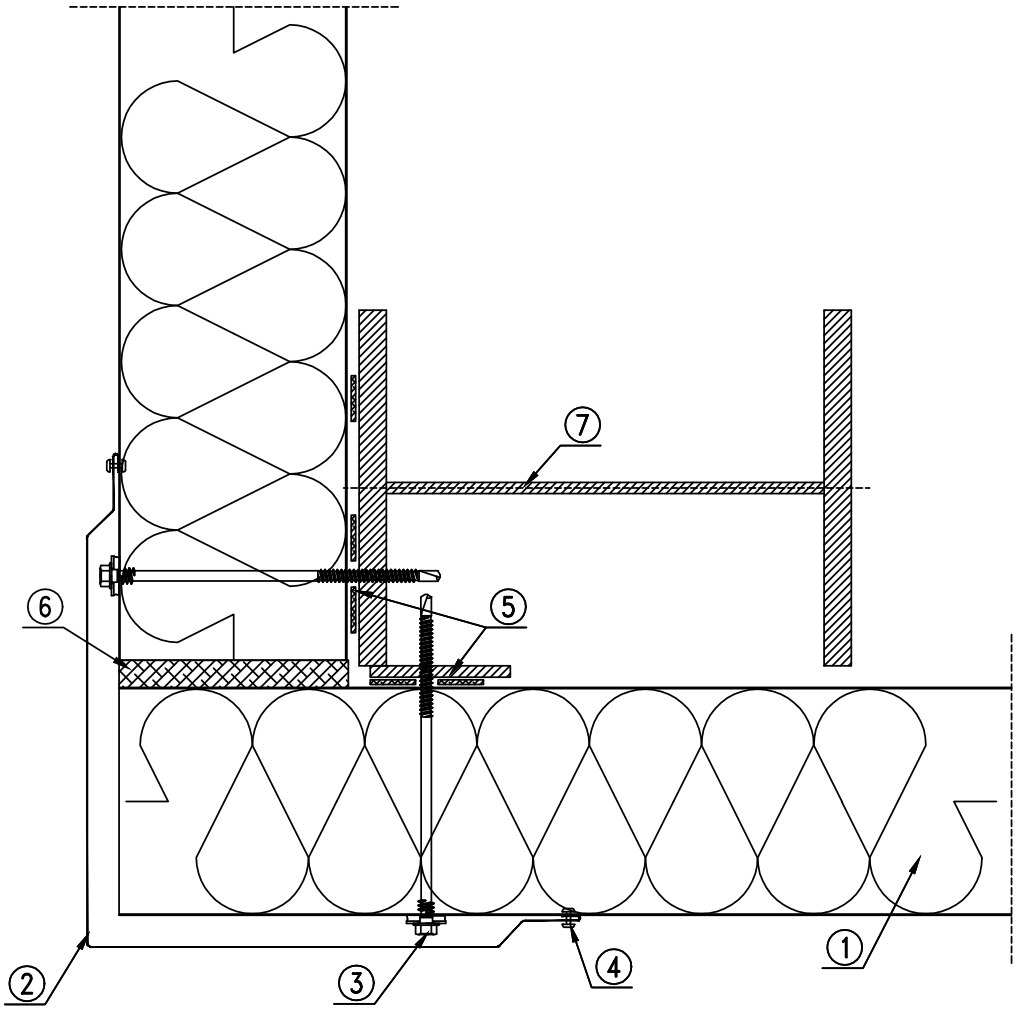
1. Wall sandwich panel PIR Standard (PU-PIR-W-ST)
2. OBR113 External corner - base flashing
3. OBR104 Internal strip flashing
4. OBR111 External corner - decorative element flashing
5. Self-drilling fastener or rivet every 300mm
6. Sandwich panel fastener
7. PES self-adhesive tape
8. Polyurethane gasket or mounting foam
9. Load bearing column + angle profile
acc. to construction design
10. Load bearing beam acc. to construction design

1.14.
ST010/1 Corner joint - horizontal panel set - Option I



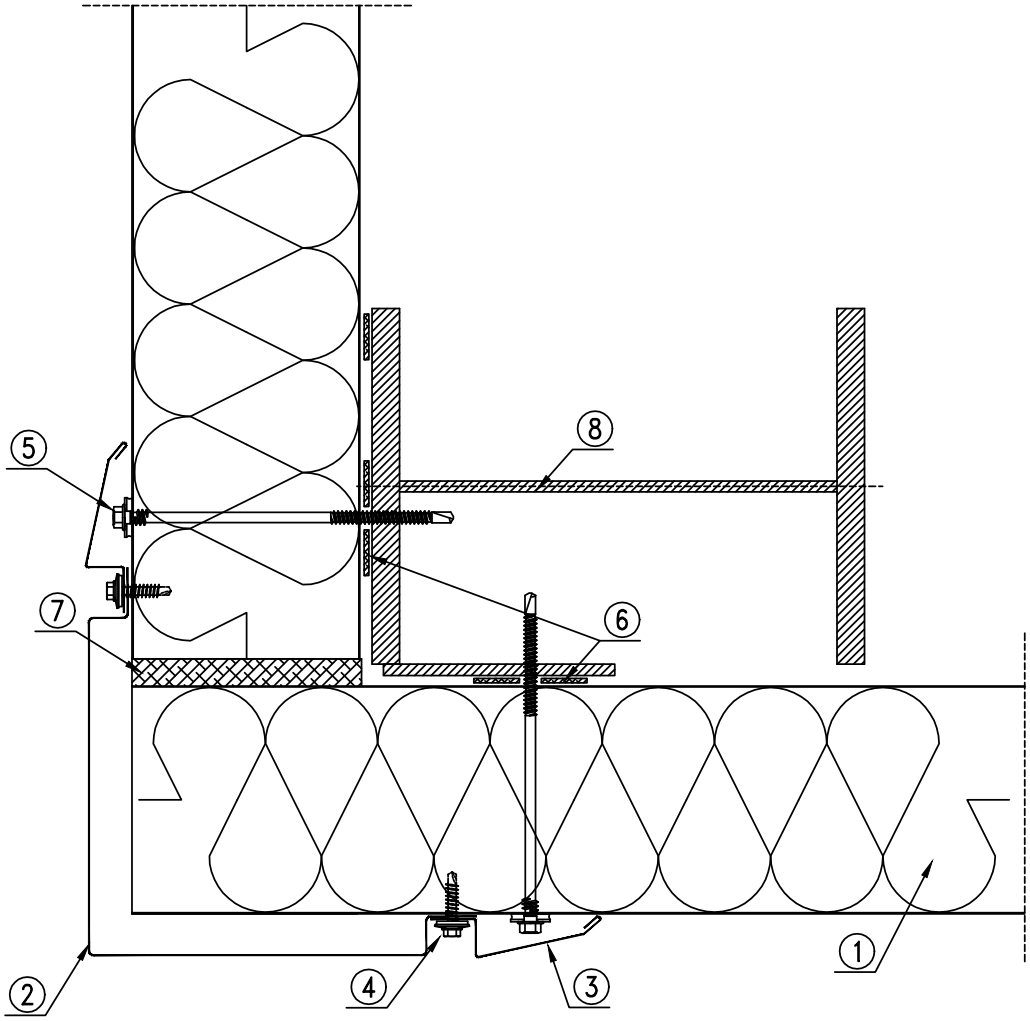
1. Wall sandwich panel PIR Standard (PU-PIR-W-ST)
2. OBR103 External corner with a hidden joint flashing
3. Sandwich panel fastener
4. Self-drilling fastener or rivet every 300mm
5. PES self-adhesive tape
6. Polyurethane gasket or mounting foam
7. Load bearing column acc. to construction design

1.15.
ST010/2 Corner joint - horizontal panel set - Option II



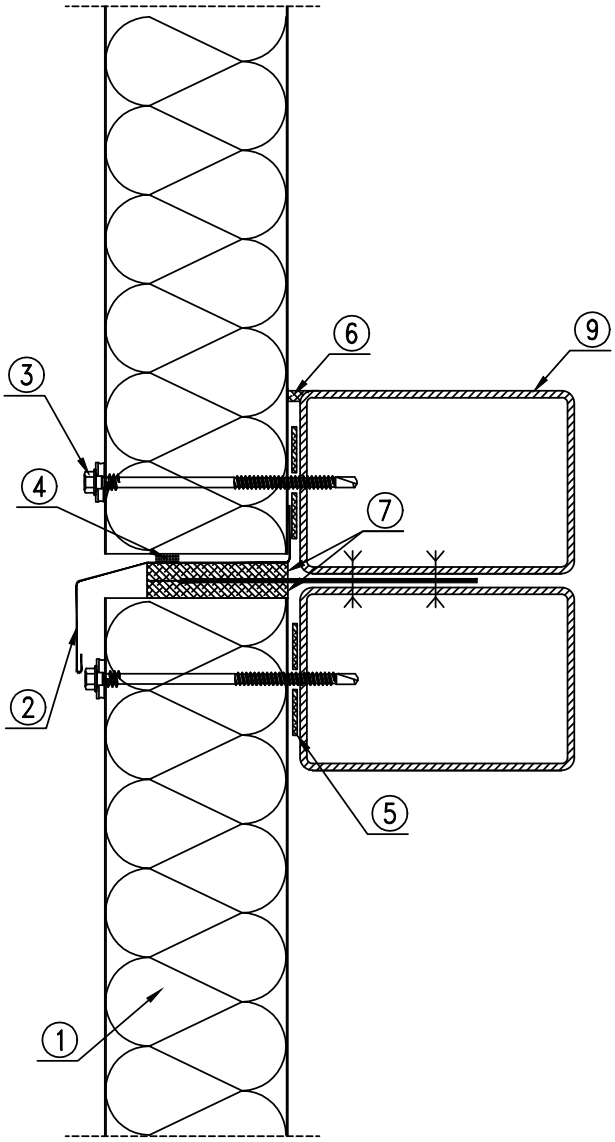
1. Wall sandwich panel PIR Standard (PU-PIR-W-ST)
2. OBR103 External corner with a hidden joint flashing
3. Sandwich panel fastener
4. Self-drilling fastener or rivet every 300mm
5. PES self-adhesive tape
6. Polyurethane gasket or mounting foam
7. Load bearing column acc. to construction design

1.16.
ST010/3 Corner joint - horizontal panel set - Option III



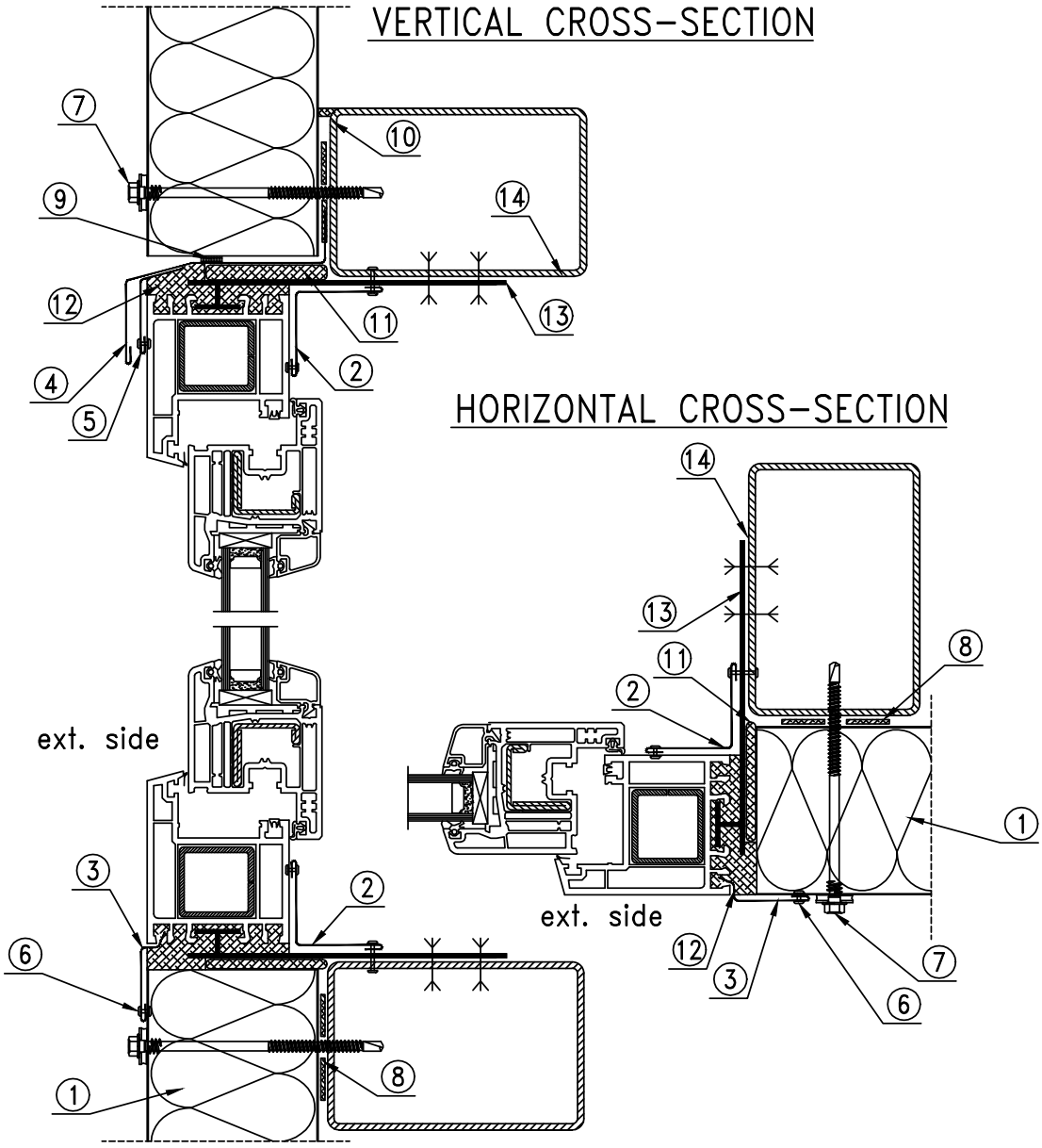
1. Wall sandwich panel PIR Standard (PU-PIR-W-ST)
2. OBR113 External corner - base flashing
3. OBR111 External corner - decorative element flashing
4. Self-drilling fastener or rivet every 300mm
5. Sandwich panel fastener
6. PES self-adhesive tape
7. Polyurethane gasket or mounting foam
8. Load bearing column acc. to construction design

1.17.
ST11 Lengthwise panel joint - vertical panel set



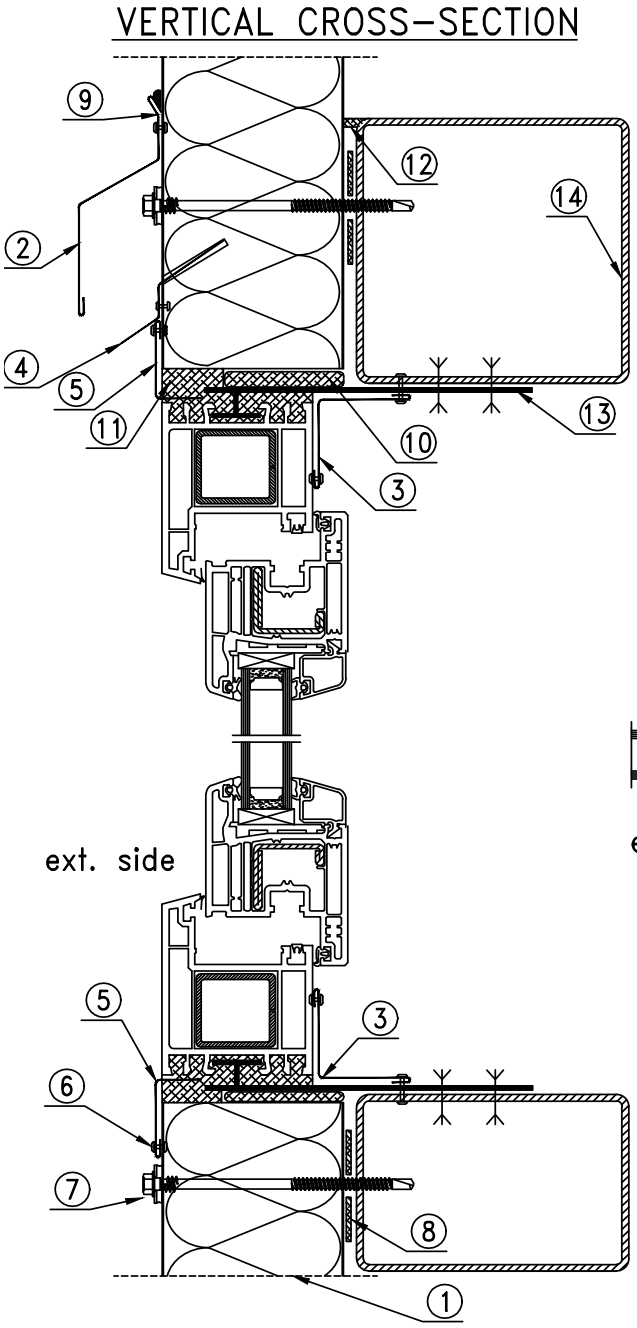
1. Wall sandwich panel PIR Standard (PU-PIR-W-ST)
2. OBR100 Standard drip cap
3. Sandwich panel fastener
4. Butyl mass or tape
5. PES self-adhesive tape
6. Butyl mass
7. Polyurethane gasket - thickness adjusted to gap width
8. Load bearing beam + flat metal bar acc. to construction design

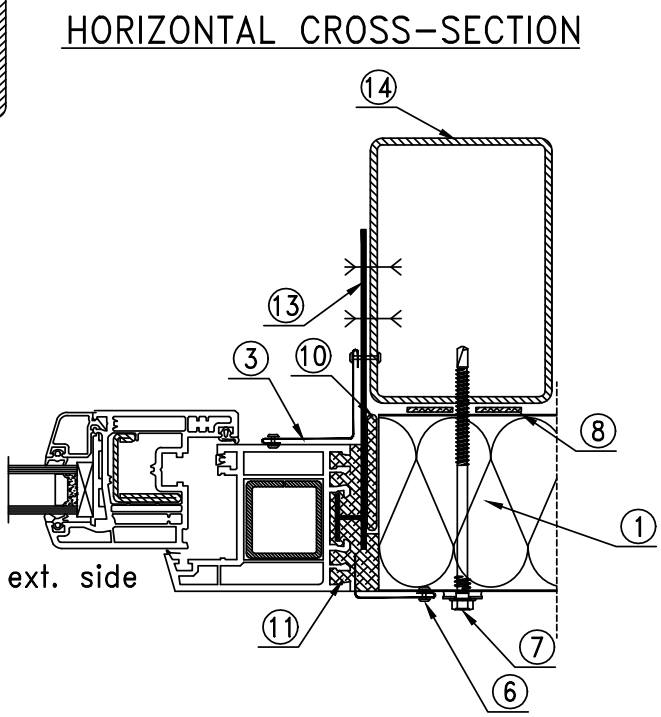
1.18.
ST12 Panels - window joint - horizontal panel set

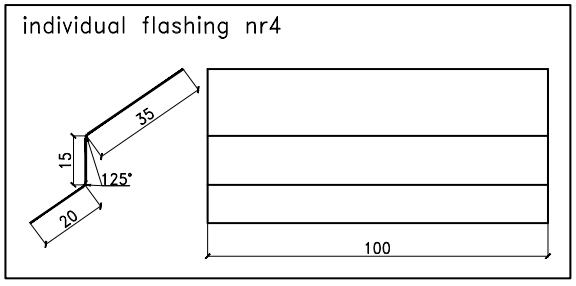


1. Wall sandwich panel PIR Standard (PU-PIR-W-ST)
2. OBR104 Internal strip flashing
3. OBR06 External window strip flashing
4. OBR100 Standard drip cap flashing
5. Individual flashing
6. Self-drilling fastener or rivet every 300mm
7. Sandwich panel fastener
8. PES self-adhesive sealing tape
9. Butyl mass or tape
10. Butyl mass
11. PURS self-adhesive sealing tape
12. Mounting foam
13. Window bar
14. Load bearing beam acc. to construction design

1.19.
ST13 Panels - window joint - vertical panel set

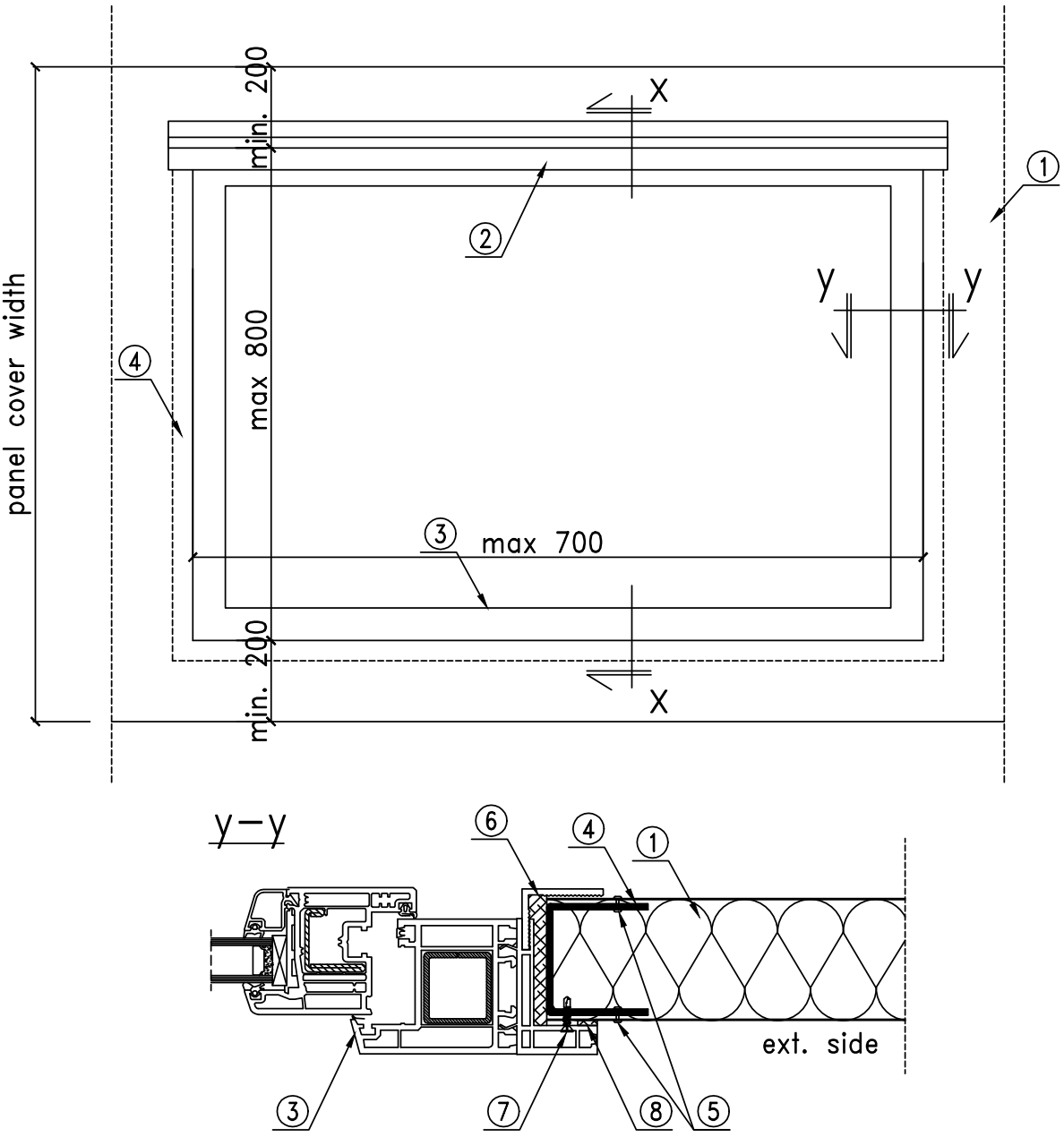






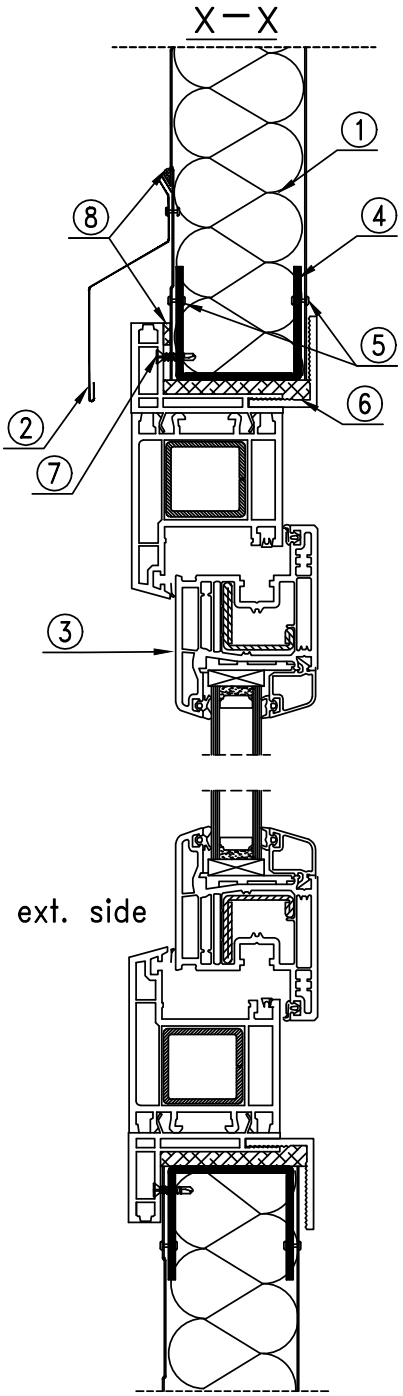
1. Wall sandwich panel PIR Standard (PU-PIR-W-ST)
2. OBR107 Over window drip cap flashing
3. OBR104 Internal strip flashing
4. Individual flashing - place in sandwich wall panel cut in area of locks
5. Individual flashing
6. Self-drilling fastener or rivet every 300mm
7. Sandwich panel fastener
8. PES self-adhesive sealing tape
9. Butyl mass or tape
10. PURS self-adhesive sealing tape
11. Mounting foam
12. Butyl mass
13. Window bar
14. Load bearing beam acc. to construction design

1.20.
ST14/1 Panels - window joint - without additional construction - Y-Y



1. Wall sandwich panel PIR Standard (PU-PIR-W-ST)
2. OBR107 Over window drip cap flashing
3. Window
4. Window C-profile frame acc. to construction design pressed between panel claddings
5. Self-drilling fastener or rivet every 300mm
6. Polyurethane gasket or mounting foam
7. Self-drilling fastener
8. Butyl mass

1.21.
ST14/2 Panels - window joint - without additional construction - X-X

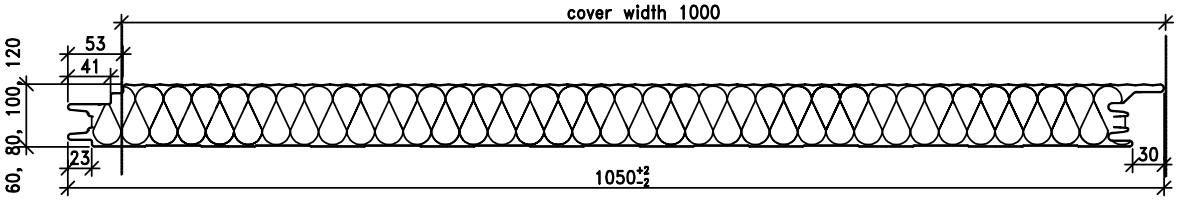


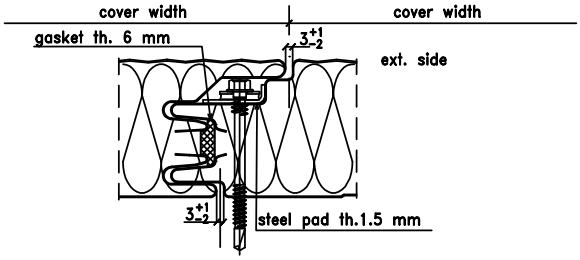
1. Wall sandwich panel PIR Standard (PU-PIR-W-ST)
2. OBR107 Over window drip cap flashing
3. Window
4. Window C-profile frame acc. to construction design pressed between panel claddings
5. Self-drilling fastener or rivet every 300mm
6. Polyurethane gasket or mounting foam
7. Self-drilling fastener
8. Butyl mass

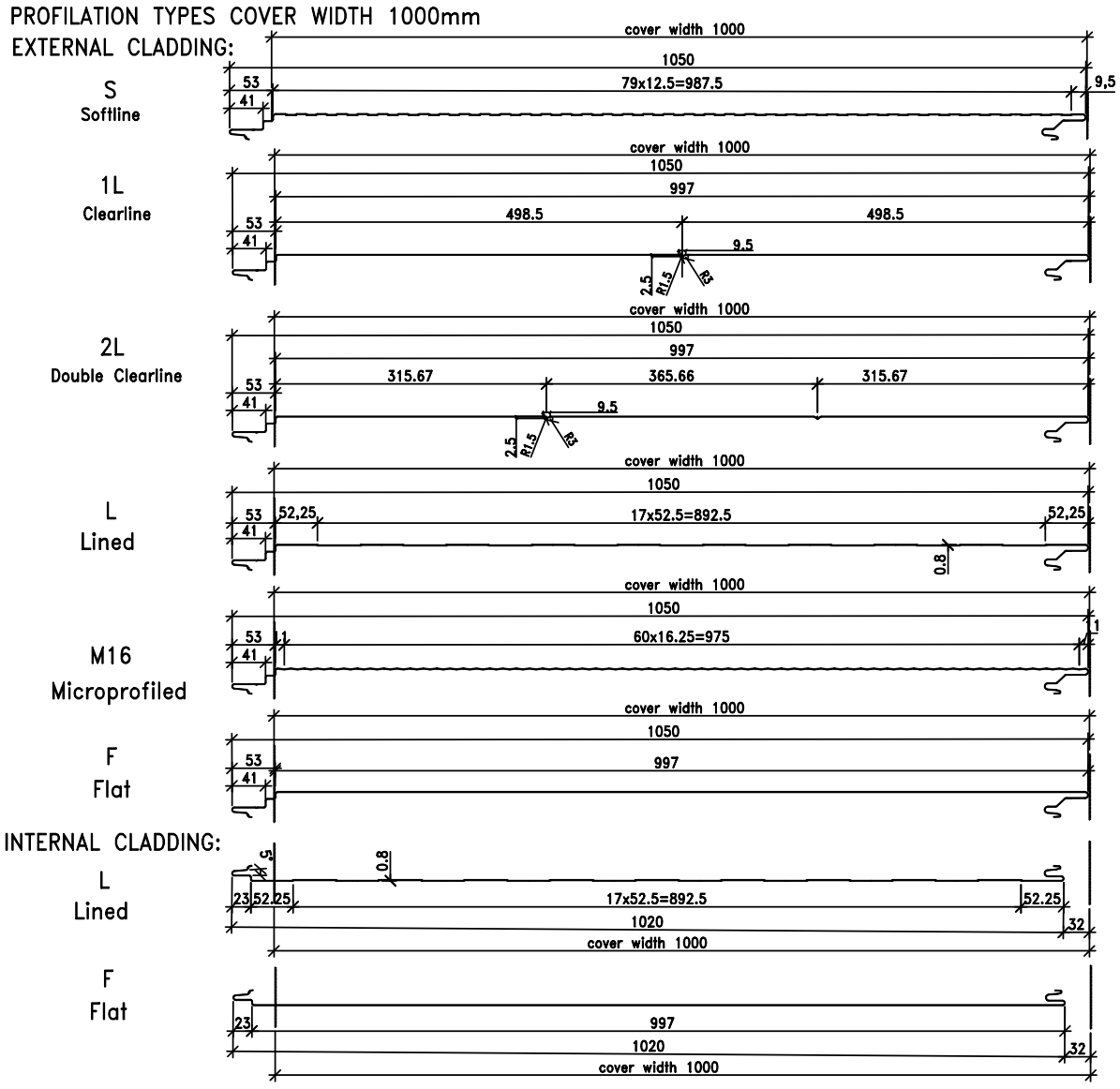
2. PIR PLUS WALL PANELS

2.1.

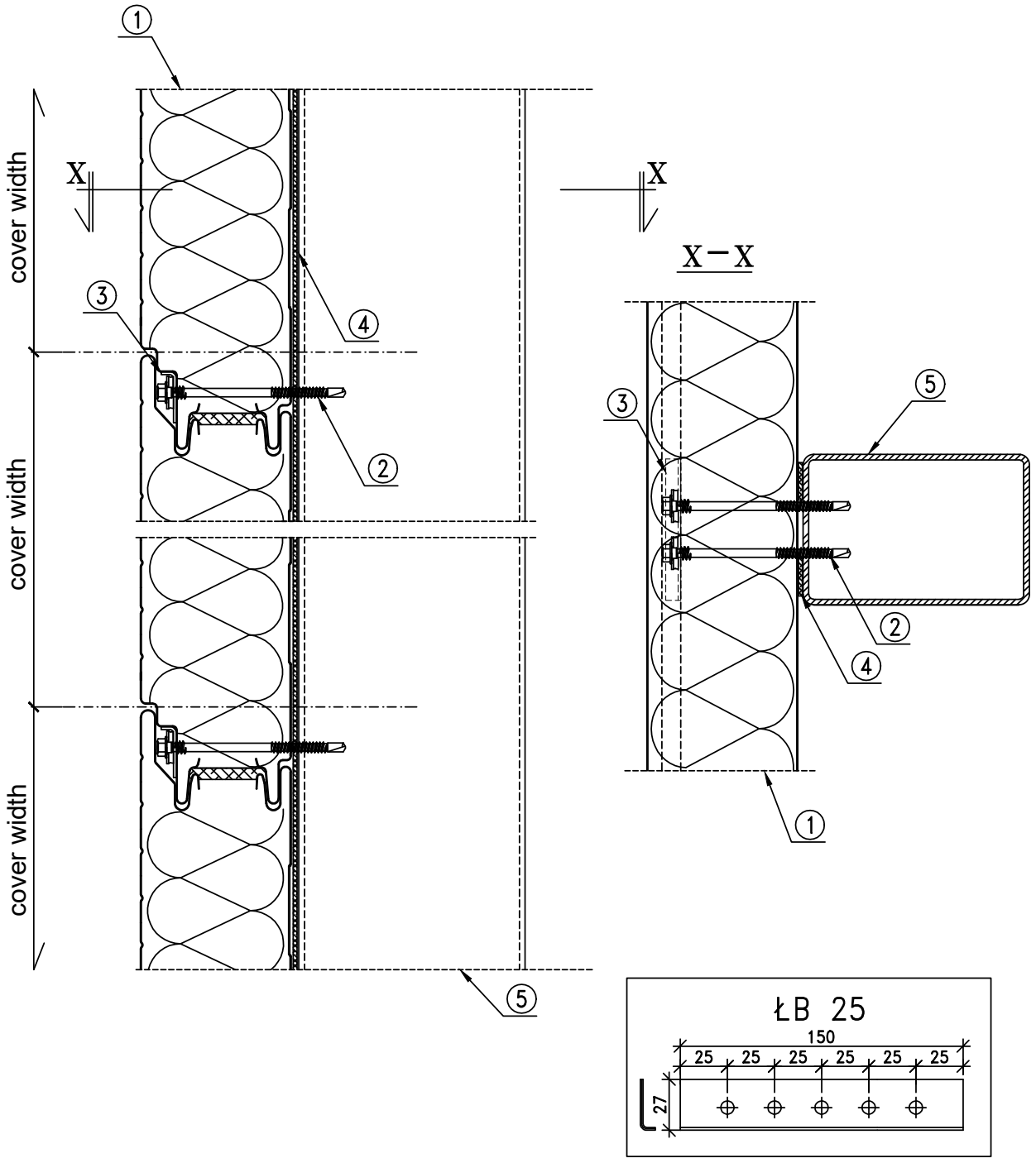
PL01 PIR PLUS wall panel - joint, profilation types - MODULE 1000





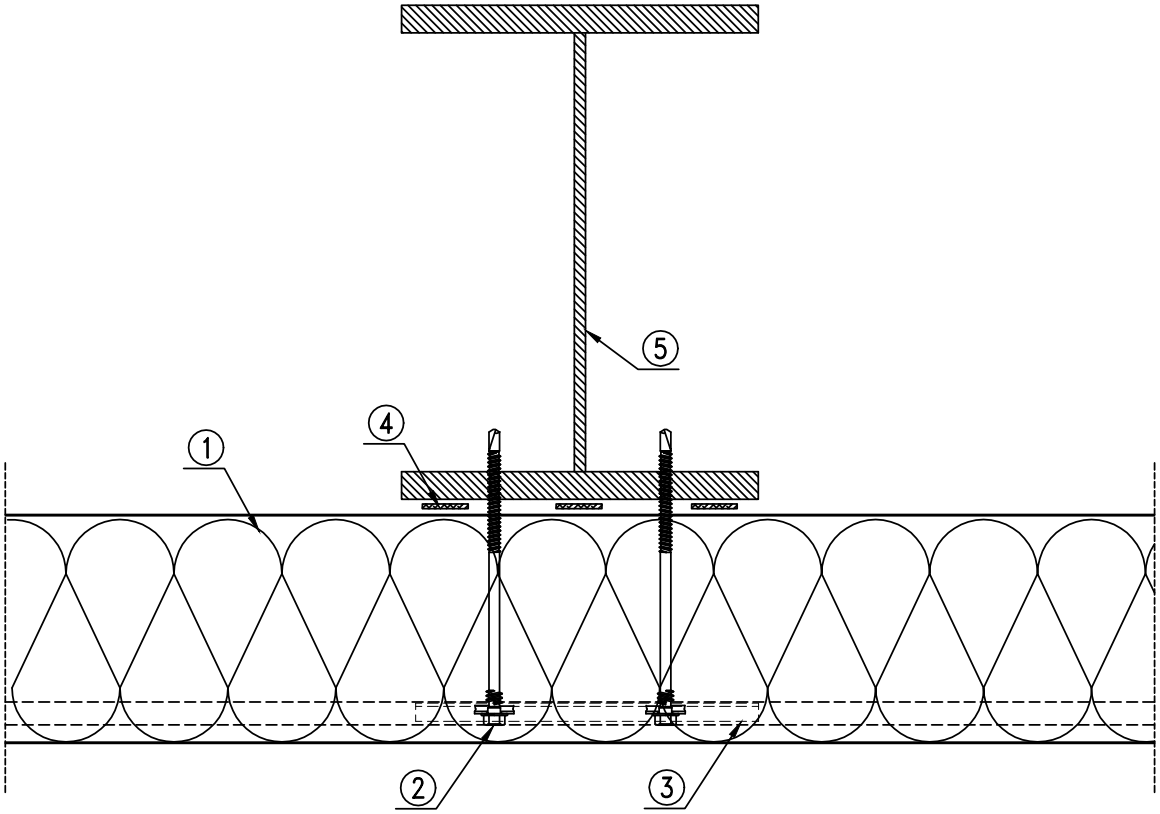


2.2.
PL02 Fastening panel - vertical panel set



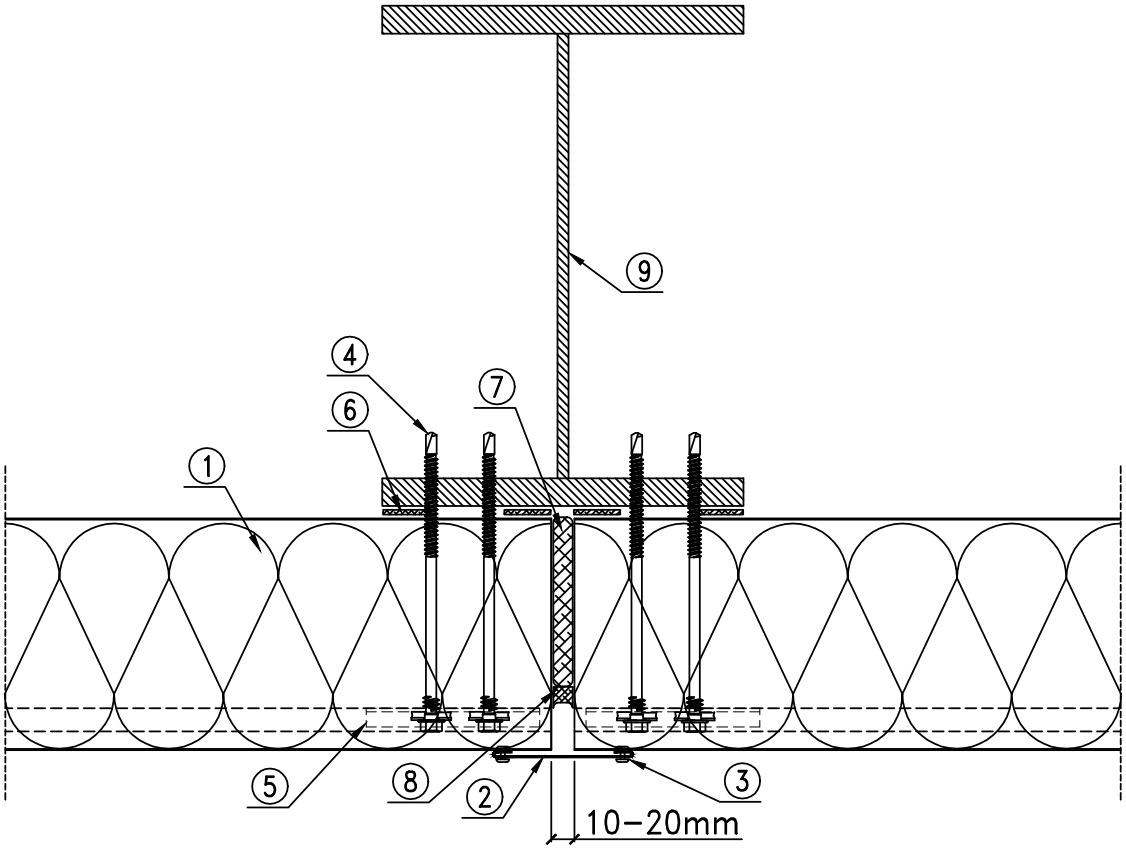
1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. Sandwich panel fastener
3. ŁB25 steel pad
4. Self-adhesive sealing tape PES
5. Load bearing profile acc. to construction design

2.3.
PL03 Fastening panel - horizontal panel set



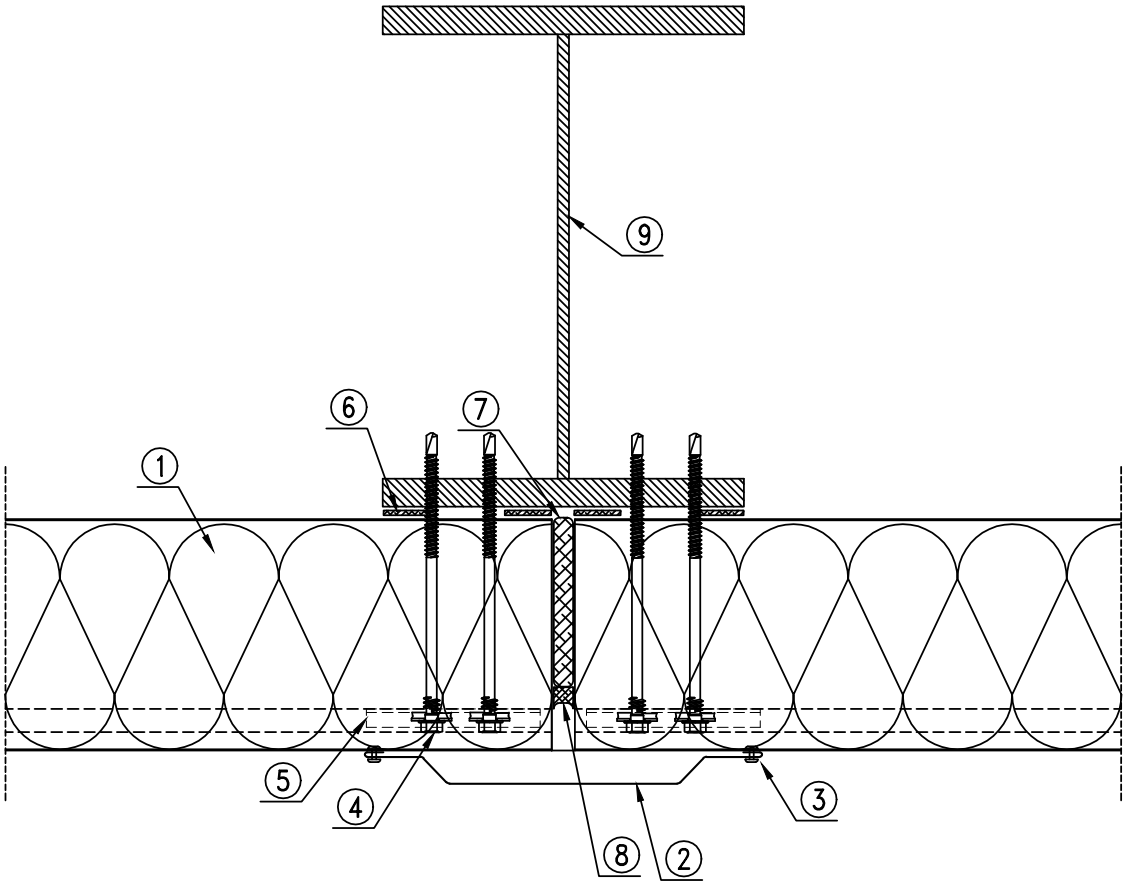
1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. Sandwich panel fastener
3. ŁB25 steel pad
4. Self-adhesive sealing tape PES
5. Load bearing column acc. to construction design

2.4.
PL04/1 Fastening panels to end support - horizontal panel set - Option I



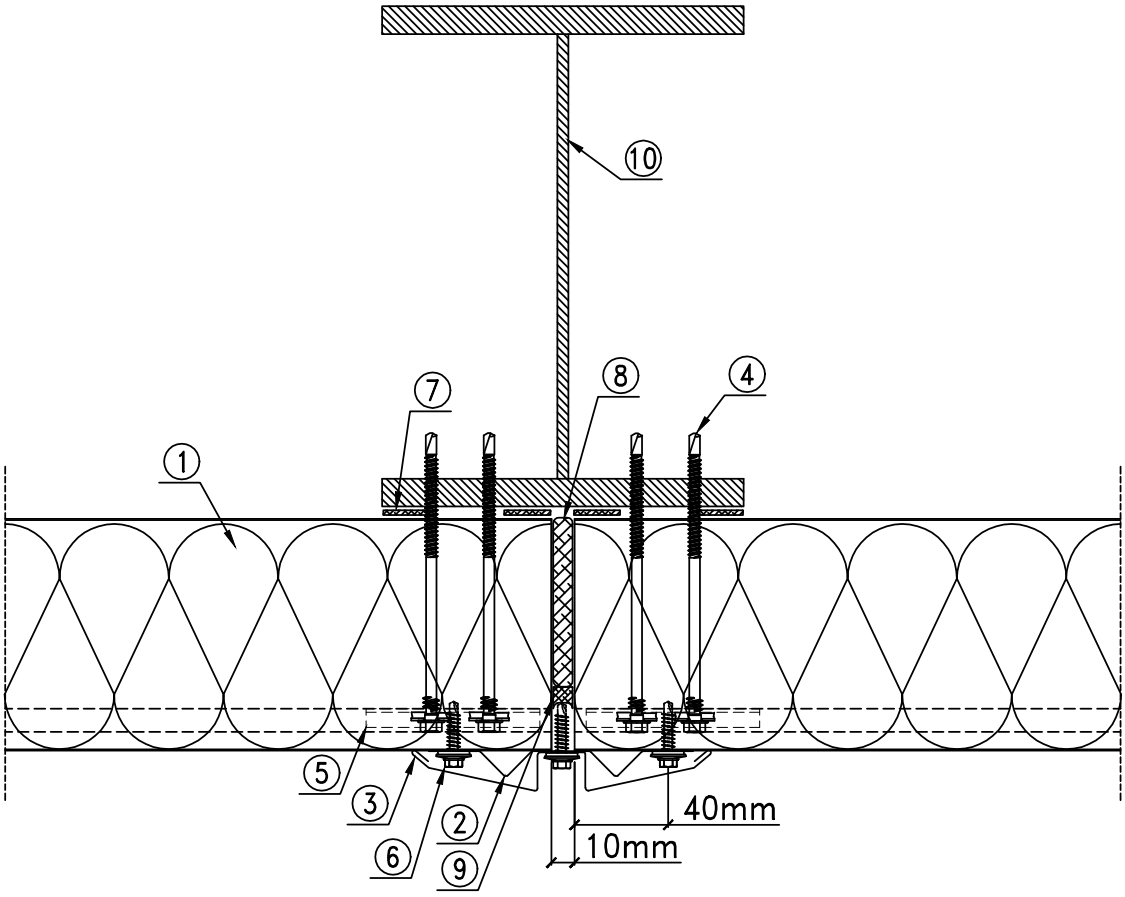
1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. OBR 106 Masking strip flashing
3. Self-drilling fastener or rivet every 300mm
4. Sandwich panel fastener
5. ŁB25 steel pad
6. PES self-adhesive sealing tape
7. Polyurethane gasket or mounting foam
8. PURS impregnated potyurethane gasket
9.Load bearing column acc. to construction design

2.5.
PL04/2 Fastening panels to end support - horizontal panel set - Option II



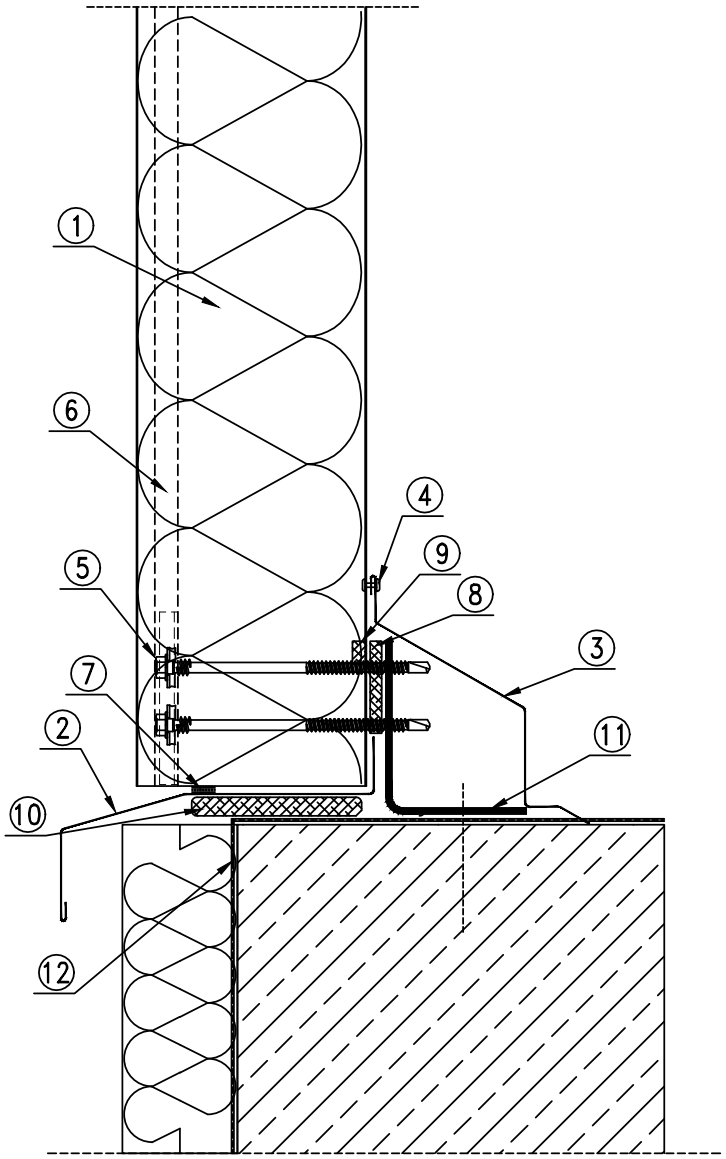
1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. OBR 105 contact area masking frame - straight flashing
3. Self-drilling fastener or rivet every 300mm
4. Sandwich panel fastener
5. ŁB25 steel pad
6. PES self-adhesive sealing tape
7. Polyurethane gasket or mounting foam
8. PURS impregnated potyurethane gasket
9. Load bearing column acc. to construction design

2.6.
PL04/3 Fastening panels to end support - horizontal panel set - Option III



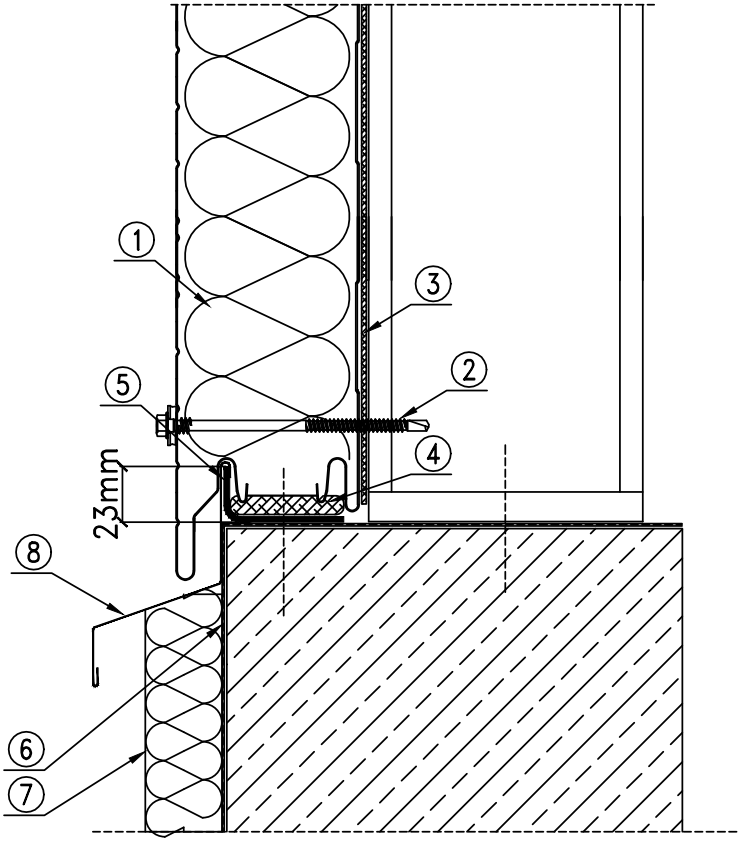
1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. OBR110 contact area masking ground beam flashing
3. OBR111 contact area masking frame - decorative element flashing
4. Sandwich panel fastener
5. ŁB 25 steel pad
6. Self-drilling fastener or rivet every 300mm
7. PES self-adhesive sealing tape
8. Polyurethane gasket or mounting foam
9. PURS self-adhesive sealing tape
10. Load bearing column acc. to construction design

2.7.
PL05 Ground beam wall panels start - vertical panel set



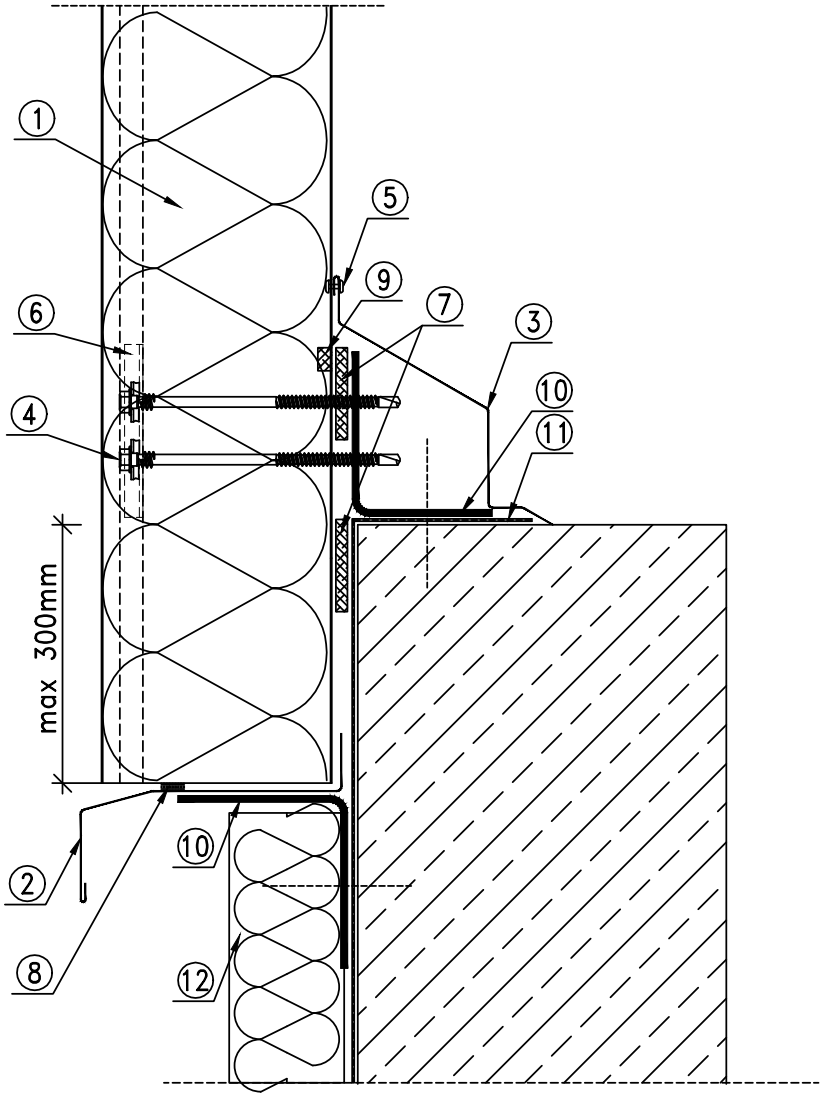
1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. OBR100 standard drip cap flashing
3. OBR101 Internal masking frame flashing
4. Self-drilling fastener or rivet every 300mm
5. Sandwich panel fastener
6. ŁB25 steel pad
7. Butyl sealing tape
8. PUS self-adhesive sealing tape
9. Butyl mass
10. PURS impregnated potyurethane gasket
11. Load bearing profile acc. to construction design
12. Hydro barrier acc. to architecture design
13. Thermal insulation + plastering acc. to architecture design

2.8.
PL06 Ground beam wall panels start - horizontal panel set



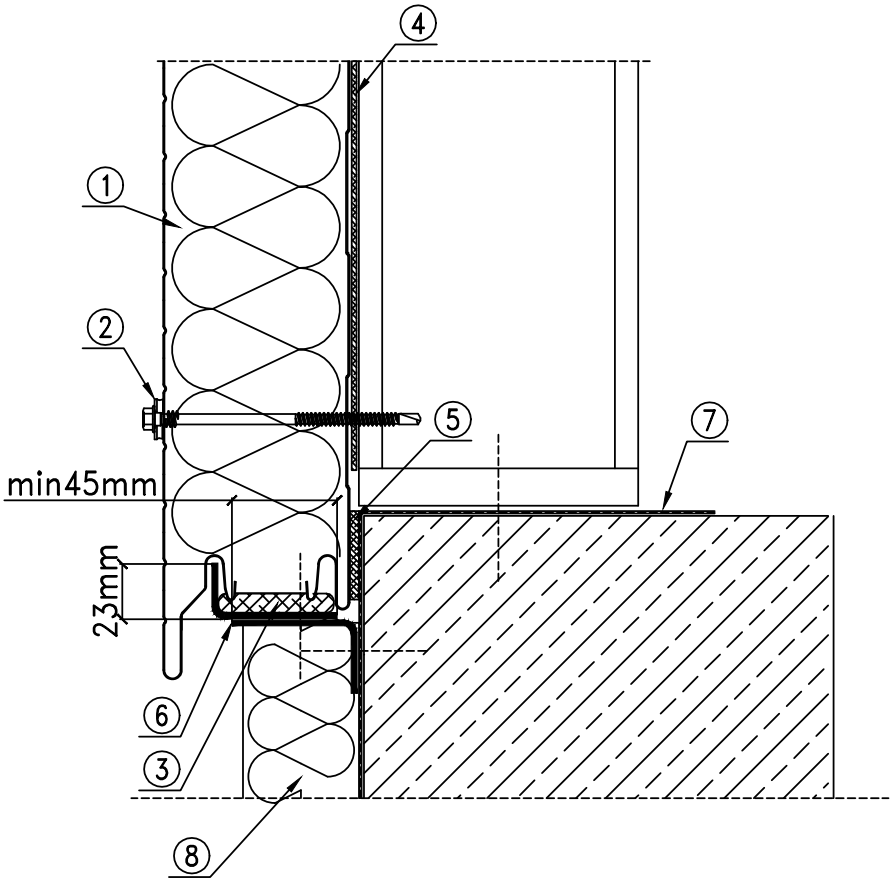
1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. Sandwich panel fastener
3. PES self-adhesive tape
4. PURS impregnated potyurethane gasket
5. Load bearing profile acc. to construction design
6. Hydro barrier acc. to architecture design
7. Thermal insulation + plastering acc. to architecture design
8. Individual flashing

2.9.
PL07 Panels start below top of ground beam - vertical panel set



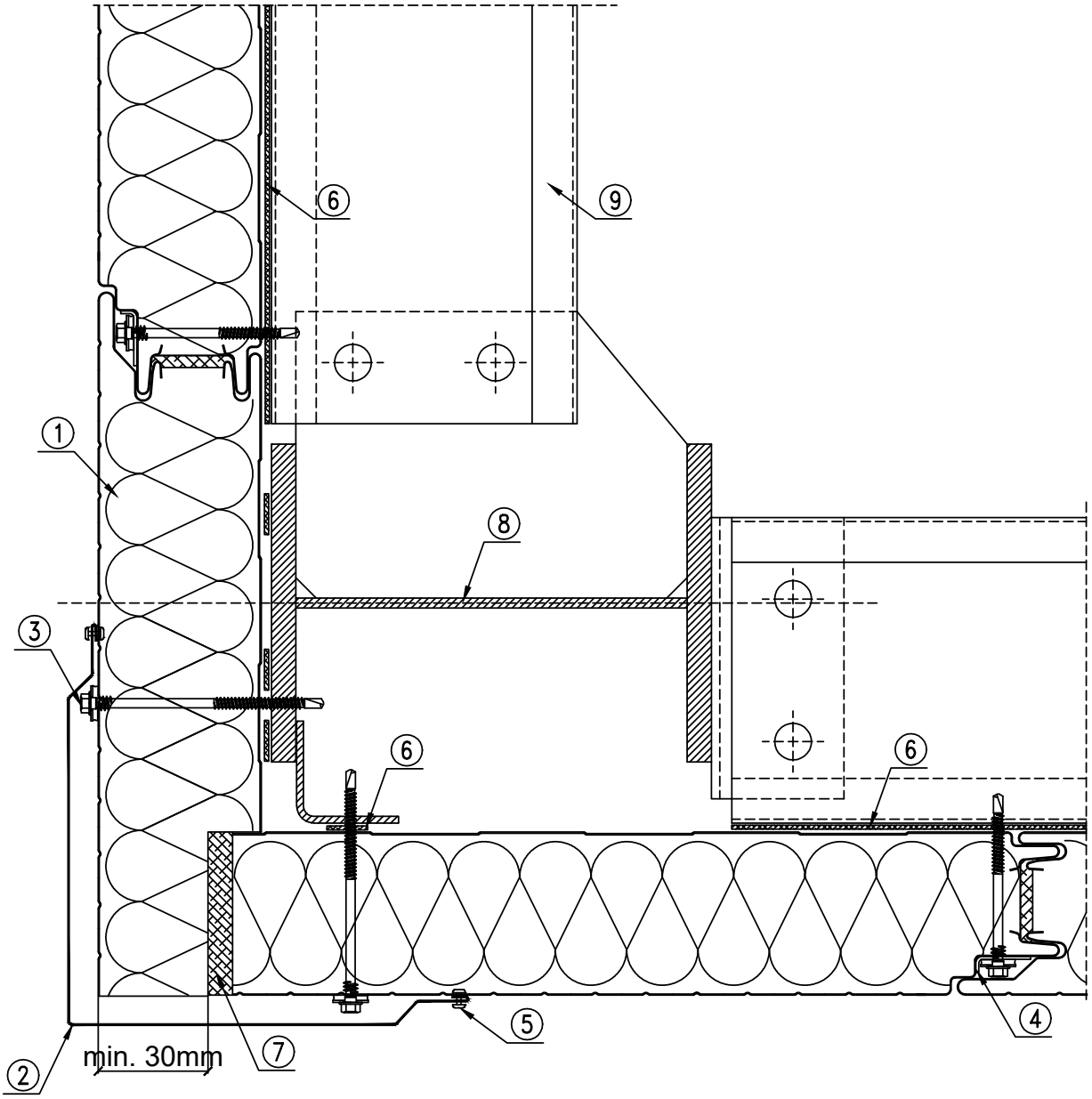
1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. OBR100 standard drip cap flashing
3. OBR101 Internal masking frame flashing
4. Sandwich panel fastener
5. Self-drilling fastener or rivet every 300mm
6. ŁB25 steel washer
7. PUS self-adhesive sealing tape
8. Butyl sealing tape
9. Butyl mass
10. Steel profile acc. to construction design
11. Hydro barrier acc. to architecture design
12. Thermal insulation + plastering acc. to architecture design

2.10.
PL08 Panels start below top of ground beam - horizontal panel set



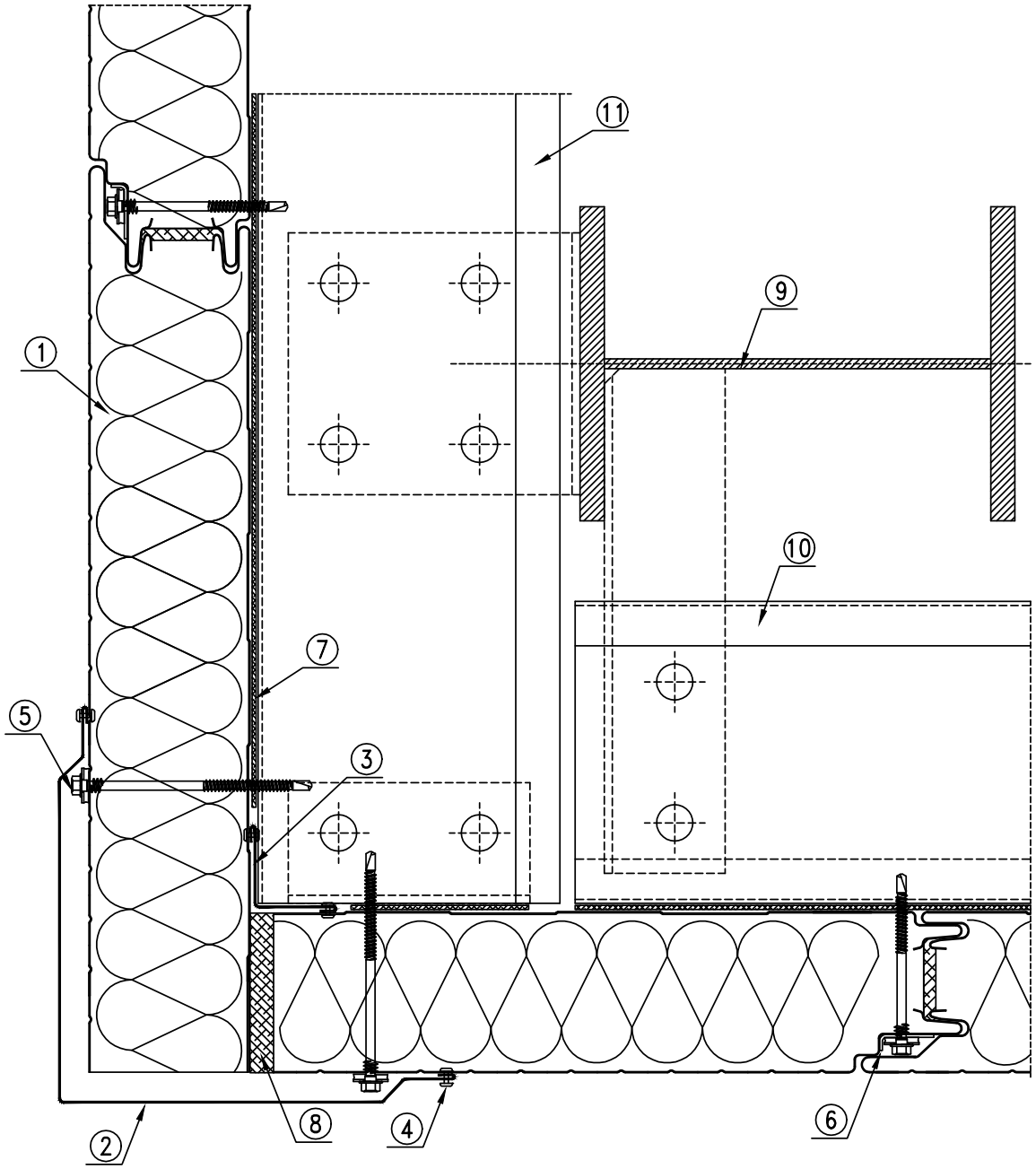
1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. Sandwich panel fastener
3. PUS self-adhesive sealing tape
4. PES self-adhesive tape
5. PURS impregnated potyurethane gasket
6. Angle profiles screwed together acc. to construction design
7. Hydro barrier acc. to architecture design
8. Thermal insulation + plastering acc. to architecture design

2.11.
PL09/1 Corner joint - vertical panel set - Option I



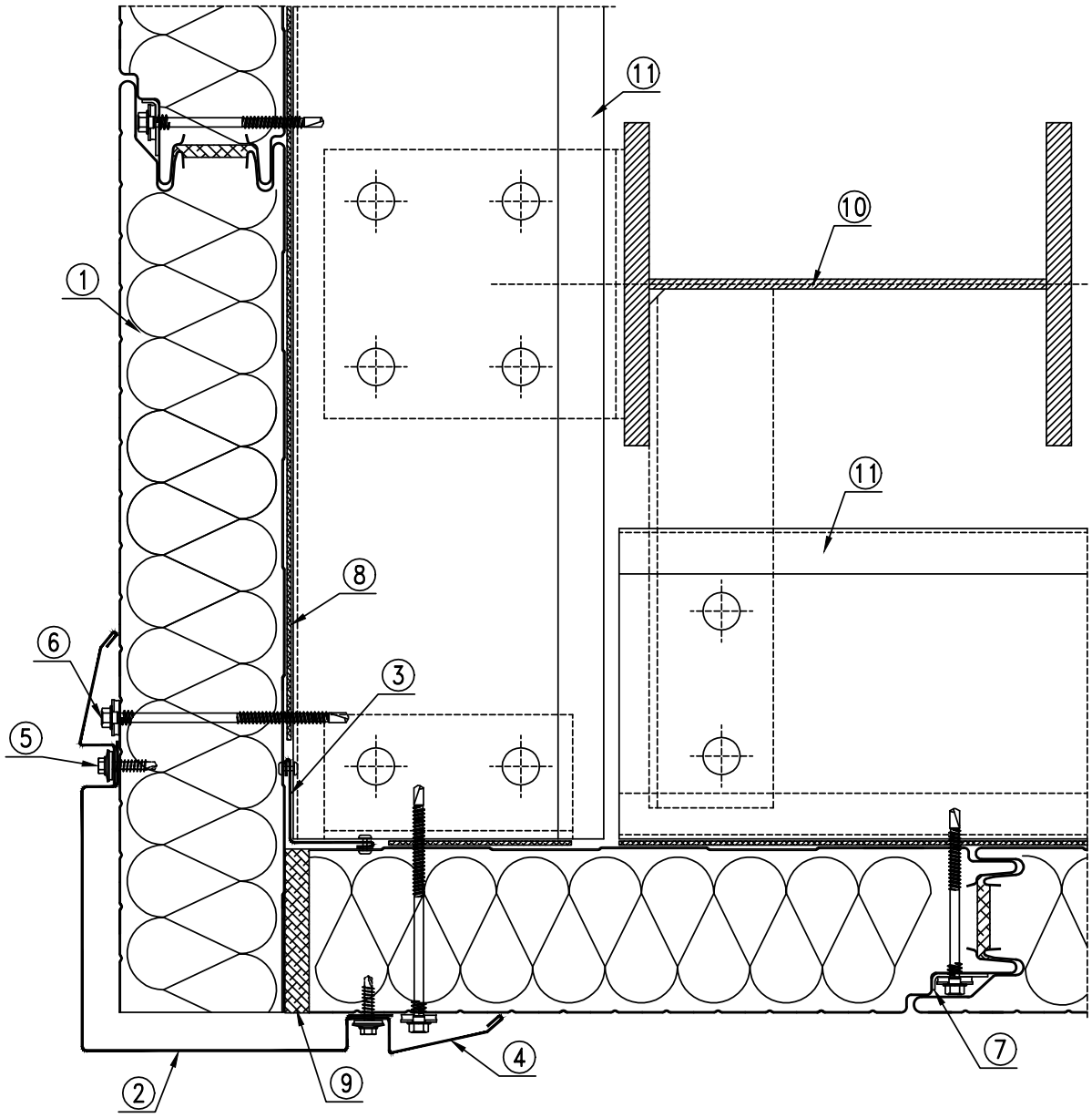
1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. OBR103 External corner with a hidden joint flashing
3. Sandwich panel fastener
4. ŁB25 steel washer
5. Self-drilling fastener or rivet every 300mm
6. PES self-adhesive tape
7. Polyurethane gasket or mounting foam
8. Load bearing column + angle profile
acc. to construction design
9. Load bearing beam acc. to construction design

2.12.
PL09/2 Corner joint - vertical panel set - Option II



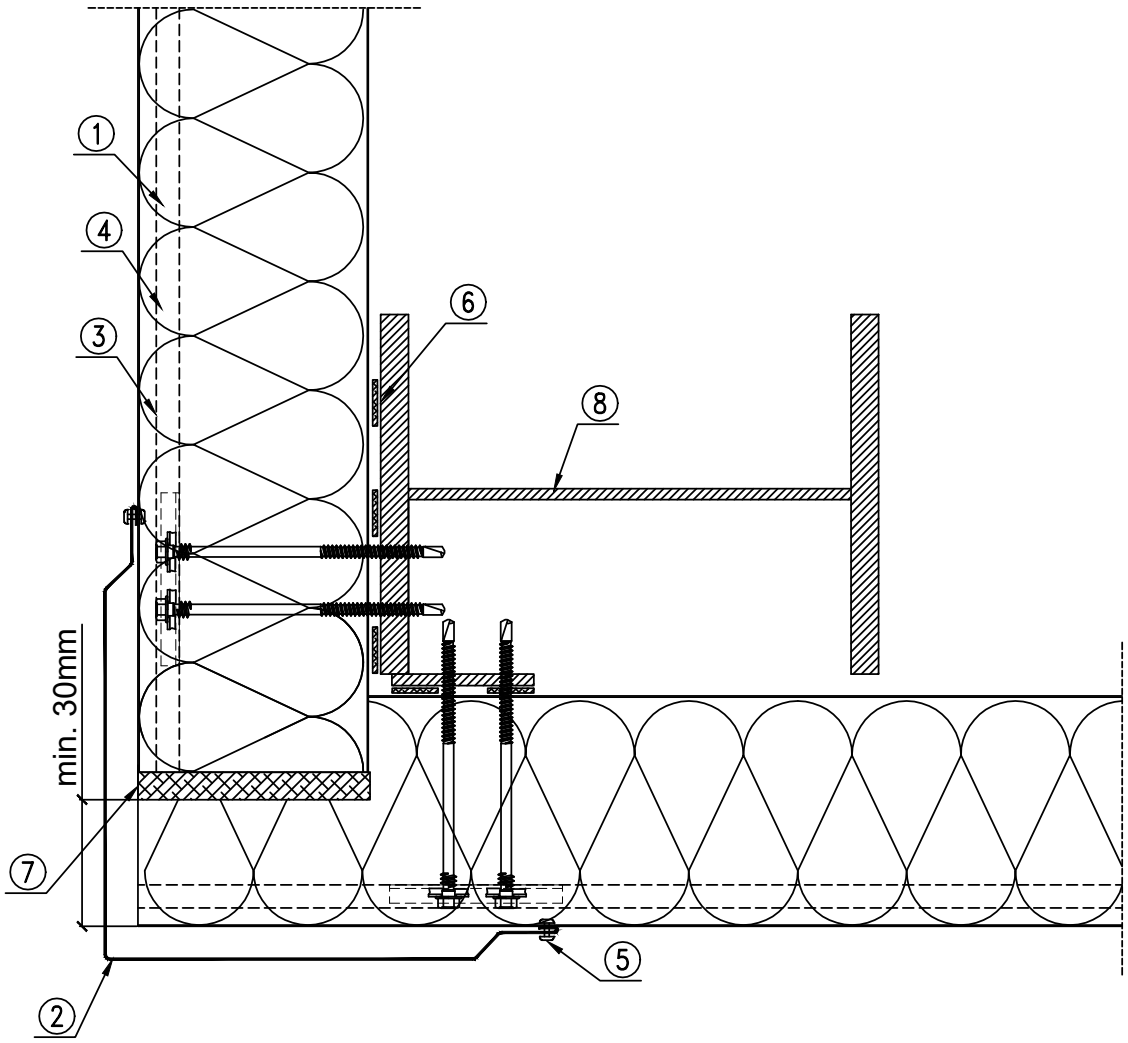
1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. OBR103 External corner with a hidden joint flashing
3. OBR104 Internal strip flashing
4. Self-drilling fastener or rivet every 300mm
5. Sandwich panel fastener
6. ŁB25 steel washer
7. PES self-adhesive tape
8. Polyurethane gasket or mounting foam
9. Load bearing column acc. to construction design
10. Load bearing beam acc. to construction design

2.13.
PL09/3 Corner joint - vertical panel set - Option III



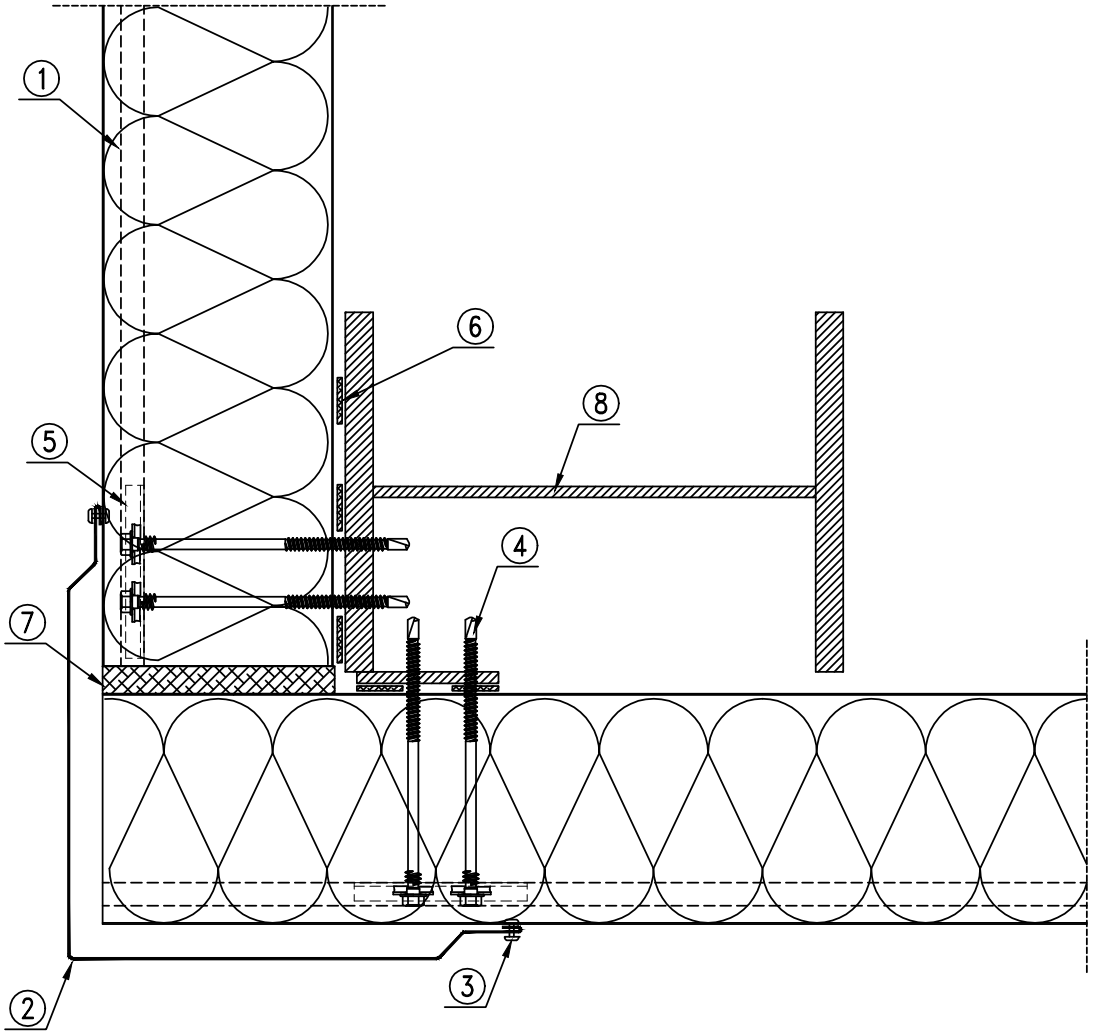
1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. OBR113 External corner - base flashing
3. OBR104 Internal strip flashing
4. OBR111 External corner - decorative element flashing
5. Self-drilling fastener or rivet every 300mm
6. Sandwich panel fastener
7. ŁB25 steel washer
8. PES self-adhesive tape
9. Polyurethane gasket or mounting foam
10. Load bearing column acc. to construction design
11. Load bearing beam acc. to construction design

2.14.
PL010/1 Corner joint - horizontal panel set - Option I



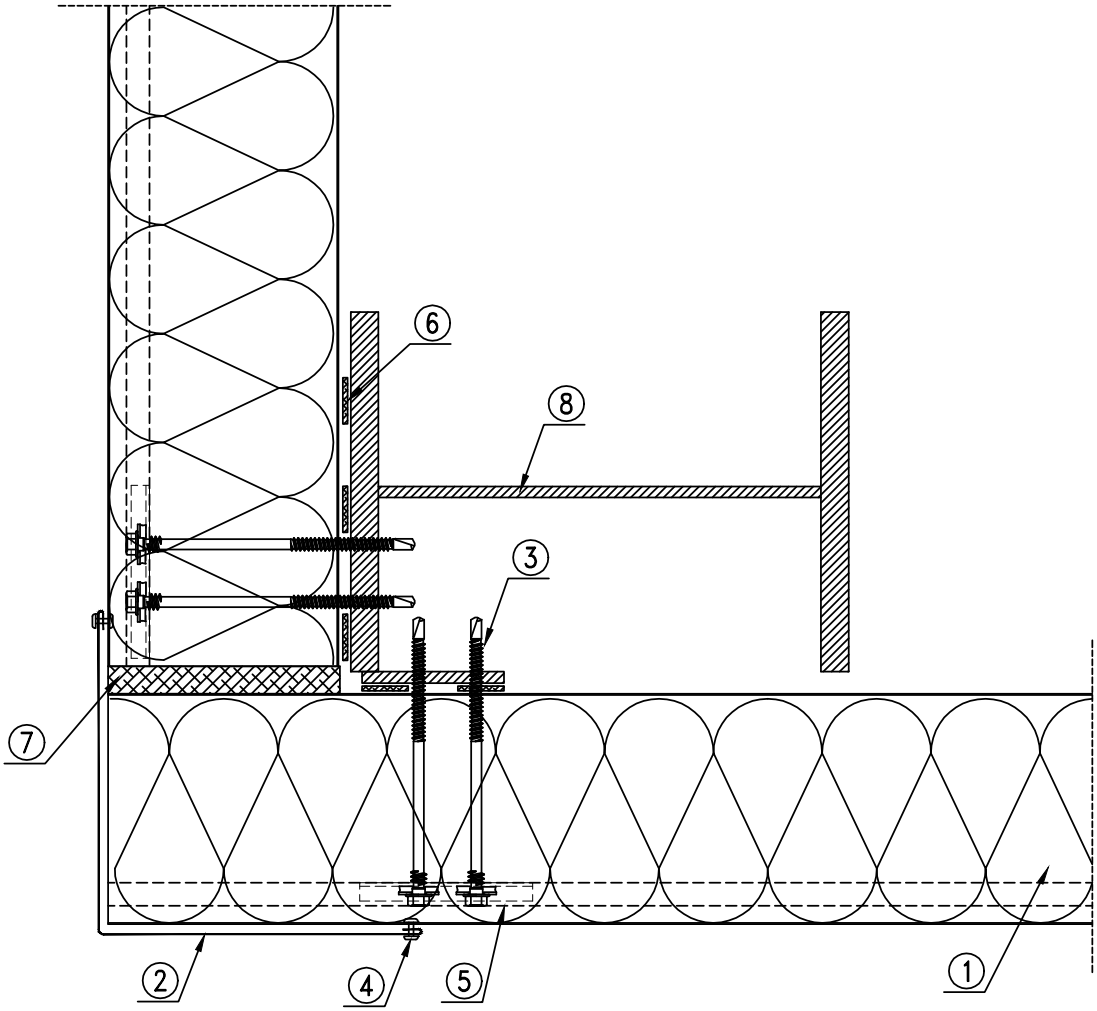
1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. OBR103 External corner with a hidden joint flashing
3. Sandwich panel fastener
4. ŁB25 steel washer
5. Self-drilling fastener or rivet every 300mm
6. PES self-adhesive tape
7. Polyurethane gasket or mounting foam
8. Load bearing column acc. to construction design

2.15.
PL010/2 Corner joint - horizontal panel set - Option II



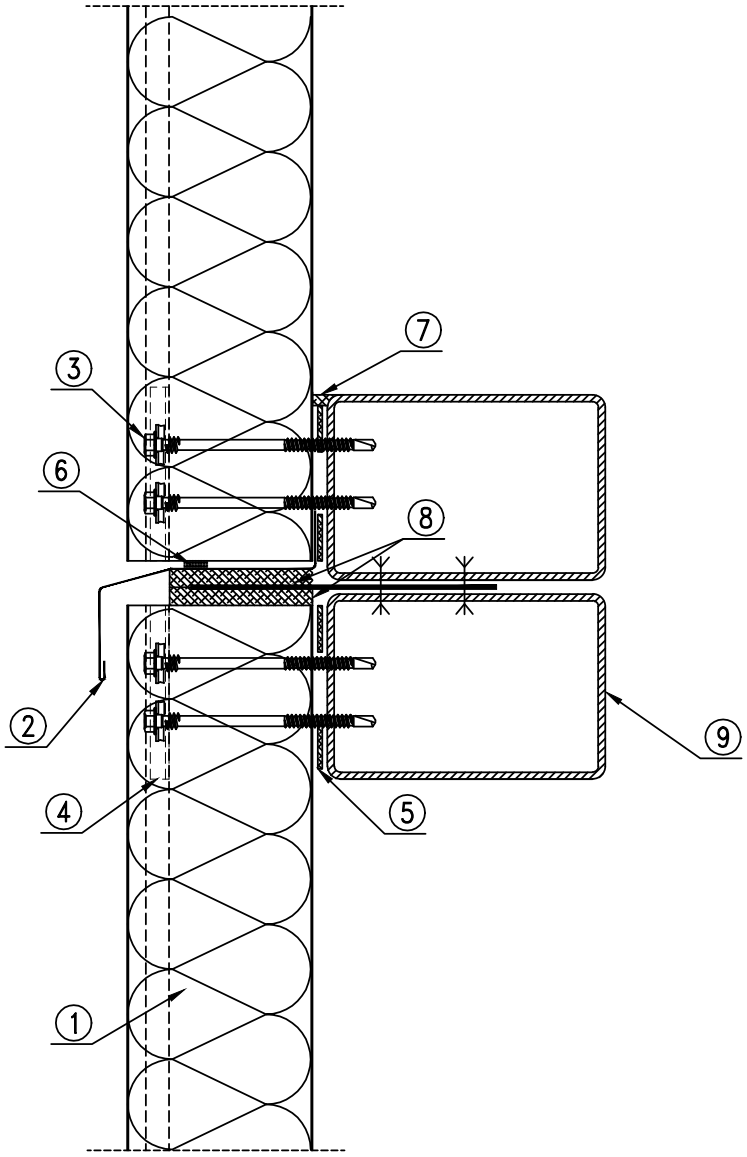
1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. OBR103 External corner with a hidden joint flashing
3. Self-drilling fastener or rivet every 300mm
4. Sandwich panel fastener
5. ŁB25 steel washer
6. PES self-adhesive tape
7. Polyurethane gasket or mounting foam
8. Load bearing column acc. to construction design

2.16.
PL010/3 Corner joint - horizontal panel set - Option III



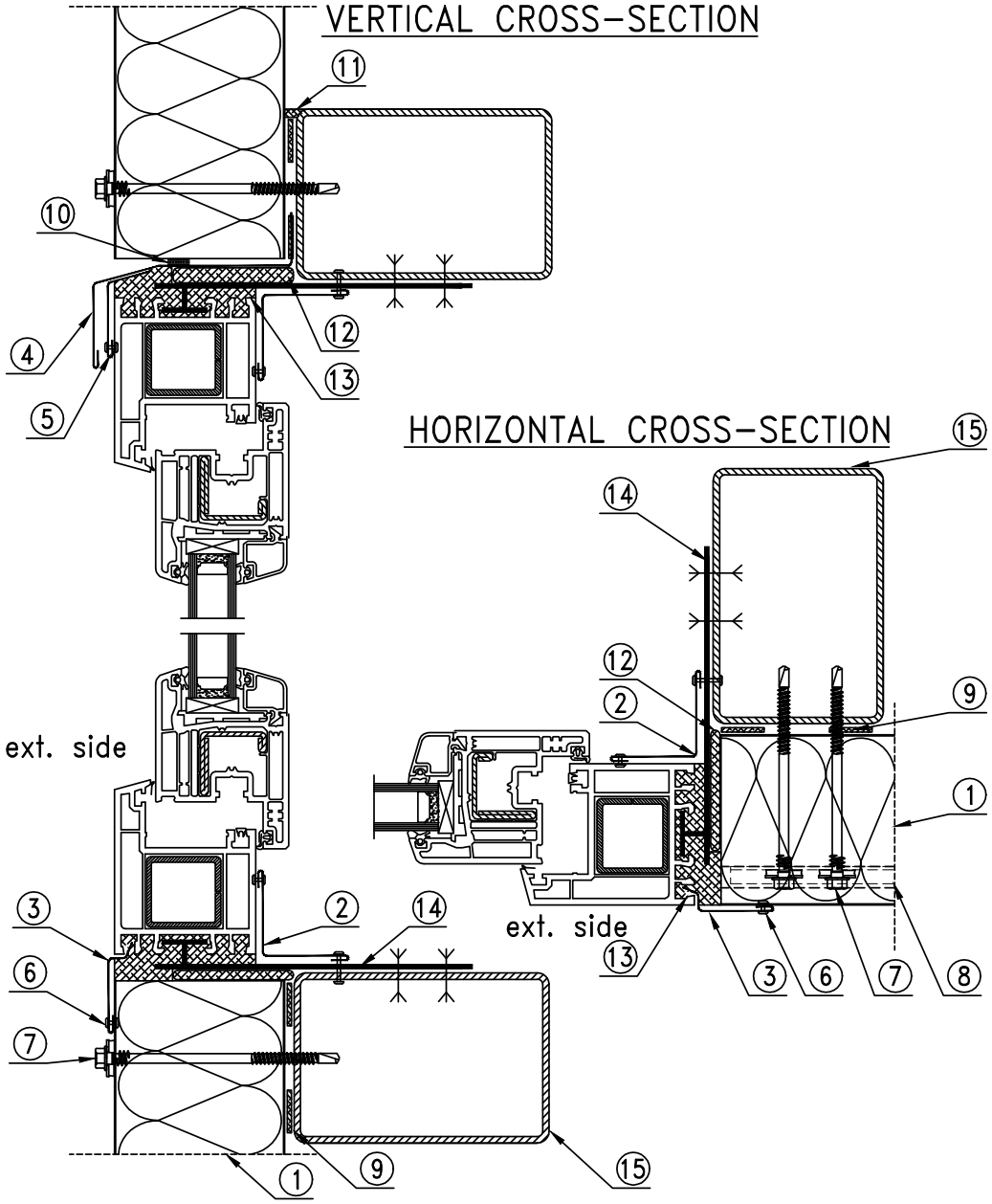
1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. OBR05 External straight corner or OBR109 External even-armed straight corner
3. Sandwich panel fastener
4. Self-drilling fastener or rivet every 300mm
5. ŁB25 steel washer
6. PES self-adhesive tape
7. Polyurethane gasket or mounting foam
8. Load bearing column acc. to construction design

2.17.
PL11 Lengthwise panel joint - vertical panel set



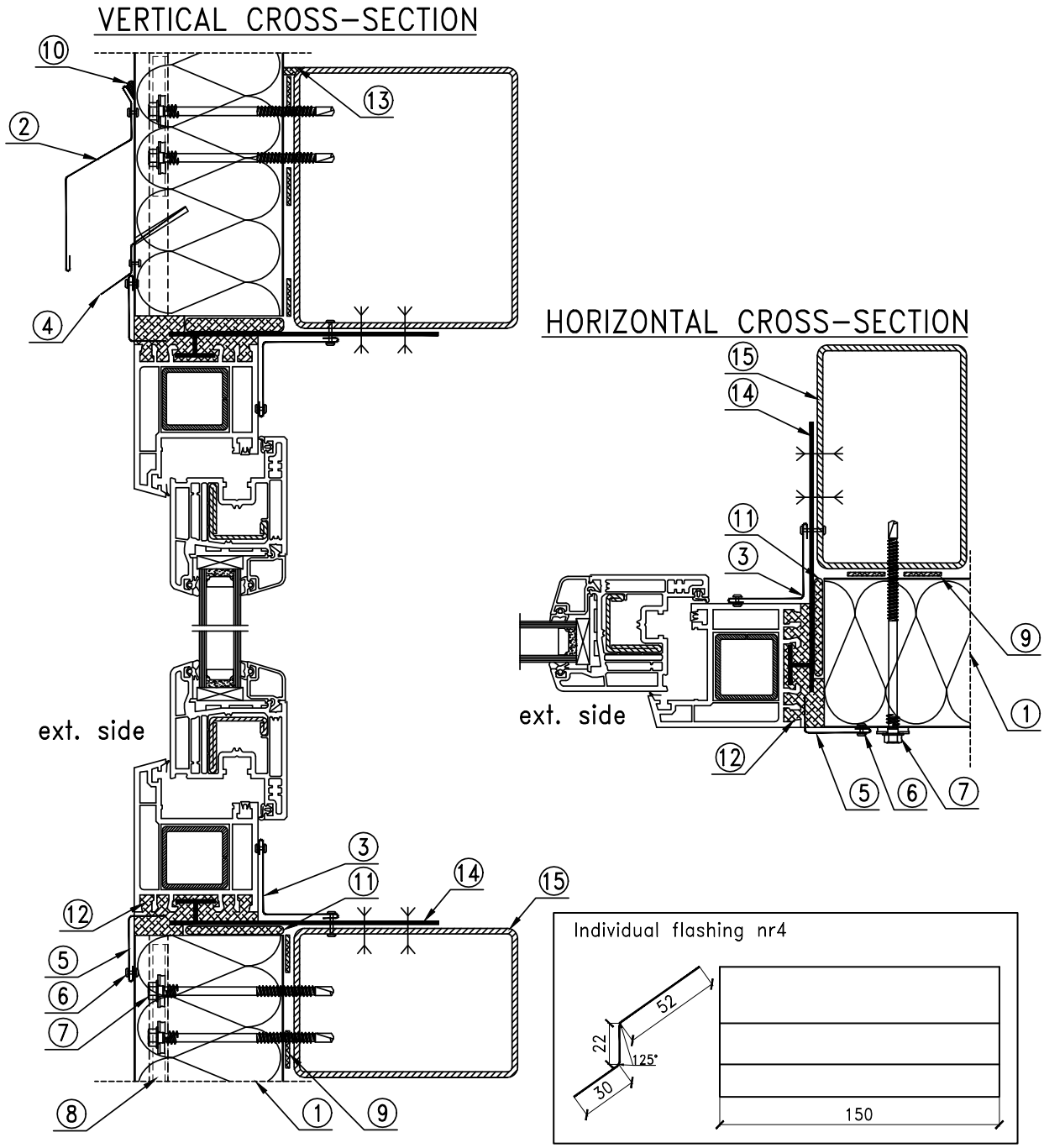
1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. OBR100 Standard drip cap
3. Sandwich panel fastener
4. ŁB25 steel washer
5. PES self-adhesive tape
6. Butyl mass or tape
7. Butyl mass
8. Polyurethane gasket - thickness adjusted to gap width
9. Load bearing beam + flat metal bar acc. to construction design

2.18.
PL12 Panels - window joint - horizontal panel set



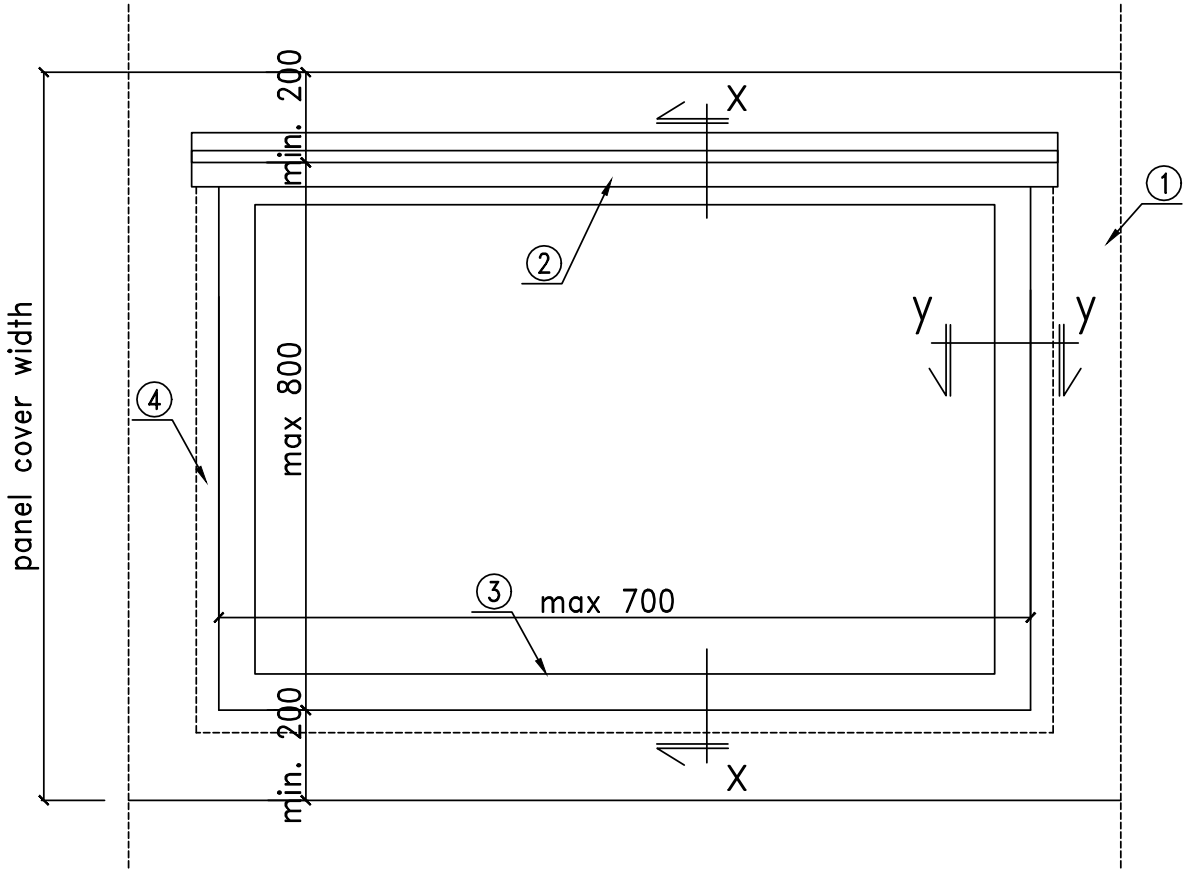
1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. OBR104 Internal strip flashing
3. OBR06 External window strip flashing
4. OBR100 Standard drip cap flashing
5. Individual flashing
6. Self-drilling fastener or rivet every 300mm
7. Sandwich panel fastener
8. ŁB25 steel washer
9. PES self-adhesive sealing tape
10. Butyl mass or tape
11. Butyl mass
12. PURS self-adhesive sealing tape
13. Mounting foam
14. Window bar
15. Load bearing beam acc. to construction design

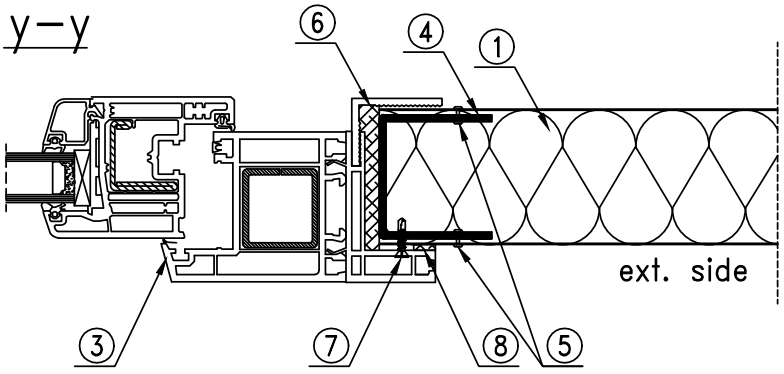
2.19.
PL13 Panels - window joint - vertical panel set



1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. OBR107 Over window drip cap flashing
3. OBR104 Internal strip flashing
4. Individual flashing - place in sandwich wall panel cut in area of locks
5. Individual flashing
6. Self-drilling fastener or rivet every 300mm
7. Sandwich panel fastener
8. ŁB25 steel washer
9. PES self-adhesive sealing tape
10. Butyl mass or tape
11. PURS self-adhesive sealing tape
12. Mounting foam
13. Butyl mass
14. Window bar
15. Load bearing beam acc. to construction design

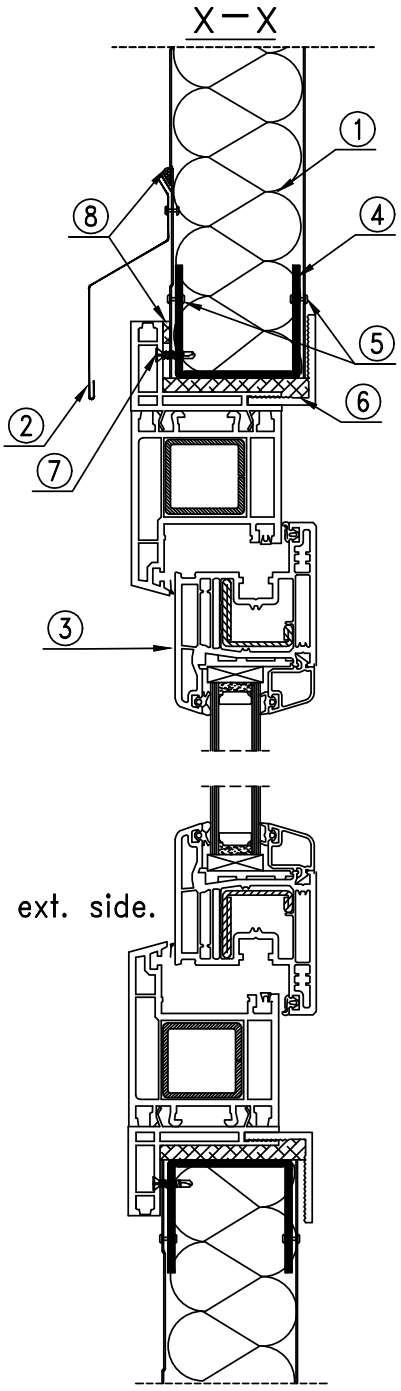
2.20.
PL14/1 Panels - window joint - without additional construction - Y-Y





1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. OBR107 Over window drip cap flashing
3. Window
4. Window C-profile frame acc. to construction design
pressed between panel claddings
5. Self-drilling fastener or rivet every 300mm
6. Polyurethane gasket or mounting foam
7. Self-drilling fastener
8. Butyl mass

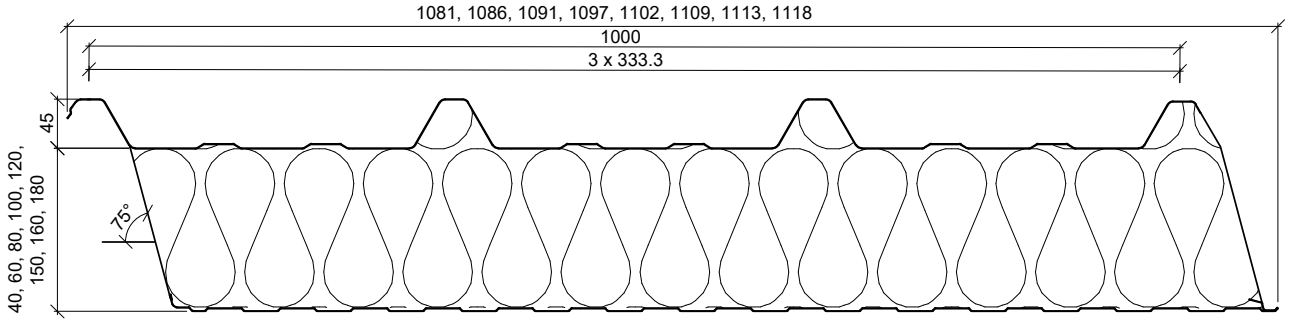
2.21.
PL14/2 Panels - window joint - without additional construction - X-X

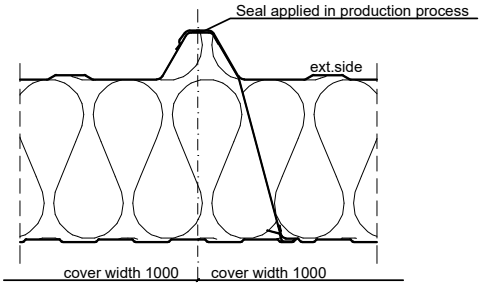


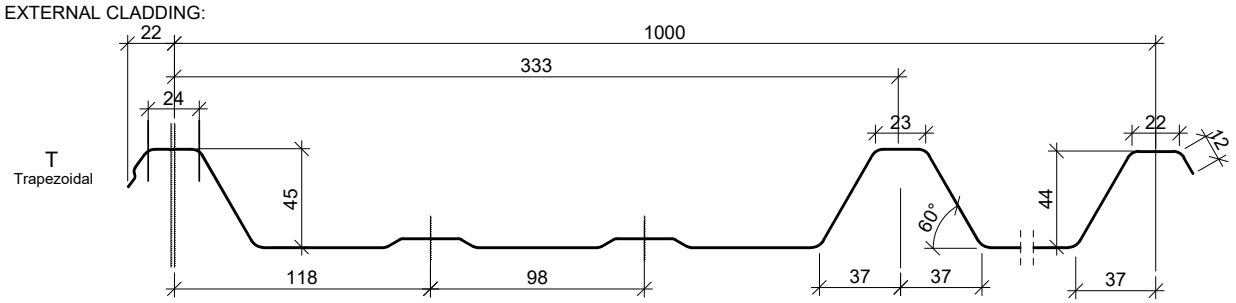
1. Wall sandwich panel PIR Plus (PU-PIR-W-PLUS)
2. OBR107 Over window drip cap flashing
3. Window
4. Window C-profile frame acc. to construction design
pressed between panel claddings
5. Self-drilling fastener or rivet every 300mm
6. Polyurethane gasket or mounting foam
7. Self-drilling fastener
8. Butyl mass

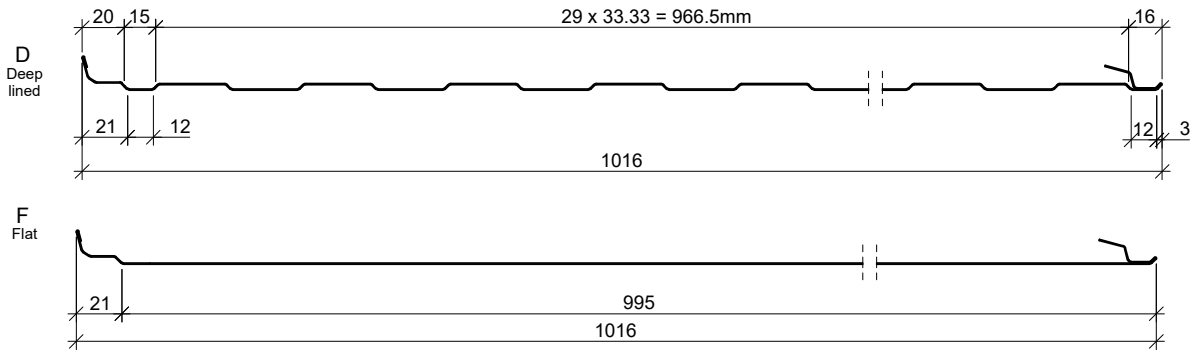
3. PIR ROOF ROOF PANEL

3.1.
R01 PIR ROOF - Joint profilation types 1000mm

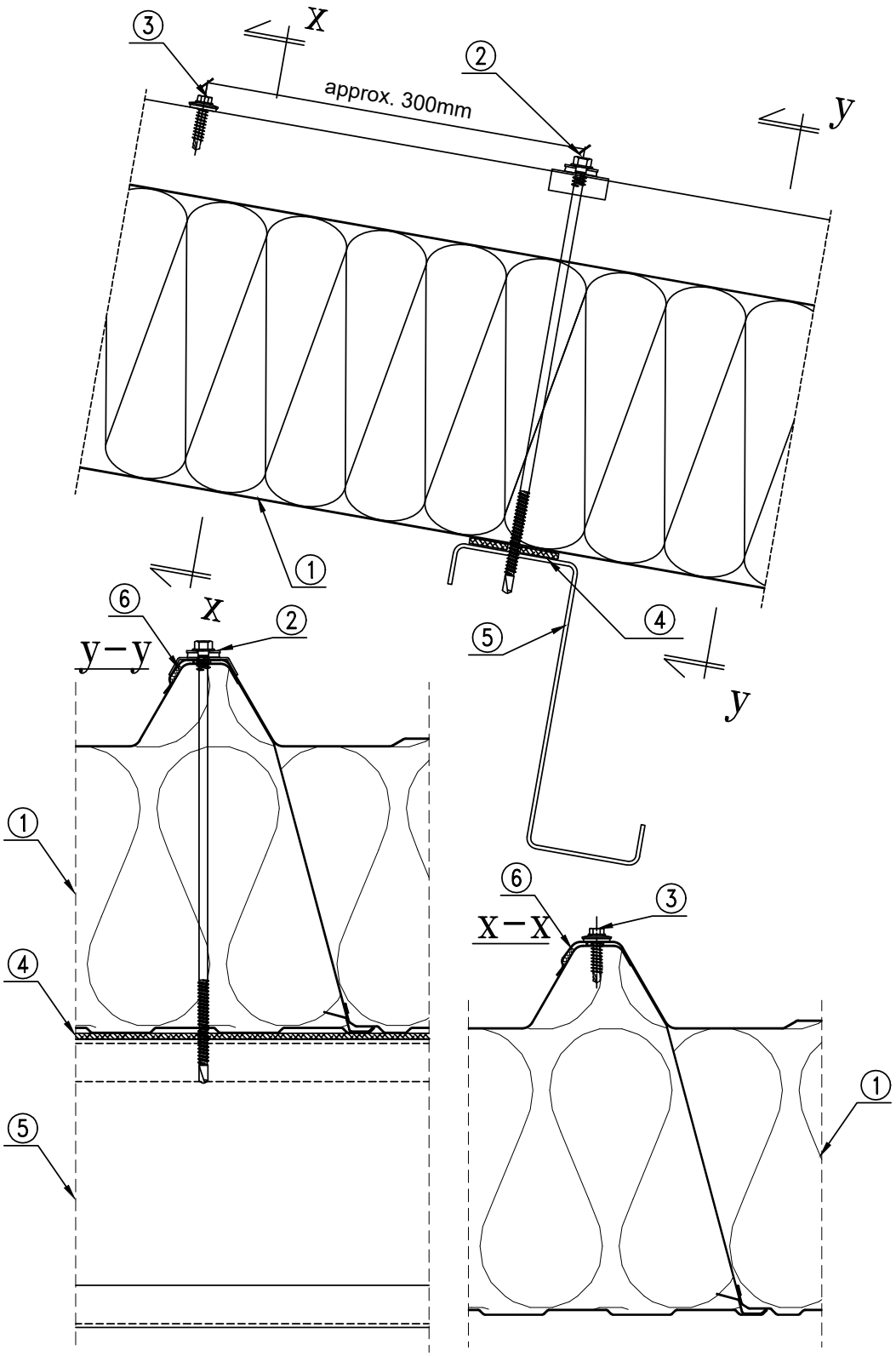






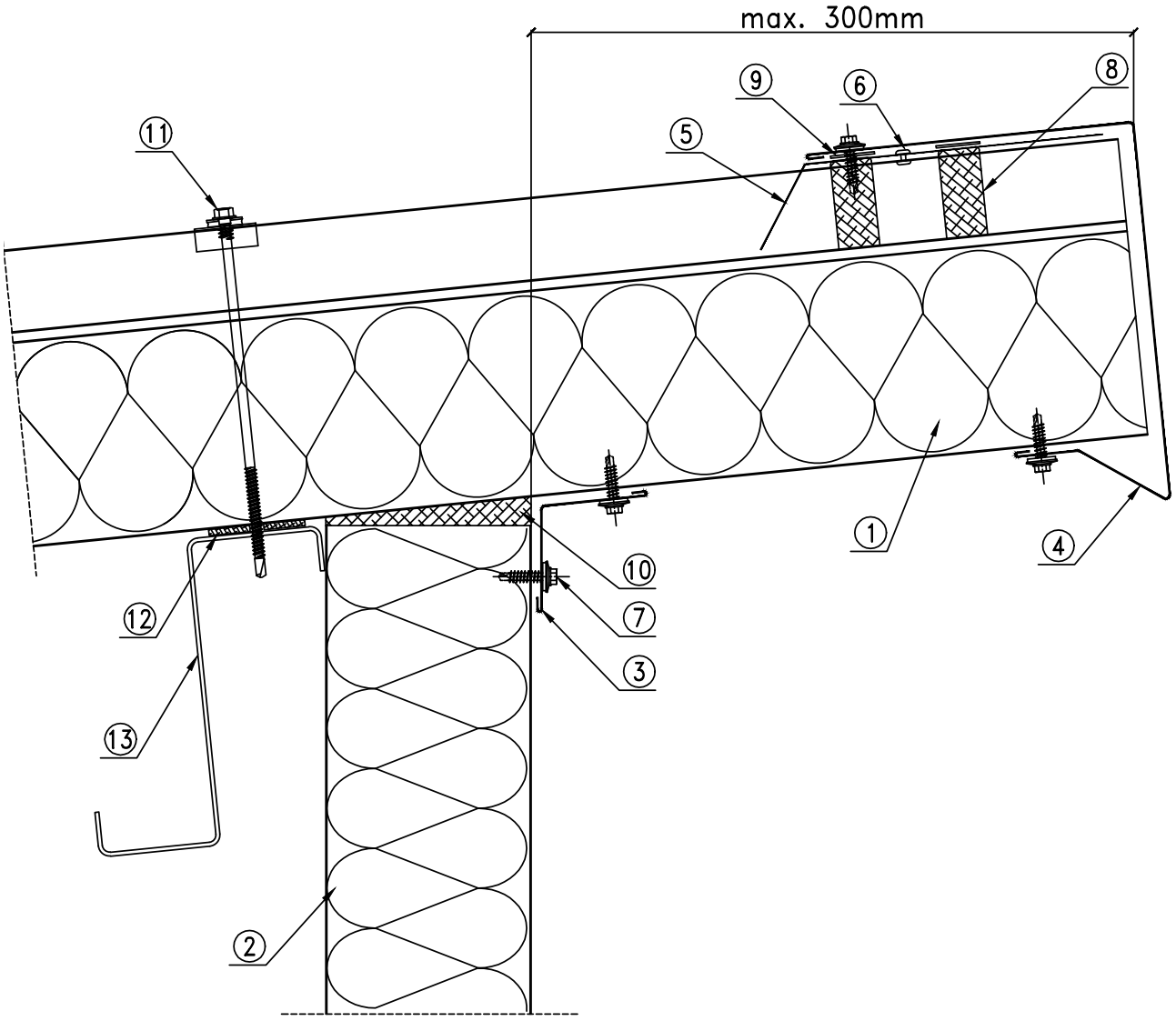


3.2.
R02 Fastening type



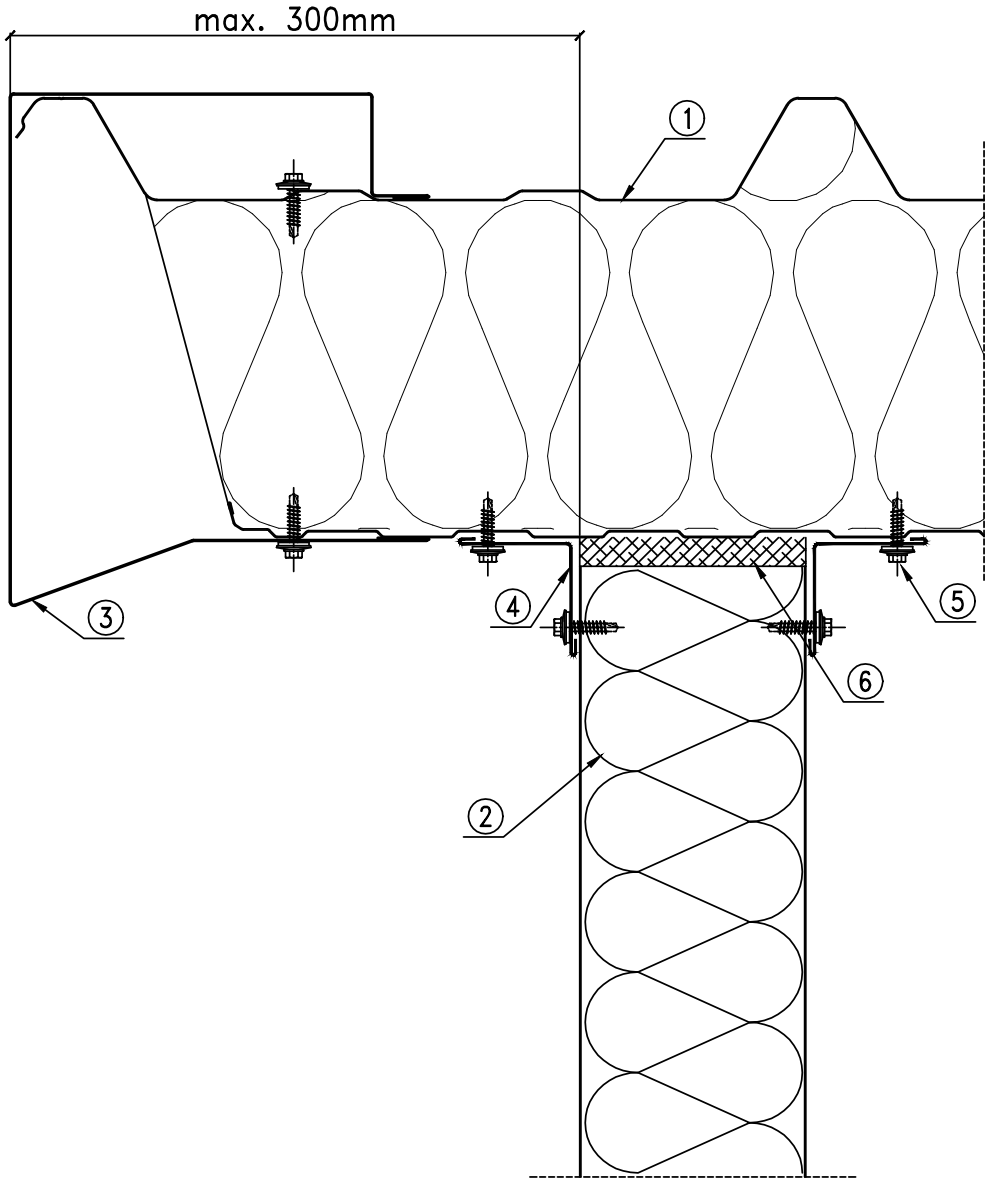
1. Roof sandwich panel PIR ROOF
2. Sandwich panel fastener with saddle pad
3. Self-drilling fastener or rivet every 300mm
4. PES self-adhesive sealing tape
5. Load bearing purlin acc. to construction design
6. Butyl mass or tape

3.3.
R03 Monopith roof



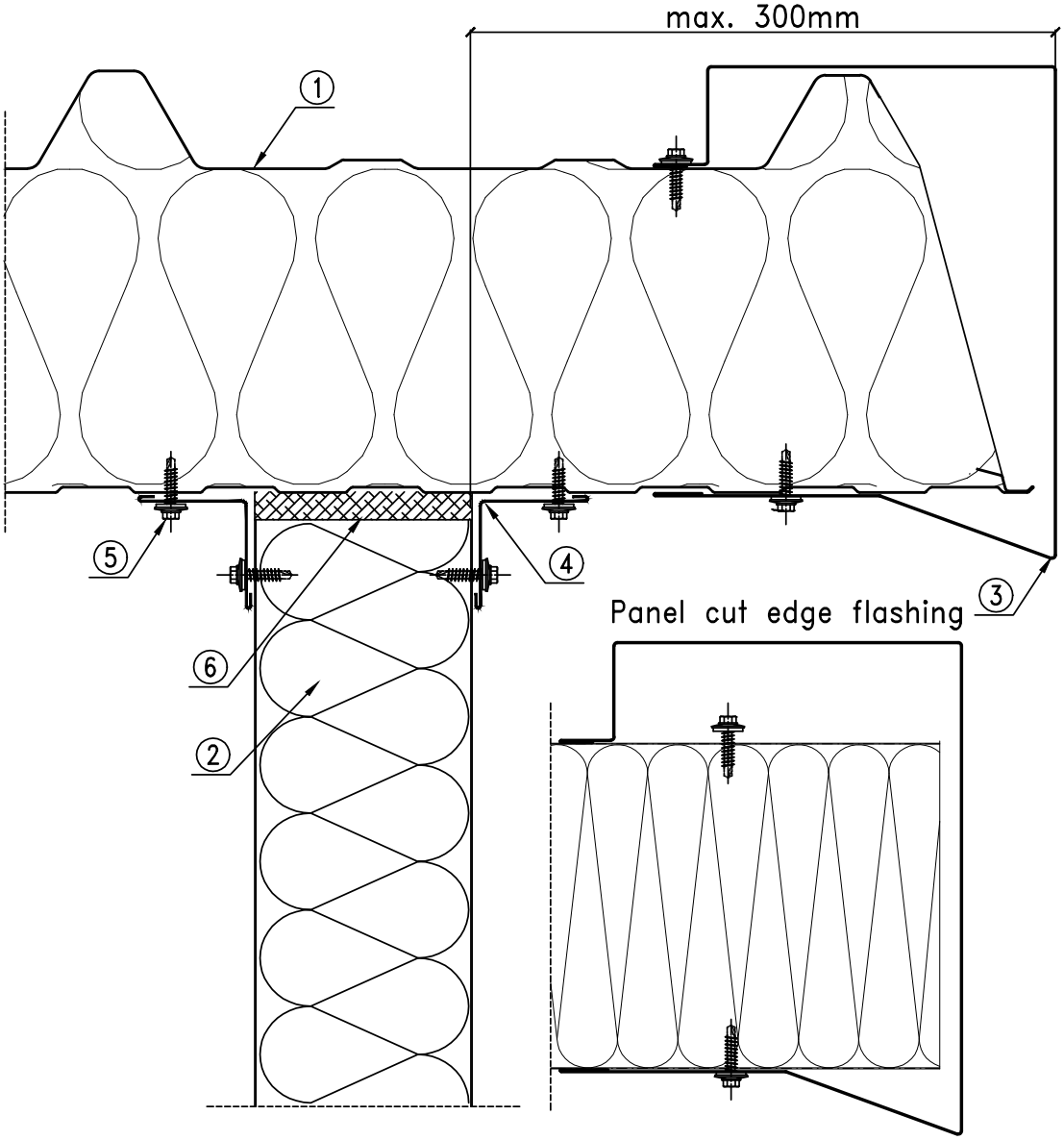
1. Roof sandwich panel PIR ROOF
2. Wall sandwich panel
3. OBR104 Internal strip flashing
4. OBR200 Roof ridge masking frame flashing
5. OBR201 Roof ridge strip flashing
6. Assembly rivet between trapezoids approximately every 100
7. Self-drilling fastener or rivet every 300mm
8. Sealing tape TUN45
9. Butyl mass or tape
10. Polyurethane gasket or mounting foam
11. Sandwich panel fastener with saddle pad
12. PES self-adhesive sealing tape
13. Load bearing purlin acc. to construction design

3.4.
R04.1 Gable wall



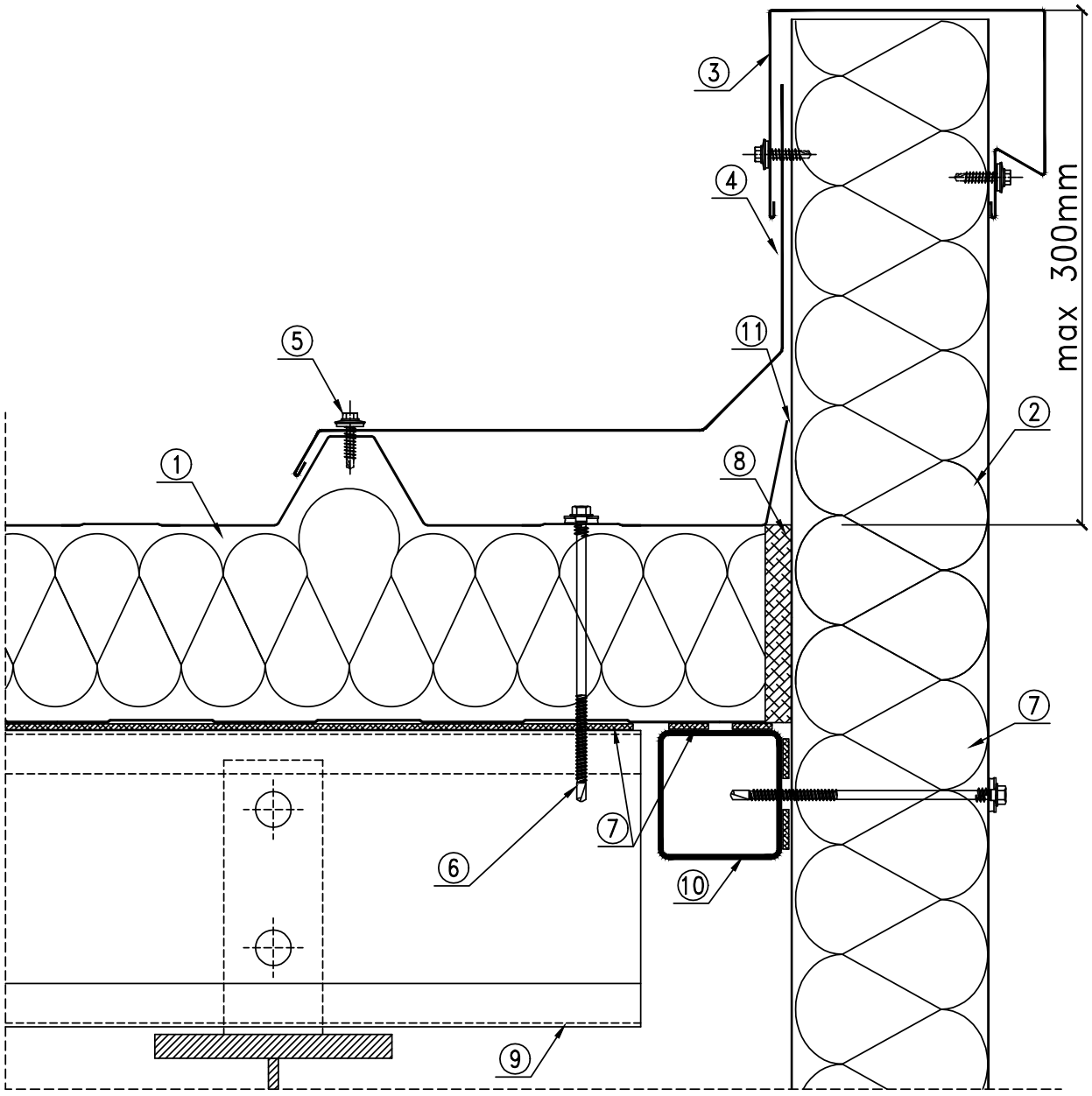
1. Roof sandwich panel PIR ROOF
2. Wall sandwich panel
3. OBR202.1 Wind blade flashing
4. OBR104 Internal strip flashing
5. Self-drilling fastener or rivet every 300mm
6. Polyurethane gasket or mounting foam

3.5.
R04.2 Gable wall



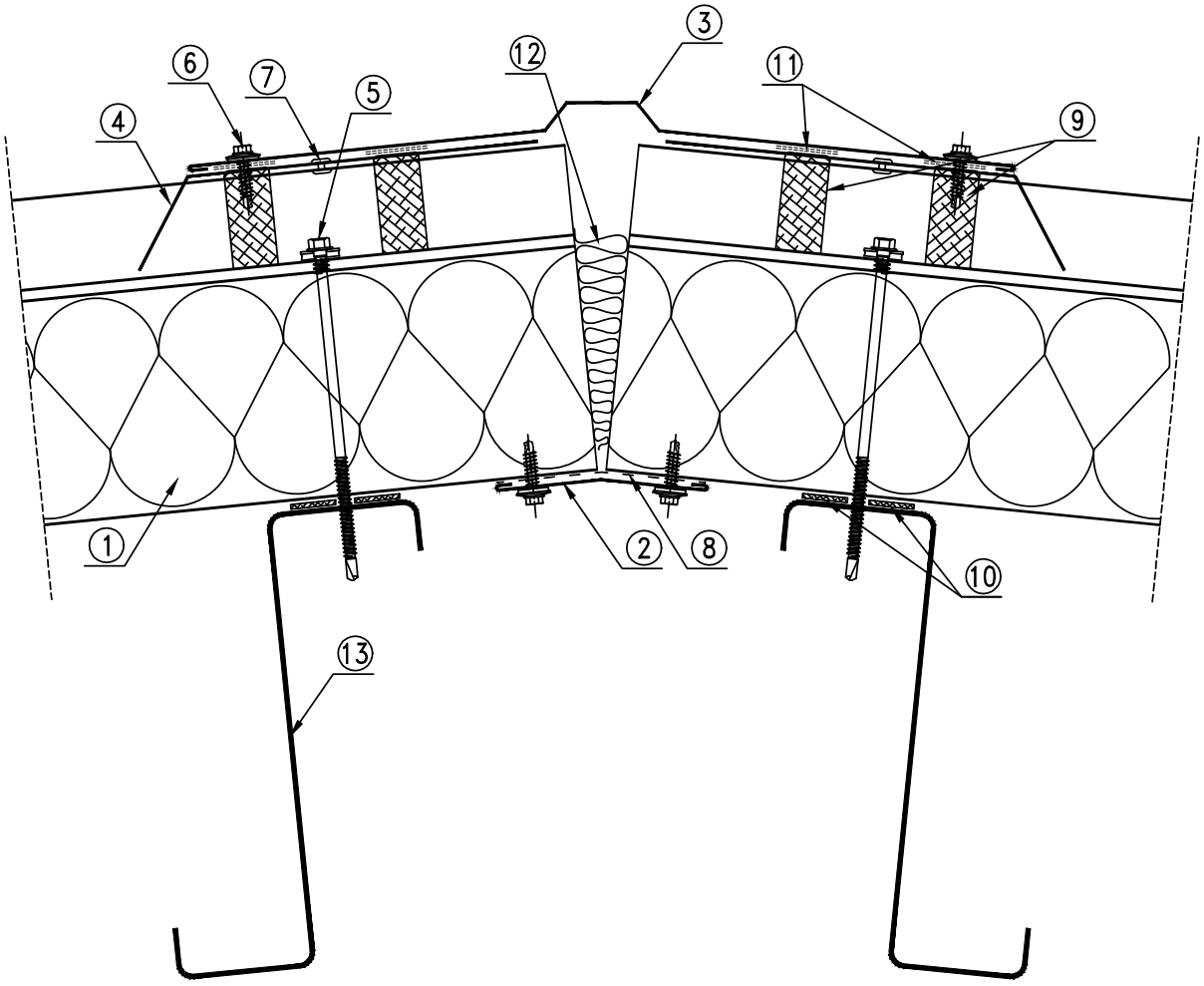
1. Roof sandwich panel PIR ROOF
2. Wall sandwich panel
3. OBR202.1 Wind blade flashing
4. OBR104 Internal strip flashing
5. Self-drilling fastener or rivet every 300mm
6. Polyurethane gasket or mounting foam

3.6.
R05 Roof parapet joint



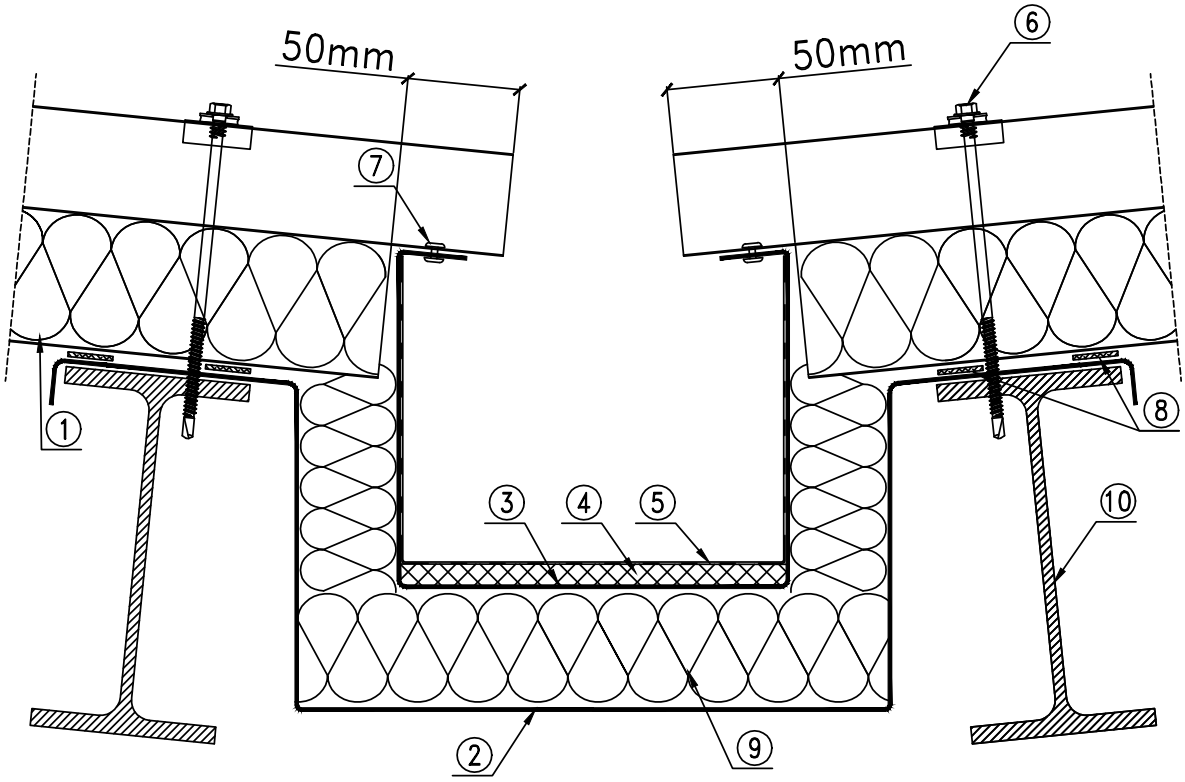
1. Roof sandwich panel PIR ROOF
2. Wall sandwich panel
3. OBR112 Paraphet eaves flashing
4. Individual flashing
5. Self-drilling fastener or rivet every 300mm
6. Sandwich panel fastener with saddle pad
7. PES self-adhesive sealing tape
8. Polyurethane gasket or mounting foam
9. Load bearing purlin acc. to construction design
10. Load bearing beam acc. to construction design
11. Bent out top cladding

3.7.
R06 Roof ridge joint



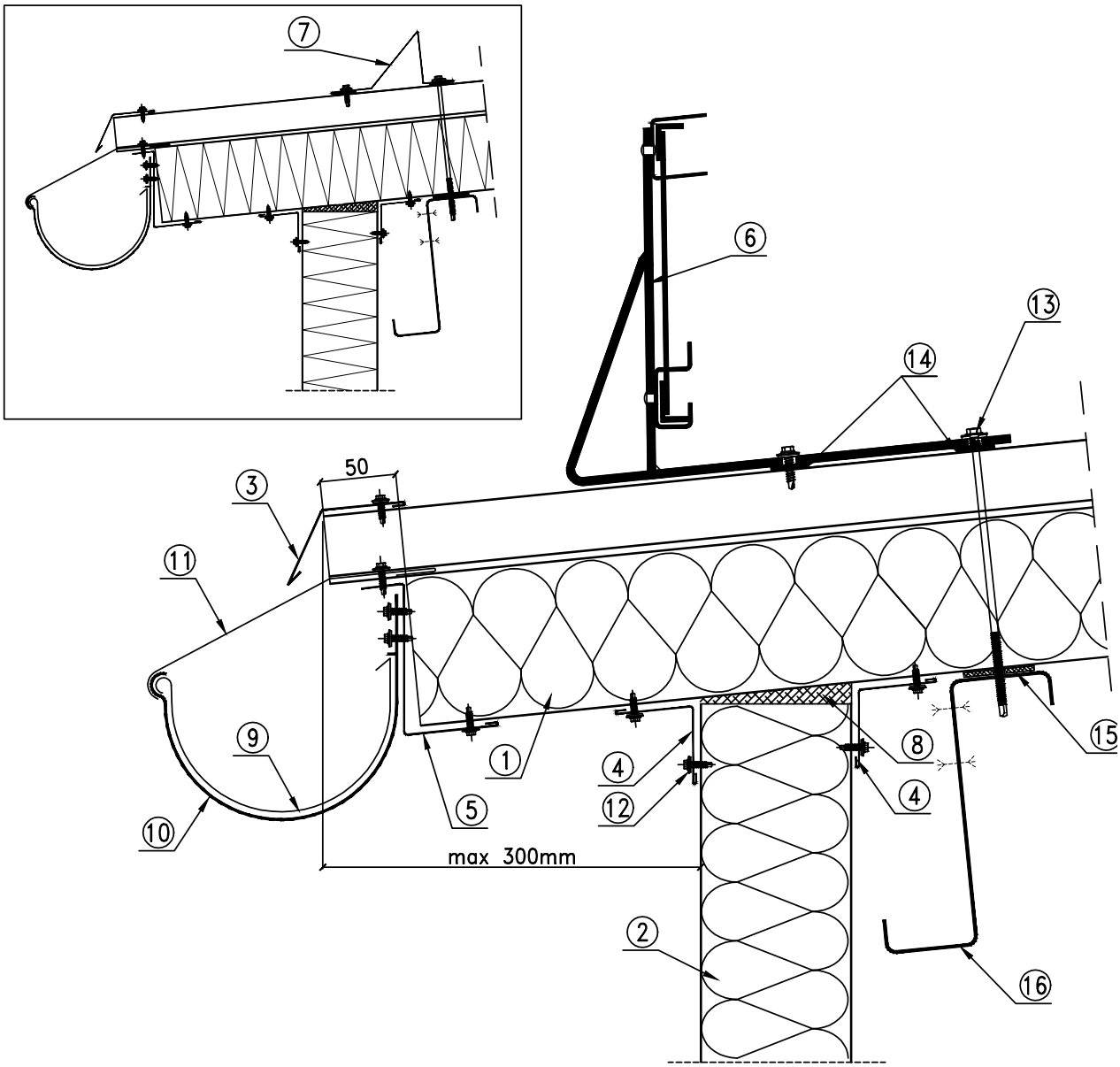
1. Roof sandwich panel PIR ROOF
2. OBR104 Internal strip flashing
3. OBR52 Profiled external roof ridge or OBR205 Flat external ridge flashing
4. OBR201 Roof ridge strip flashing
5. Sandwich panel fastener
6. Self-drilling fastener or rivet every 300mm
7. Assembly rivet approximately every 1000mm
8. Alubutyl tape
9. Sealing tape TUN45
10. PES self-adhesive sealing tape
11. Butyl sealing tape
12. Polyurethane gasket or mounting foam
13. Load bearing purlin acc. to construction design

3.8.
R07 Internal gutter joint



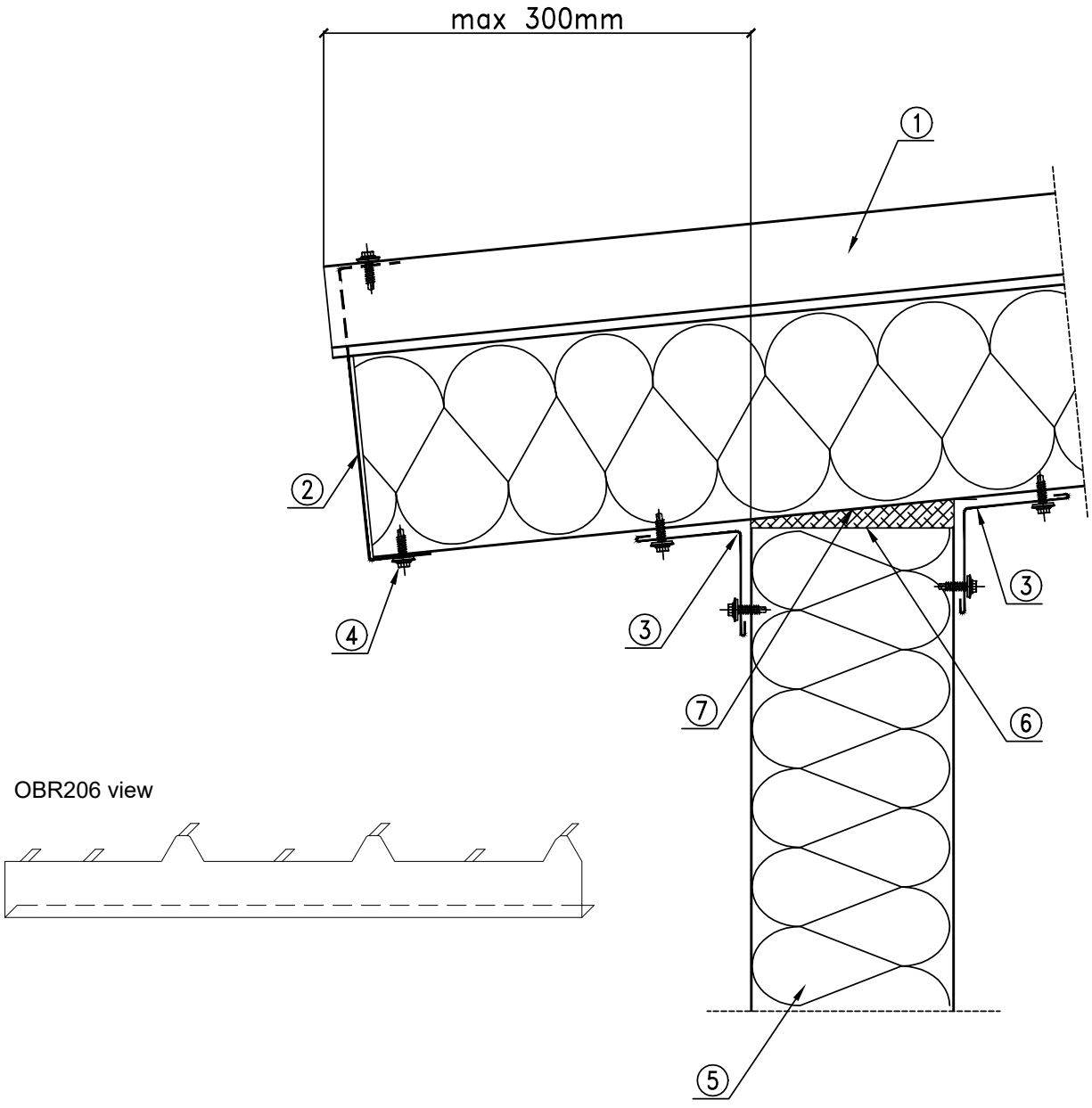
1. Roof sandwich panel PIR ROOF with cutback
2. Internal gutter top profile
3. Internal gutter bottom profile
4. Heating mat
5. Roof membrane
6. Sandwich panel fastener with saddle pad
7. Self-drilling fastener or rivet every 300mm
8. PES self-adhesive sealing tape
9. Gutter thermal insulation
10. Load bearing purlin acc. to construction design

3.9.
R08.1 Roof eave joint



1. Roof sandwich panel PIR ROOF
2. Wall sandwich panel
3. OBR203 Over gutter drip cap flashing
4. OBR104 Internal strip flashing
5. OBR62 Eaves masking frame flashing
6. Snowguard
7. OBR204 Snowguard flashing
8. Polyurethane gasket or mounting foam
9. Gutter acc. to architecture design
10. Gutter bracket
11. Gutter strap
12. Self-drilling fastener or rivet every 300mm
13. Sandwich panel fastener with saddle pad
14. Butyl mass or tape
15. PES self-adhesive sealing tape
16. Load bearing purlin acc. to construction design

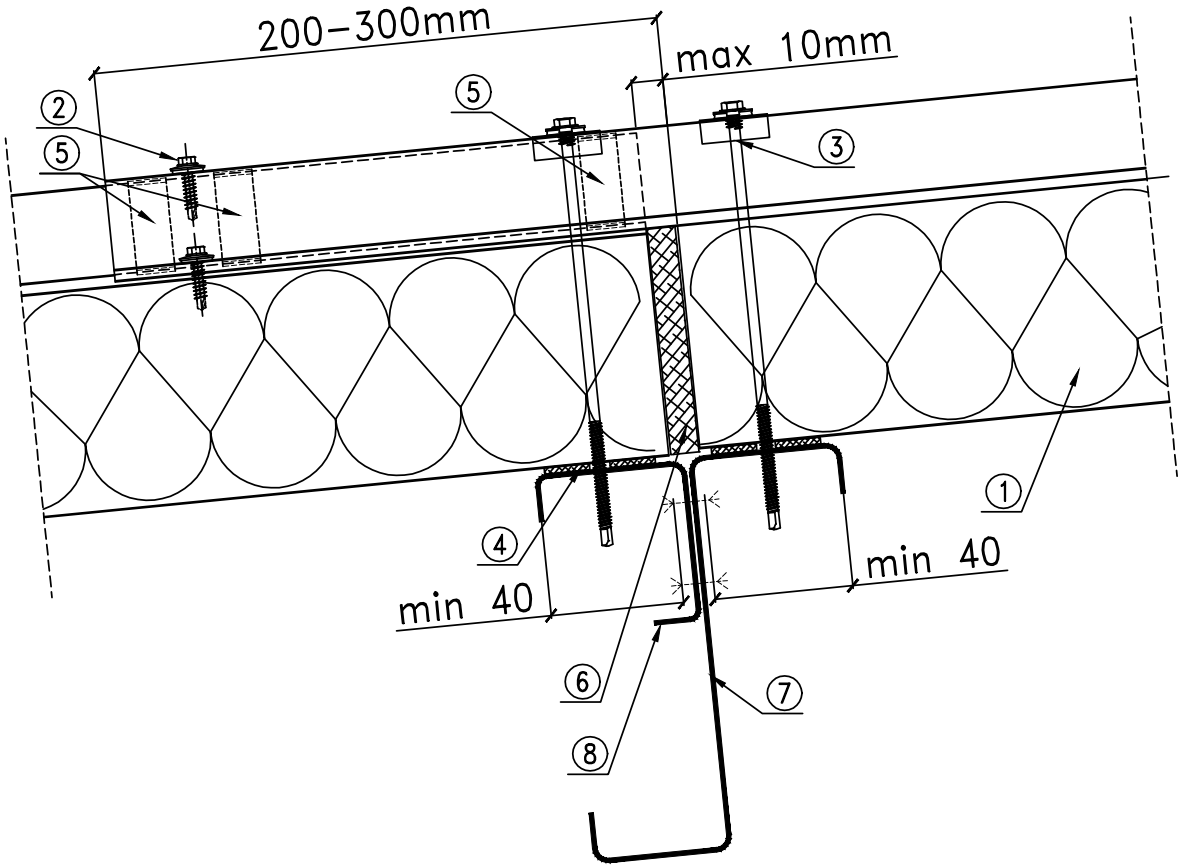
3.10.
R08.2 Roof eave joint



1. Roof sandwich panel PIR ROOF
2. OBR206 Clousure flashing
3. OBR104 Internal strip flashing
4. Self-drilling fastener or rivet every 300mm
5. Wall sandwich panel
6. Polyurethane gasket or mounting foam

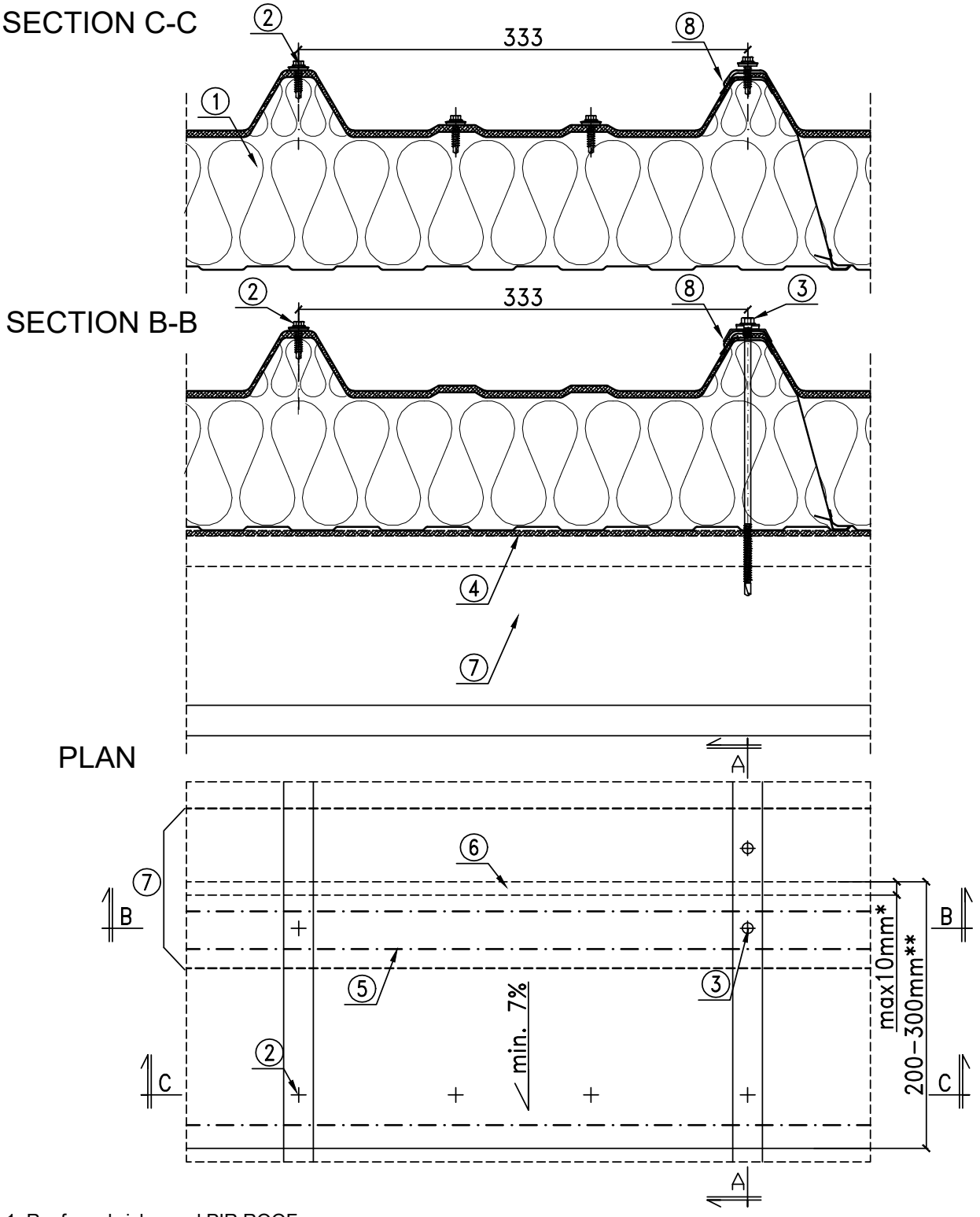
3.11.
R09.1 Panel lenthwise joint

SECTION A-A



1. Roof sandwich panel PIR ROOF
2. Self-drilling fastener or rivet
3. Sandwich panel fastener with saddle pad
4. PES self-adhesive sealing tape
5. Butyl sealing tape
6. Polyurethane gasket or mounting foam
7. Load bearing purlin acc. to construction design
8. Load bearing purlin width extension profile
acc. to construction design (to ensure both panels support)

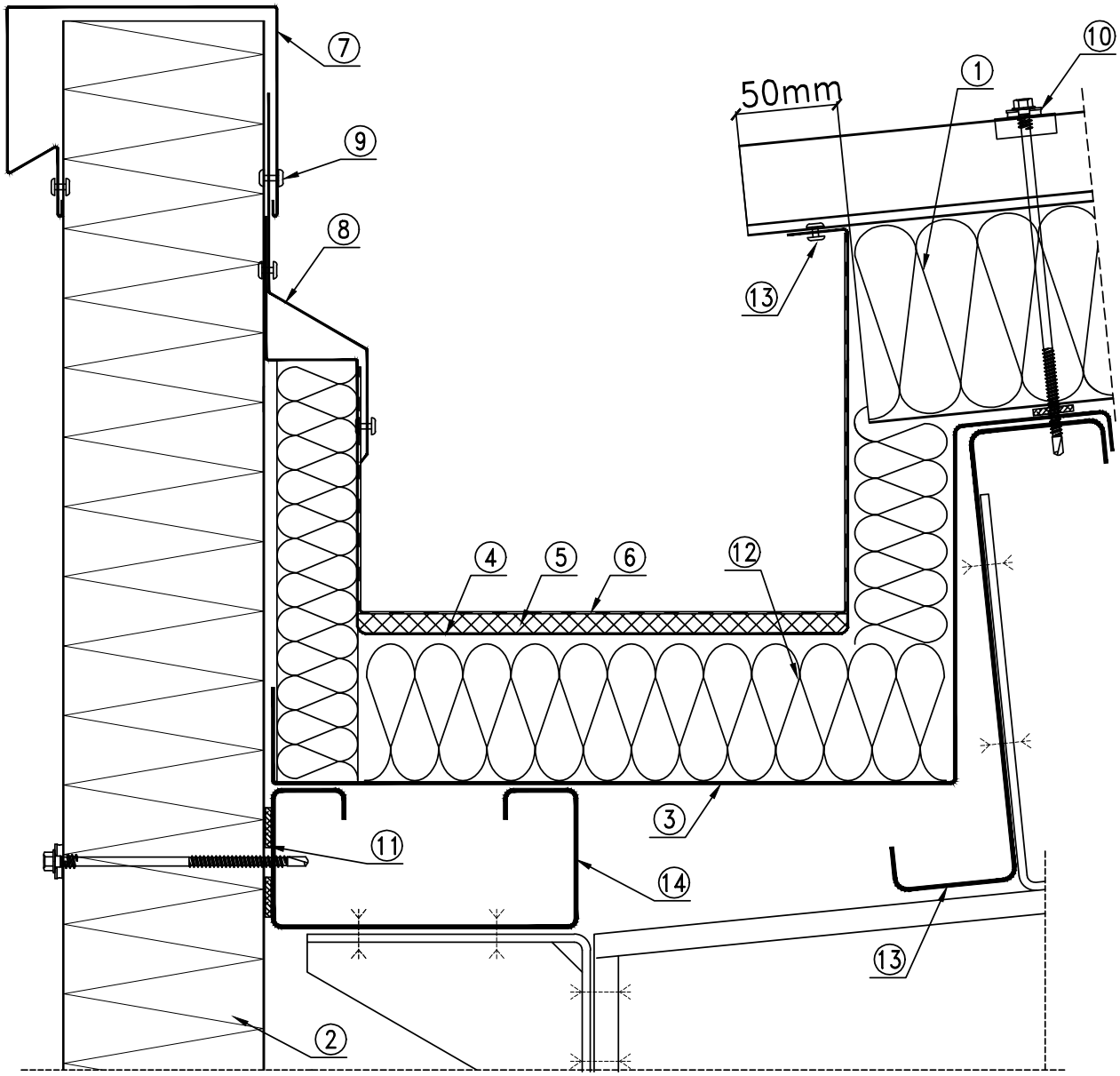
3.12.
R09.2 Panel lenthwise joint



1. Roof sandwich panel PIR ROOF
2. Self-drilling fastener or rivet
3. Sandwich panel fastener with saddle pad
4. PES self-adhesive sealing tape
5. Butyl sealing tape
6. Polyurethane gasket or mounting foam
7. Load bearing purlin acc. to construction design
8. Butyl mass

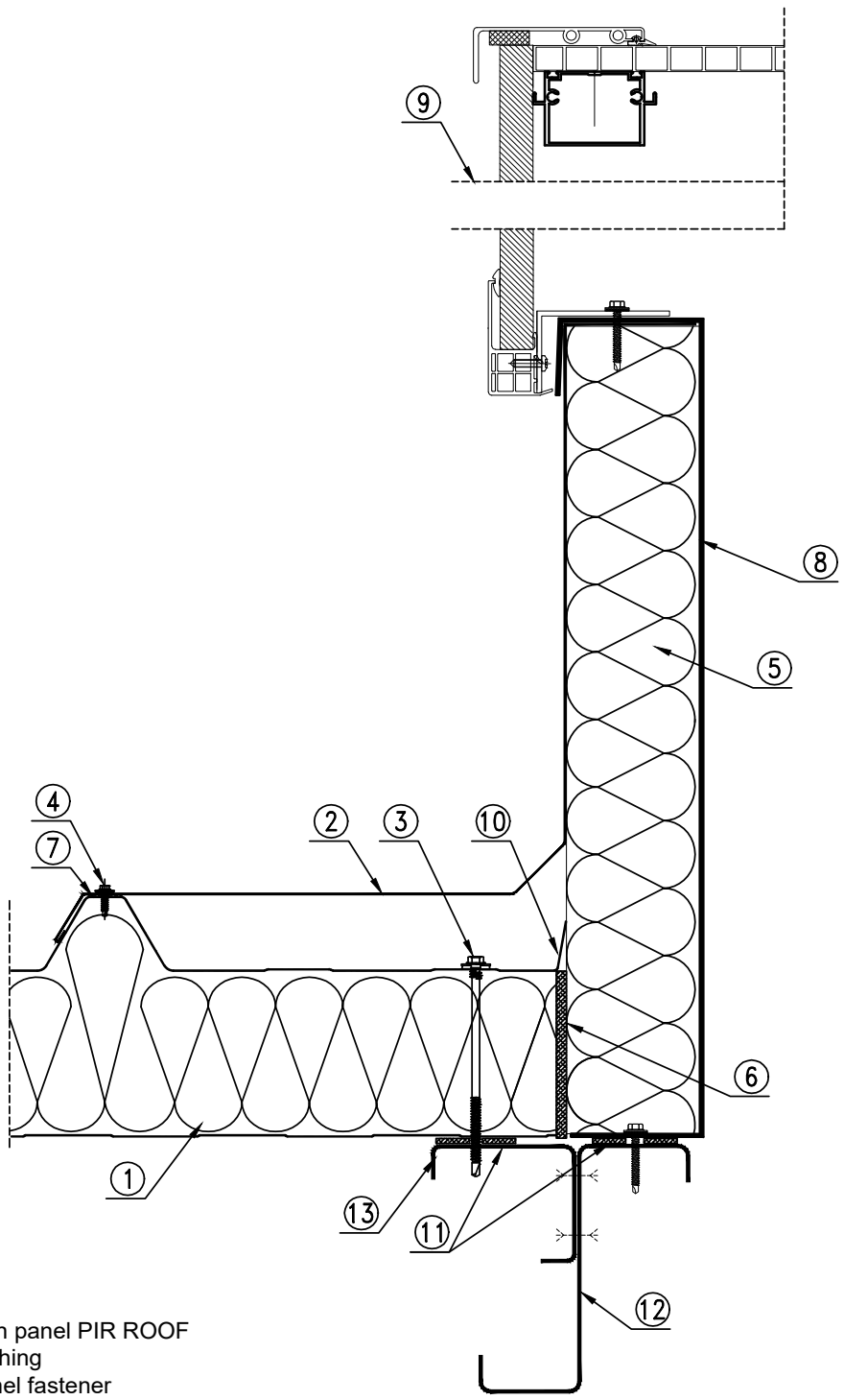
*gap between joining panels
** cutback length

3.13.
R10 Internal gutter joint near parapet



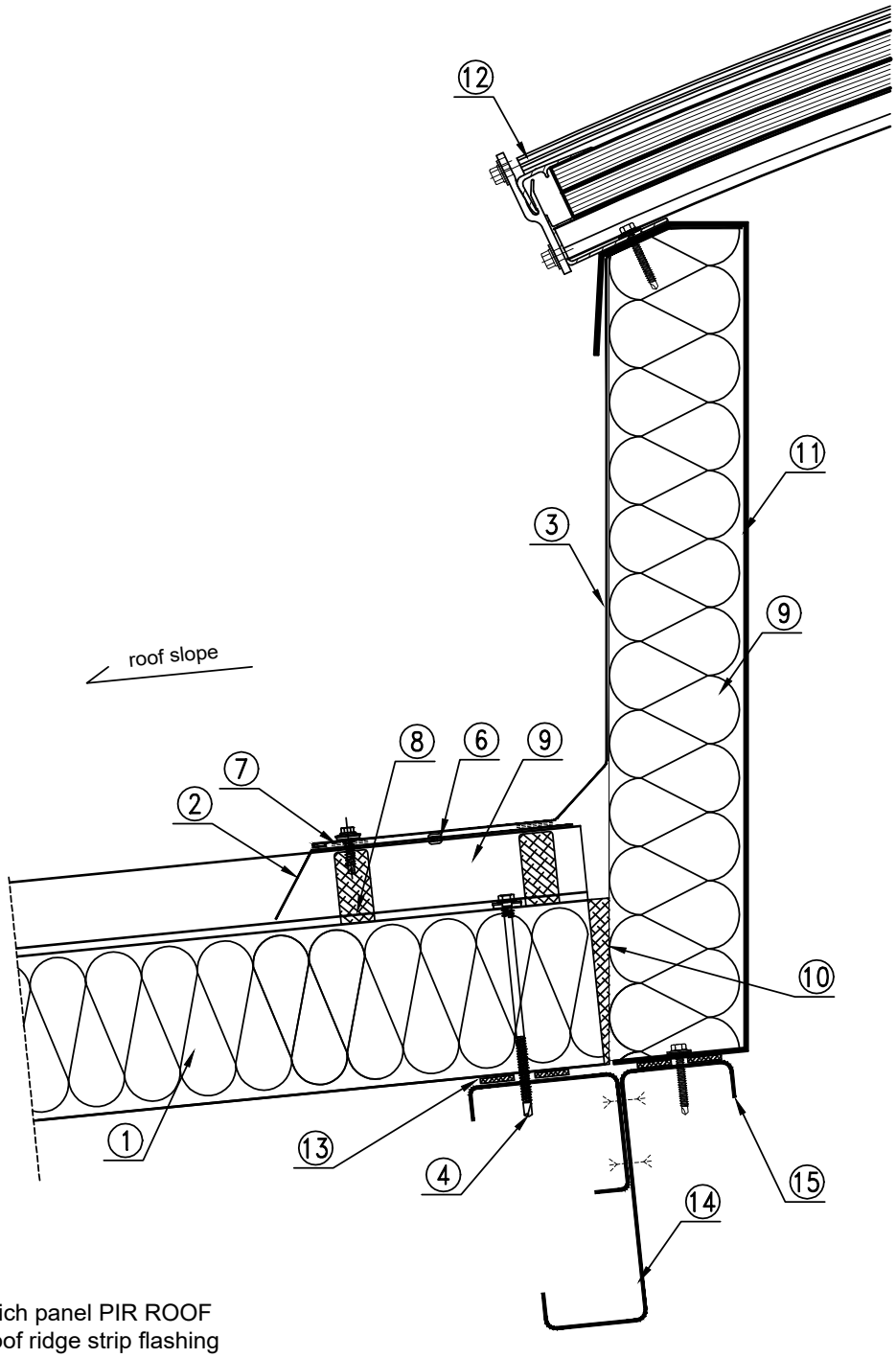
1. Roof sandwich panel PIR ROOF with cutback
2. Wall sandwich panel
3. Internal gutter top profile
4. Internal gutter bottom profile
5. Heating mat
6. Roof membrane
7. OBR112 Parapet eave flashing
8. Individual flashing
9.Self-drilling fastener or rivet every 300mm
10. Sandwich panel fastener with saddle pad
11. PES self-adhesive sealing tape
12. Gutter thermal insulation
13. Load bearing purlin acc. to construction design
14. Load bearing gutter support profile
acc. to construction design

3.14.
R11.1 Roof ridge skylight-longitudal section



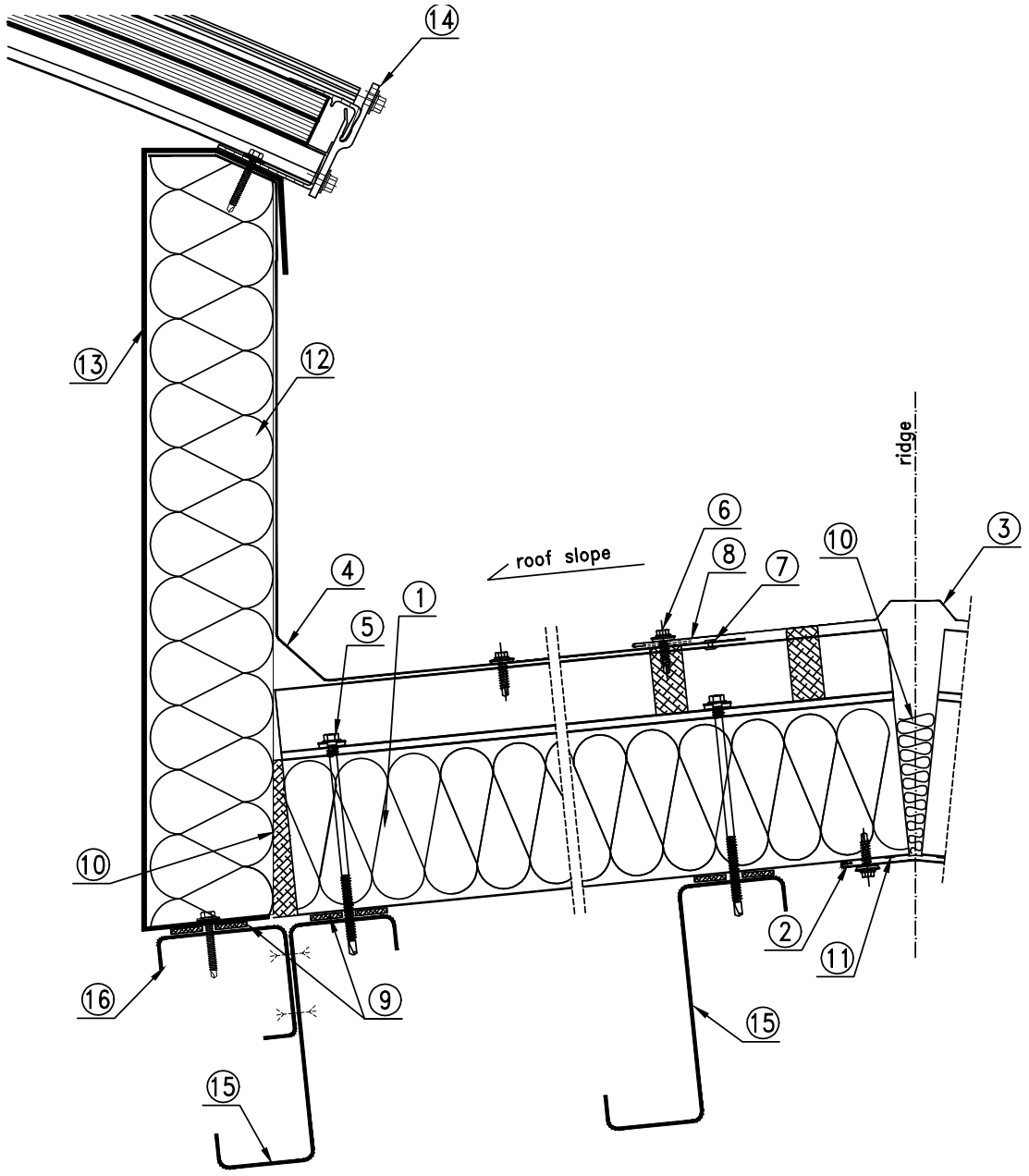
1. Roof sandwich panel PIR ROOF
2. Individual flashing
3. Sandwich panel fastener
4. Self-drilling fastener or rivet every 300mm
5. Thermal insulation np. THERMANO
6. Polyurethane gasket or mounting foam
7. Butyl tape
8. Skylight base for example Kingspan LB Basic
9. Skylight for example Kingspan LB Basic
10. Bent out top cladding
11. PES self-adhesive sealing tape
12. Load bearing purlin acc. to construction design
13. Load bearing purlin width extension profile
acc. to construction design (to ensure skylight support)

3.15.
R11.2 Roof ridge skylight-cross section



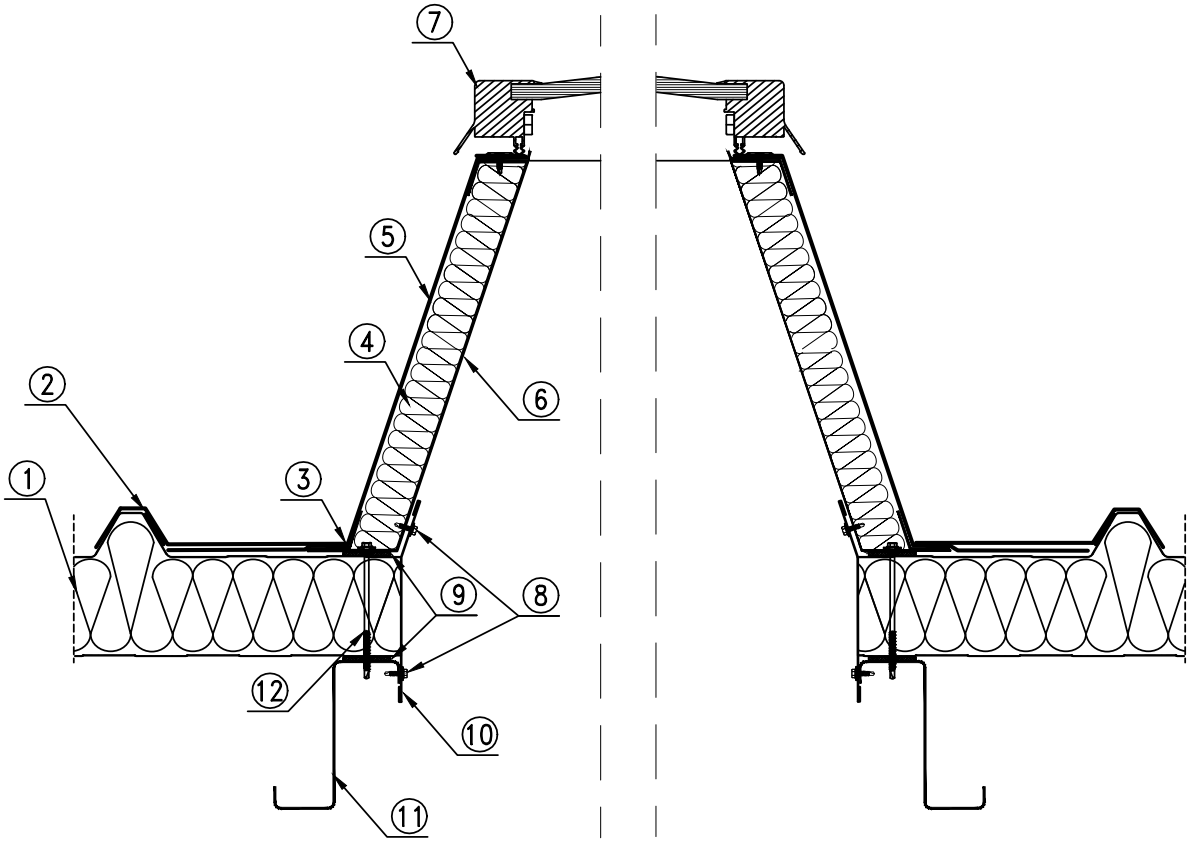
1. Roof sandwich panel PIR ROOF
2. OBR201 Roof ridge strip flashing
3. Individual flashing
4. Sandwich panel fastener
5. Self-drilling fastener or rivet every 300mm
6. Assembly rivet approximately every 1000mm
7. Butyl tape
8. Sealing tape TUN45
9. Thermal insulation np. THERMANO
10. Polyurethane gasket or mounting foam
11. Skylight base for example Kingspan LB Basic
12. Skylight for example Kingspan LB Basic
13. PES self-adhesive sealing tape
14. Load bearing purlin acc. to construction design
15. Load bearing purlin width extension profile
acc. to construction design (to ensure skylight support)

3.16.
R12 Roof skylight near ridge



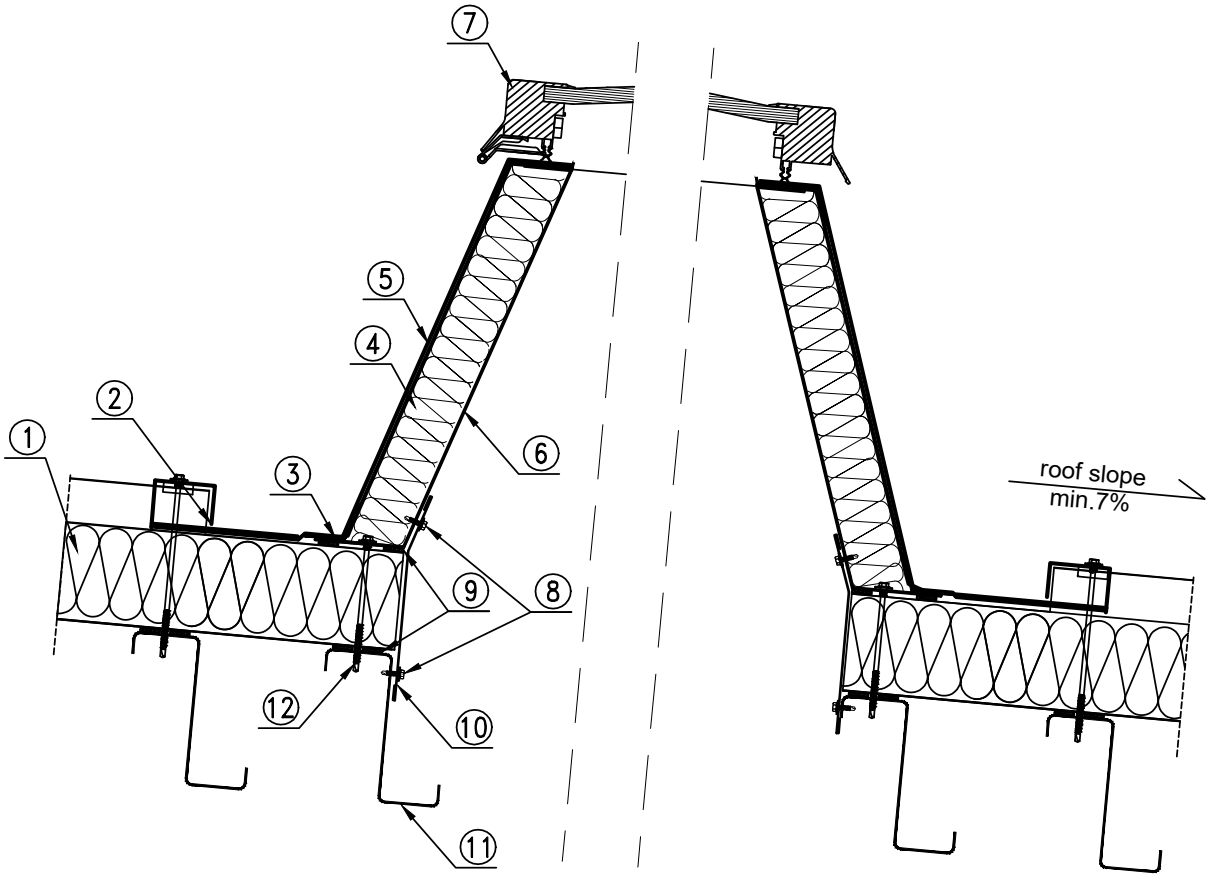
1. Roof sandwich panel PIR ROOF
2. OBR104 Internal strip flashing
3. OBR52 Profiled external roof ridge or OBR205 Flat external ridge flashing
4. Individual flashing - up to roof ridge
5. Sandwich panel fastener
6. Self-drilling fastener or rivet every 300mm
7. Assembly rivet approximately every 1000mm
8. Butyl tape
9. PES self-adhesive sealing tape
10. Polyurethane gasket or mounting foam
11. Alubutyl tape
12. Thermal insulation np. THERMANO
13. Skylight base for example Kingspan LB Basic
14. Skylight for example Kingspan LB Basic
15. Load bearing purlin acc. to construction design
16. Load bearing purlin width extension profile
acc. to construction design (to ensure skylight support)

3.17.
R13.1 Roof skylight - longitudal section



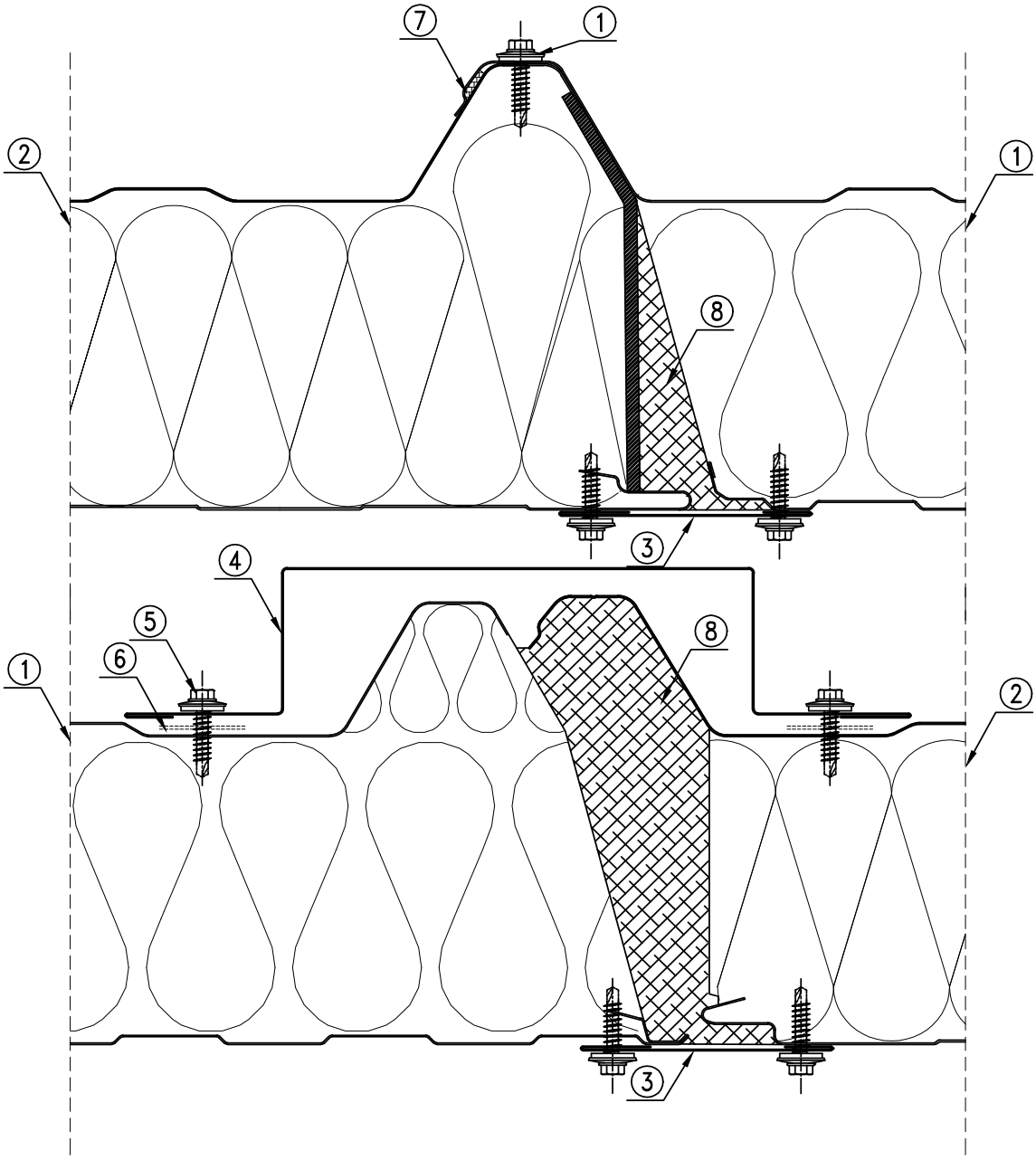
1. Roof sandwich panel PIR ROOF
2. Liquid foil
3. PCV coated steel angle profile
4. Thermal insulation np. THERMANO
5. PCV membrane glued to skylight base and insulation
6. Skylight for example Kingspan LK PCs
7. Skylight base for example Kingspan LK PC
8. Self-drilling fastener
9. PES self-adhesive sealing tape
10. Individual flashing
11. Load bearing purlin acc. to construction design
12. Sandwich panel fastener

3.18.
R13.2 Roof skylight-cross section



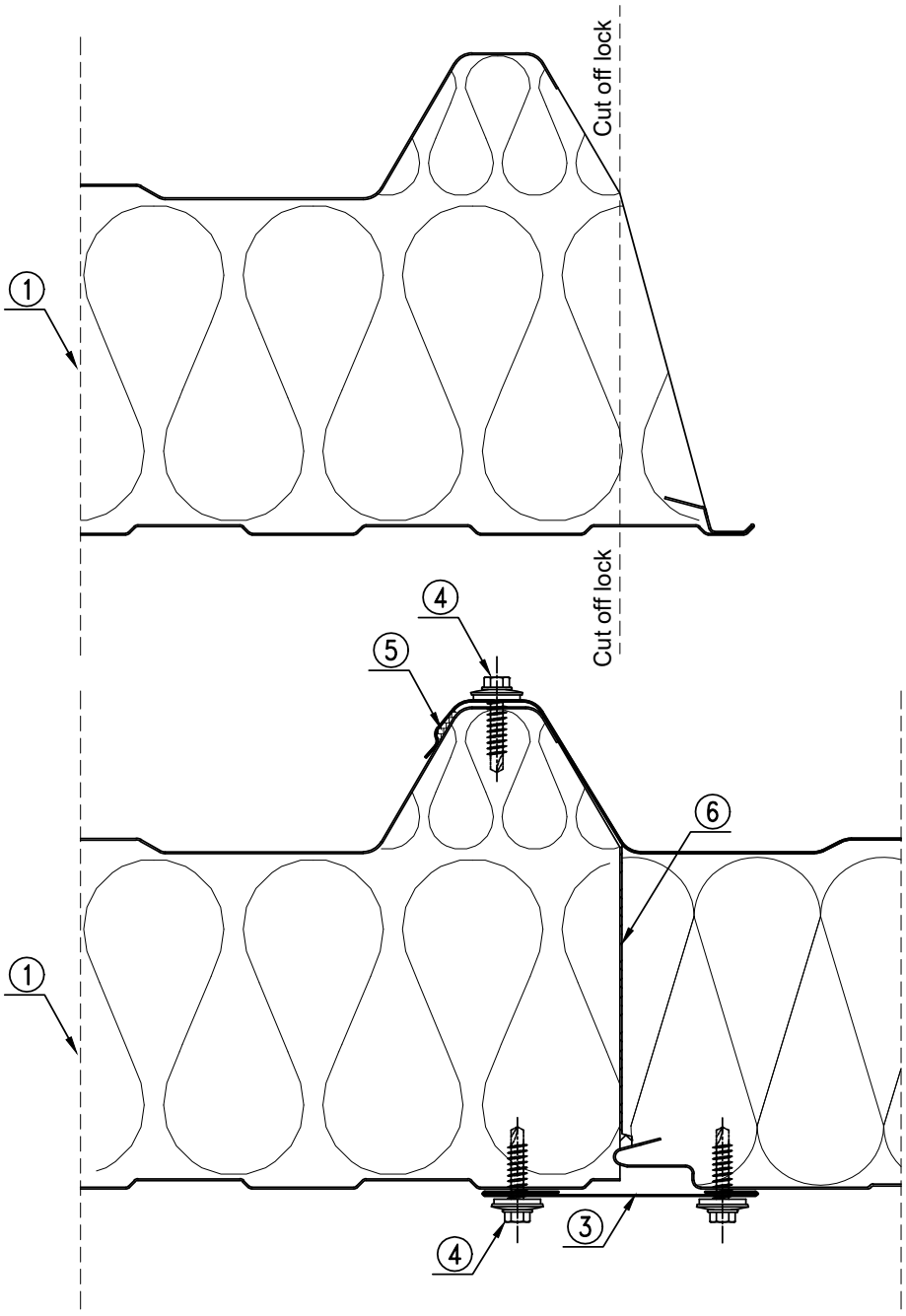
1. Roof sandwich panel PIR ROOF
2. Liquid foil
3. PCV coated steel angle profile
4. Thermal insulation np. THERMANO
5. PCV membrane glued to skylight base and insulation
6. Skylight for example Kingspan LK PCs
7. Skylight base for example Kingspan LK PC
8. Self-drilling fastener
9. PES self-adhesive sealing tape
10. Individual flashing
11. Load bearing purlin acc. to construction design
12. Sandwich panel fastener

3.19.
R14.1 Straight lock panel joint



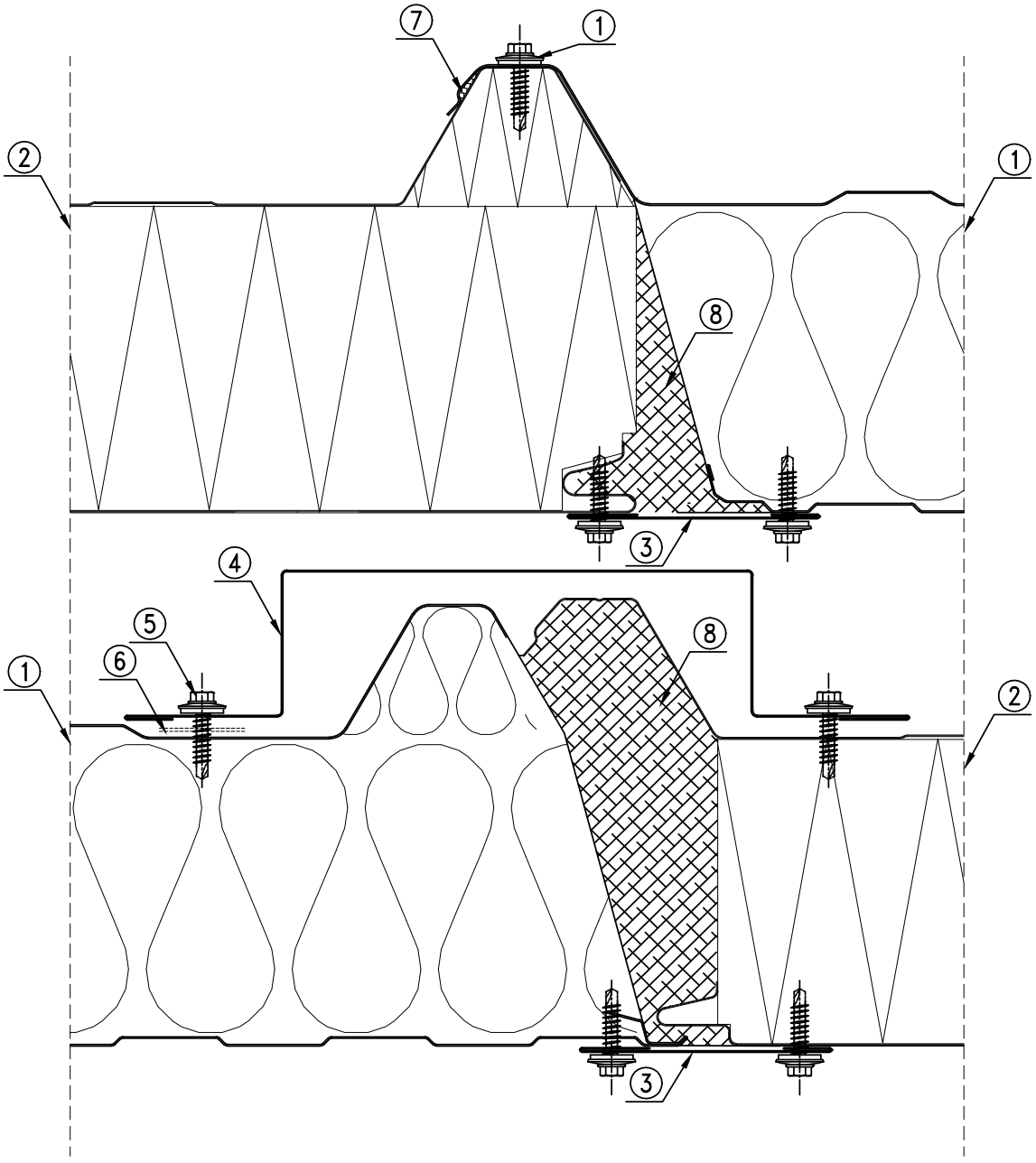
1. Roof sandwich panel PIR ROOF
2. Roof panel with straight lock
3. OBR106/2 masking lath flashing
4. OBR208 flashing
5. Self-drilling fastener or rivet every 300mm
6. Butyl tape
7. Butyl mass
8. Fireproof mouting foam

3.20.
R14.2 Straight lock panel joint



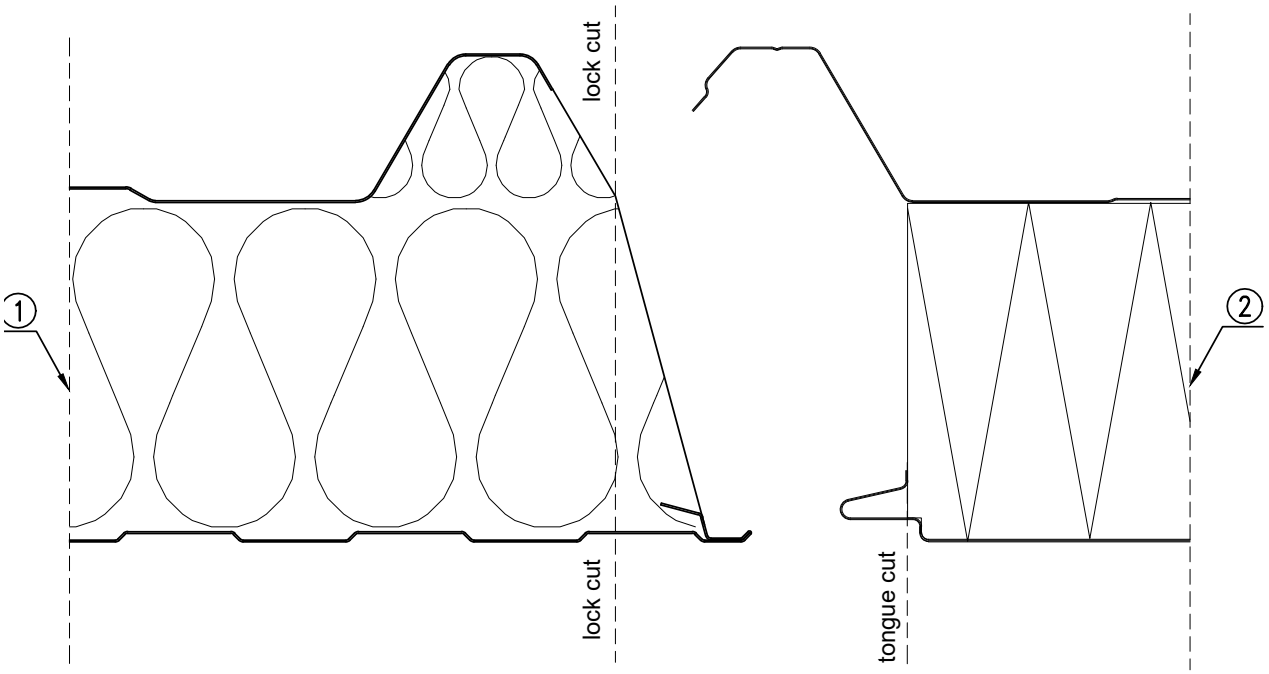
1. Roof sandwich panel PIR ROOF with cut off lock
2. Roof panel with straight lock
3. OBR106/2 masking lath flashing
4. Self-drilling fastener or rivet every 300mm
5. Butyl tape
6. Fireproof mounting foam

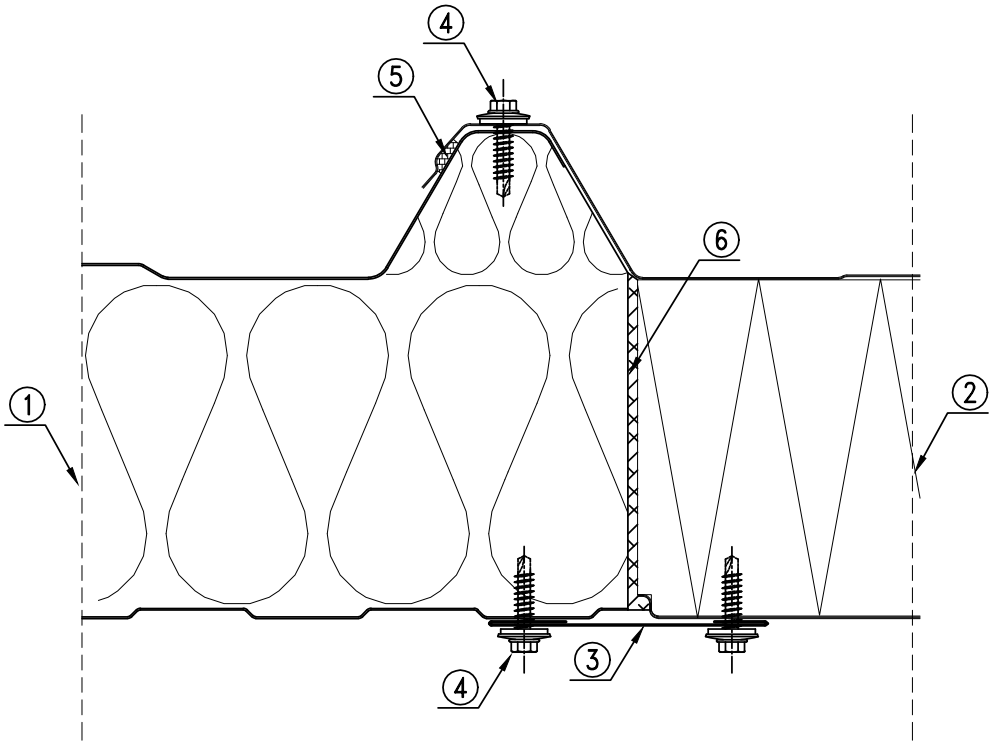
3.21.
R15.1 MW ROOF joint



1. Roof sandwich panel PIR ROOF
2. Roof sandwich panel MW ROOF
3. OBR106/2 masking lath flashing
4. OBR208 flashing
5. Self-drilling fastener or rivet every 300mm
6. Butyl tape
7. Butyl mass
8. Fireproof mouting foam

3.22.
R15.2 MW ROOF joint





1. Roof sandwich panel PIR ROOF with cut off lock
2. Roof sandwich panel MW ROOF with cut off lock
3. OBR106/2 masking lath flashing
4. Self-drilling fastener or rivet every 300mm
5. Butyl mass
6. Fireproof mouting foam

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